End-to-End CI/CD Pipeline for a Calculator REST API

Detailed implementation report

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CSE 816 - Software Production Engineering

GitHub https://github.com/kavyagupta3011/calculator-devops

Docker Hub https://hub.docker.com/r/kavyag3011/calculator-devops

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1 Introduction

The project is a calculator REST API written in Java 21 with Spring Boot and Maven, together with the automation that takes a code change from a developer's laptop to a running, health-checked deployment without manual steps.

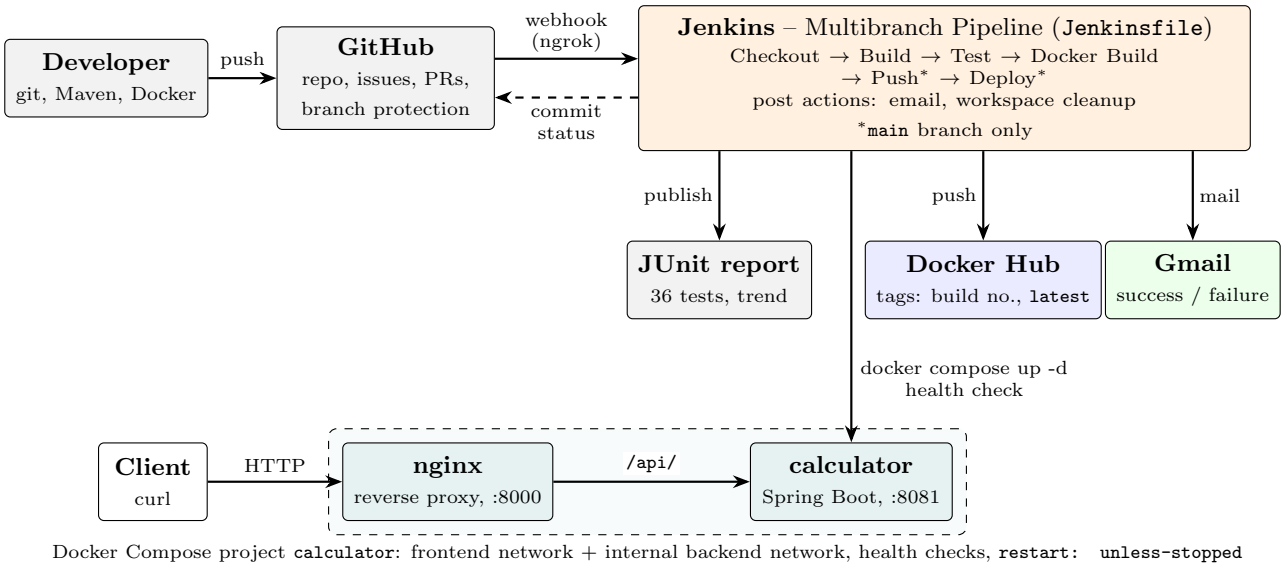


Figure 1: Architecture of the pipeline.

Area	Tools and versions
Application	Java 21 (Temurin 21.0.12), Spring Boot 3.5.6, Spring Web, Spring Boot Actuator, Maven 3.6.3
Testing	JUnit 5, Spring MockMvc, AssertJ (36 tests in v1.0.0, 43 after adding modulo)
Version control	Git 2.37, GitHub (issues, labels, milestones, project board, pull requests, branch protection)
CI/CD	Jenkins 2.568.3 on Java 21, Multibranch Pipeline, declarative `Jenkinsfile`, ngrok for the webhook
Containers	Docker 20.10.21, Docker Hub
Deployment	Docker Compose v5.6.0, Nginx 1.27

Table 1: Technology stack.

The work was organised in phases, each tracked as a GitHub issue and delivered through its own branch and pull request:

Phase	Branch	PR / issue	Result
0	–	–	Build machine prepared (Docker access, plugins, credentials, email)
1	`main`	issue #1	Repository, templates, labels, milestones, board, branch protection
2	`feature/calculator-api`	PR #8, issue #2	REST API, validation, 36 tests (merge commit)
3	`feature/docker`	PR #9, #10, issue #3	Multi-stage Dockerfile, Docker Hub (rebase, squash)
4	`feature/compose`	PR #11, #12, issue #4	Compose stack with Nginx (merge conflict, rebase and merge)
5	`feature/jenkins-pipeline`	PR #13, issue #5	Jenkins pipeline, first automatic deployment
5	`refactor/error-message`	PR #14, issue #15	Failing build caught, change reverted
6	`docs/readme-report`	PR #16, issue #6	Complete README; pipeline re-run
7	`main`	tag `v1.0.0`	Release v1.0.0 on GitHub and Docker Hub
8	`feature/modulo`	issue #17	Modulo endpoint added and deployed end to end (squash)

Table 2: Phases of the work.

2 Build machine setup

Every pipeline step runs as the `jenkins` system user, so the tools had to work for that user and not only for the developer account.

2.1 Docker Compose plugin

`docker compose` (Compose v2 syntax) failed with *"compose is not a docker command"*: only the standalone `docker-compose` binary from an earlier lab was installed in `/usr/local/bin`. The Docker CLI looks for plugins in `cli-plugins` directories, so the binary was linked into the system-wide directory, which makes the plugin available to every user including `jenkins`:

```
sudo mkdir -p /usr/local/lib/docker/cli-plugins
sudo ln -s "$(readlink -f "$(which docker-compose)")" /usr/local/lib/docker/cli-plugins/docker-compose
docker compose version           # Docker Compose version v5.6.0
sudo -u jenkins docker compose version
```

2.2 Docker access for the jenkins user

The Docker daemon only accepts commands on `/var/run/docker.sock` from root and members of the `docker` group. Without membership every `docker` step in the pipeline fails with `permission denied`. Group membership is read when a process starts, so Jenkins was restarted afterwards.

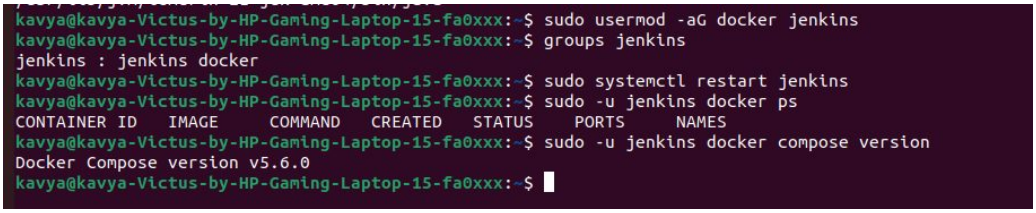


Figure 2: `jenkins` added to the `docker` group; `docker ps` and `docker compose` now work as that user.

2.3 Build node

A permanent SSH agent (an Ubuntu container without Maven or Docker) remained from earlier lab work with its usage set to *Use this node as much as possible*, so builds could be scheduled on it. It was restricted to jobs that ask for its label; the built-in node keeps two executors and runs this pipeline.

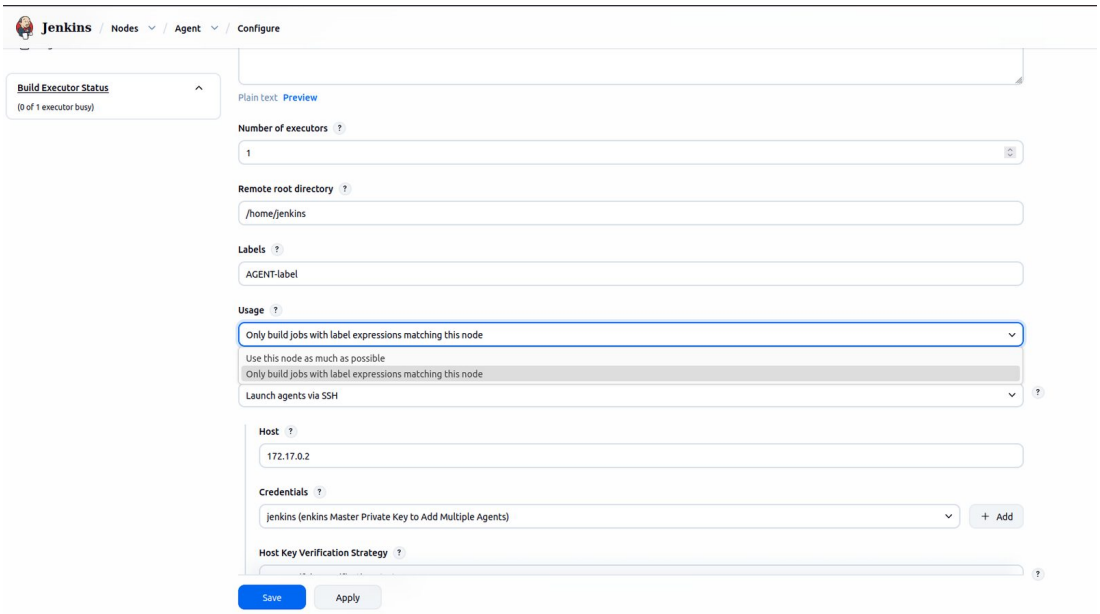


Figure 3: Agent node restricted to *Only build jobs with label expressions matching this node.*

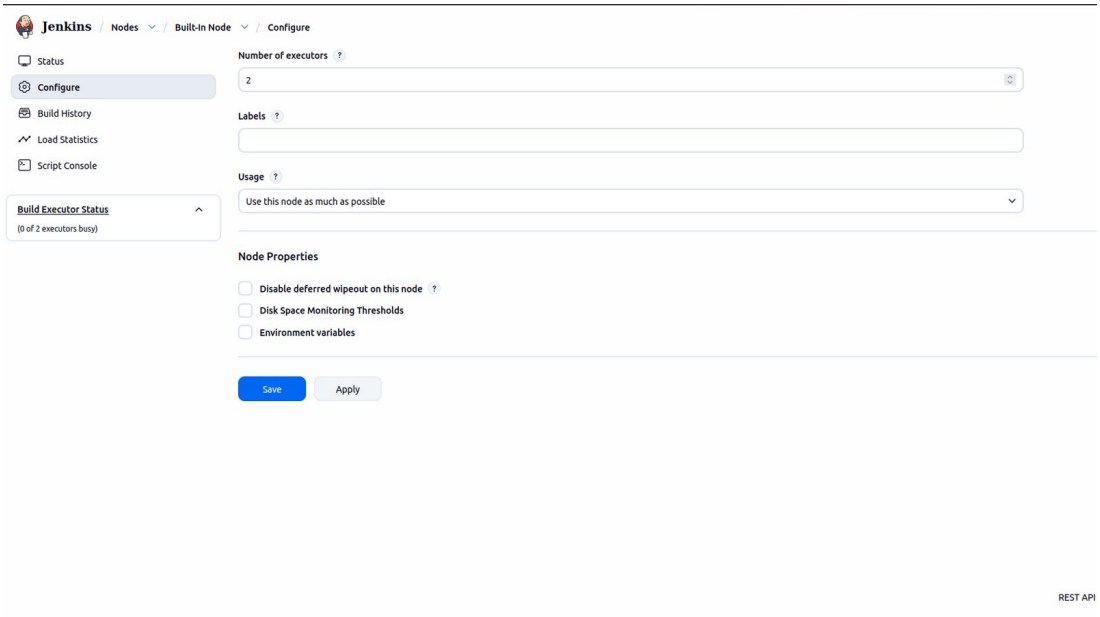


Figure 4: Built-in node with two executors.



Figure 5: Node list: the agent is offline and unused; the built-in node runs the builds.

2.4 Plugins

A Groovy script in the Script Console listed the plugins the pipeline depends on. Docker Pipeline (`docker.build`, `docker.withRegistry`) and Pipeline: Stage View were missing and were installed.



Figure 6: Script Console check for the required plugins.

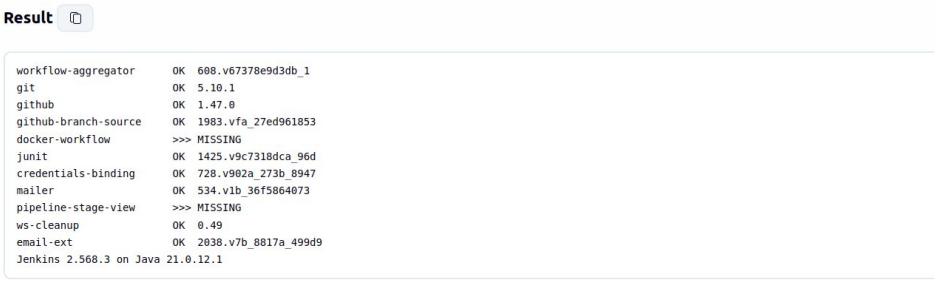


Figure 7: Before: `docker-workflow` and `pipeline-stage-view` missing.

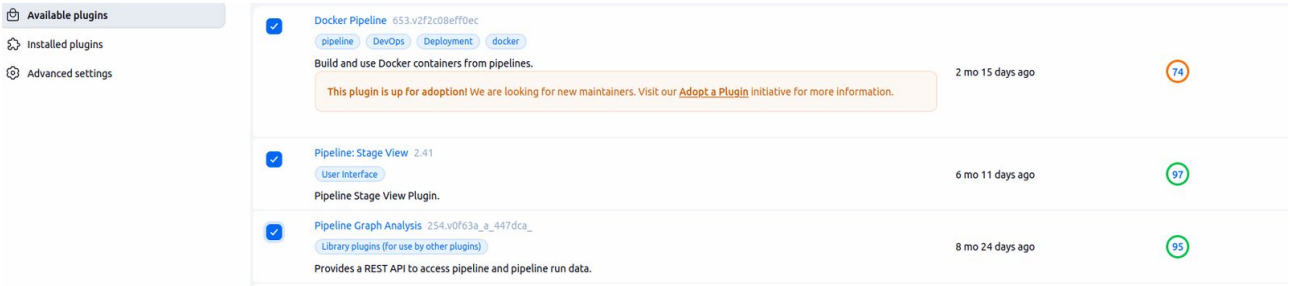


Figure 8: Installing Docker Pipeline and Pipeline: Stage View.

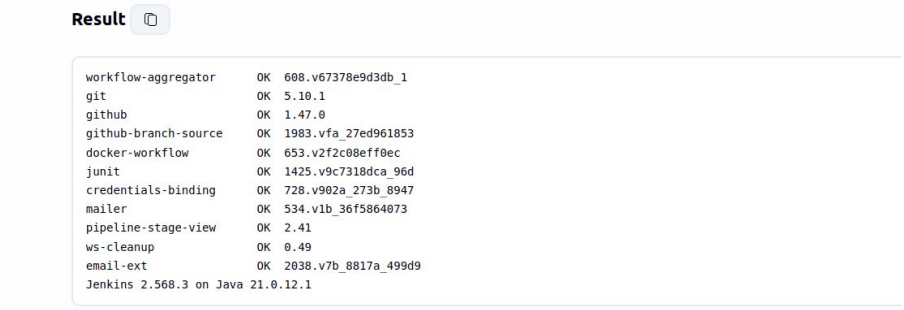


Figure 9: After: all plugins present; Jenkins 2.568.3 on Java 21.0.12.1.

2.5 Credentials

Secrets are stored encrypted in the Jenkins credentials store and referenced from the `Jenkinsfile` only by ID, so the public repository contains none. Personal access tokens with limited scope were used instead of account passwords.

ID	Kind	Use
`DockerHubCred`	username + Docker Hub access token (read/write)	Push stage
`github-pat`	username + GitHub token (`repo`, `admin:repo_hook`)	Multibranch job scanning the repository
`github-pat-secret`	secret text, same token	GitHub server configuration: webhook handling and commit statuses

Table 3: Jenkins credentials.

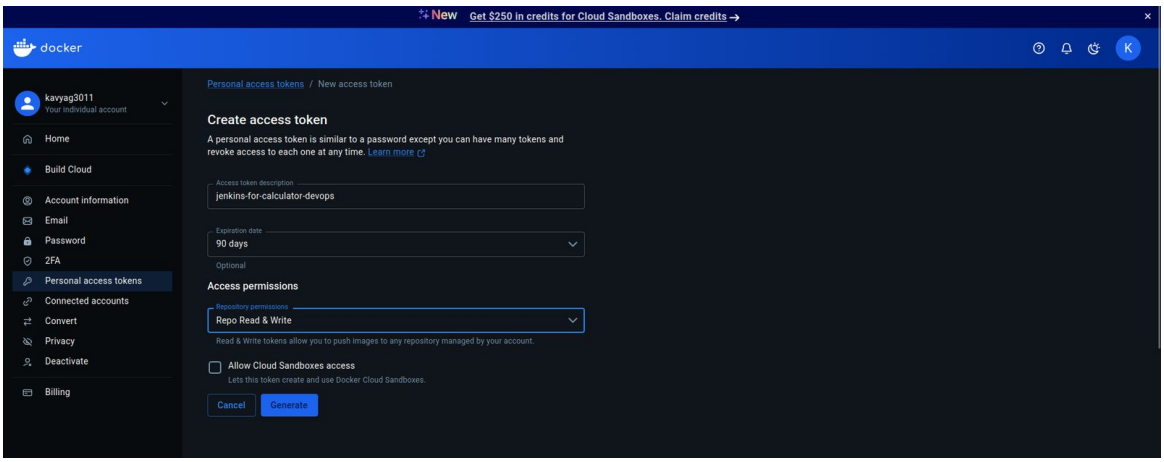


Figure 10: Docker Hub access token with read and write scope.

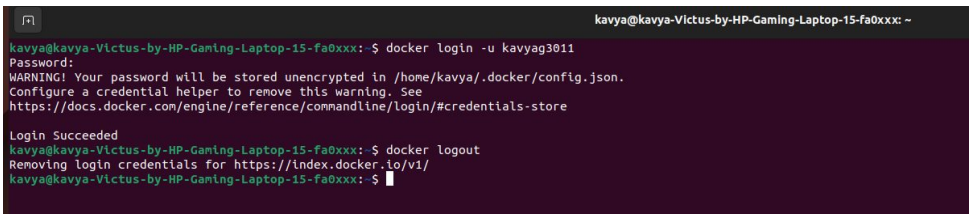


Figure 11: Token tested with `docker login`, followed by `docker logout` to remove the stored copy.

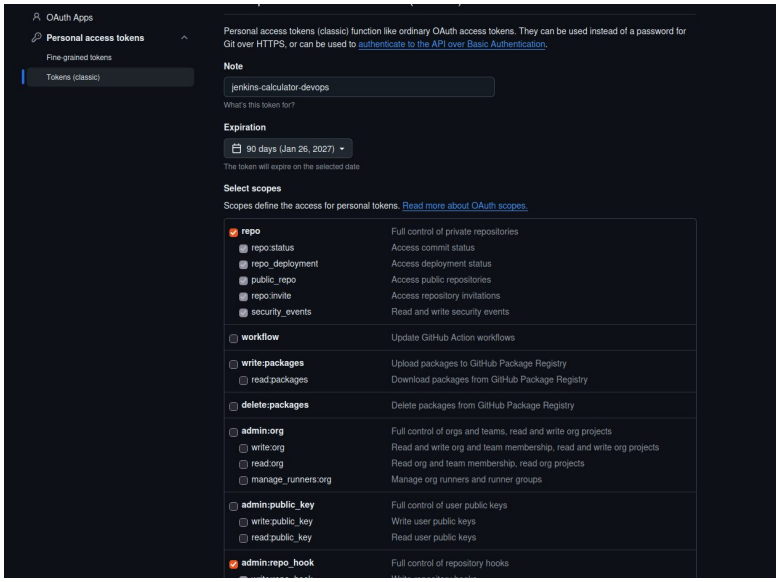


Figure 12: GitHub classic token scopes: `repo` and `admin:repo_hook`.

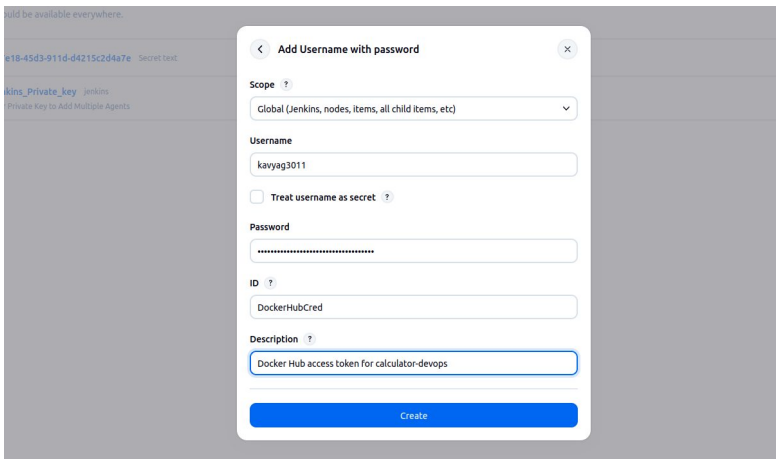


Figure 13: Adding `DockerHubCred`.

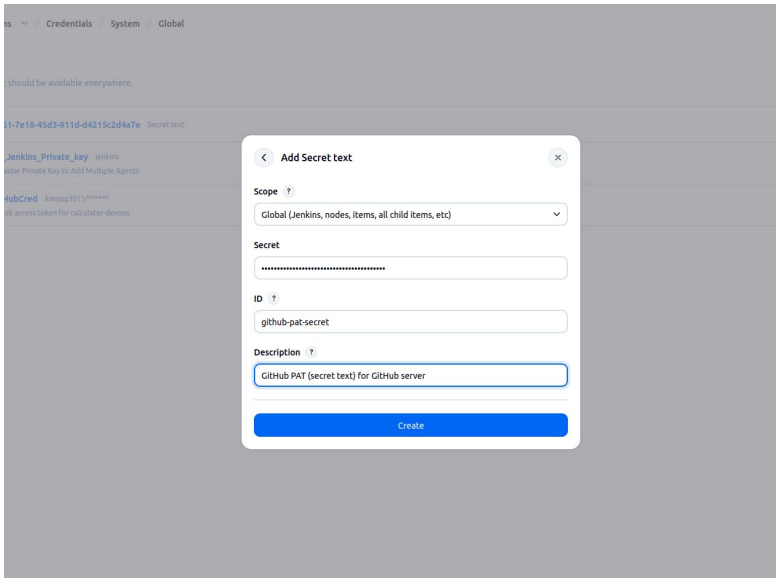


Figure 14: Adding `github-pat-secret`.

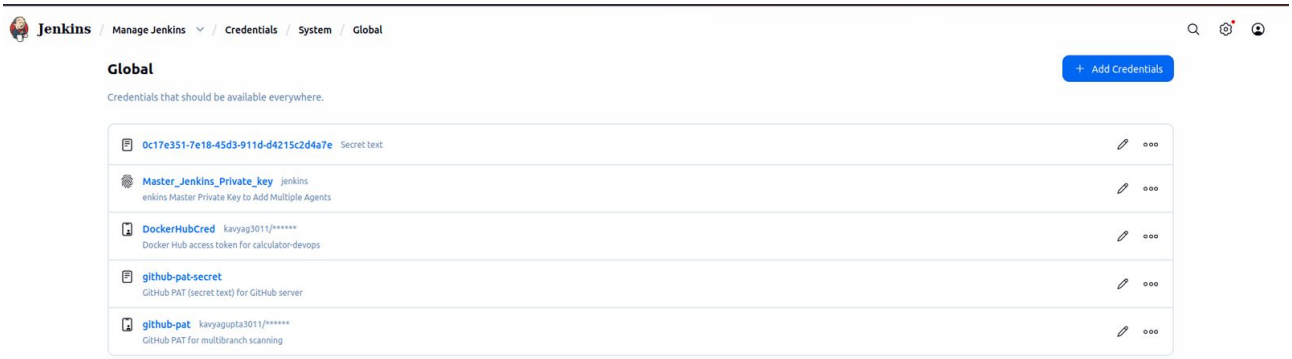


Figure 15: Global credentials; secret values are masked.

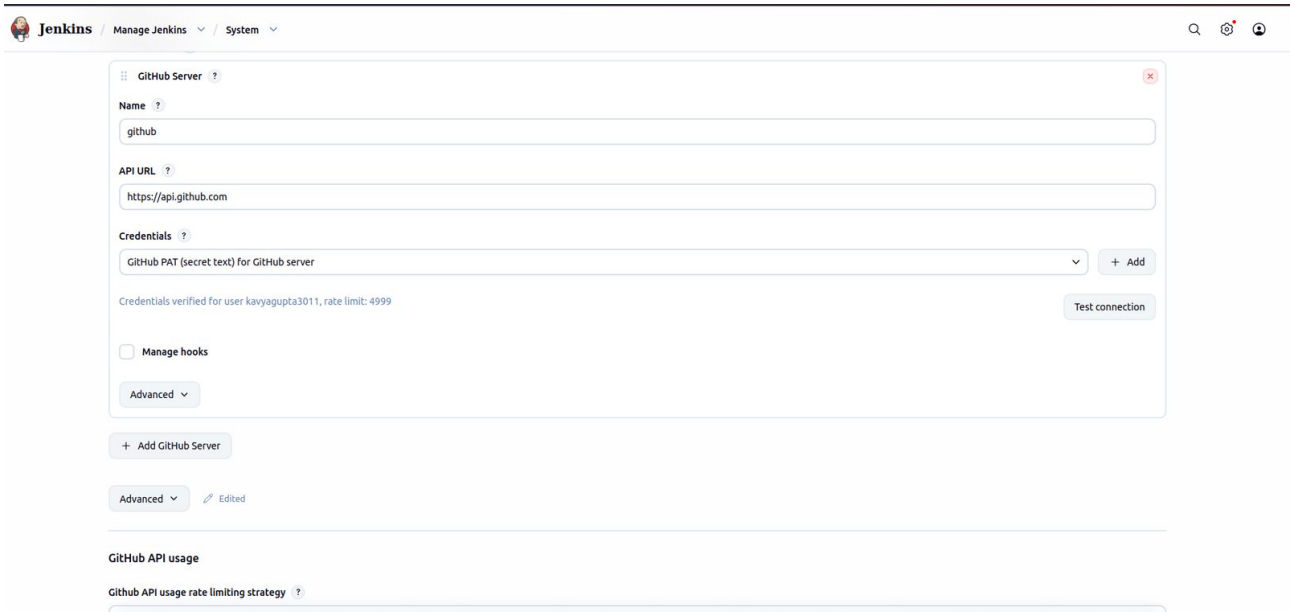


Figure 16: GitHub server configuration: credentials verified for `kavyagupta3011`.

2.6 Email

The Mailer plugin was configured for Gmail SMTP on port 465 with SSL and a Google app password; a test message confirmed delivery.



Figure 17: Email notification settings and test.

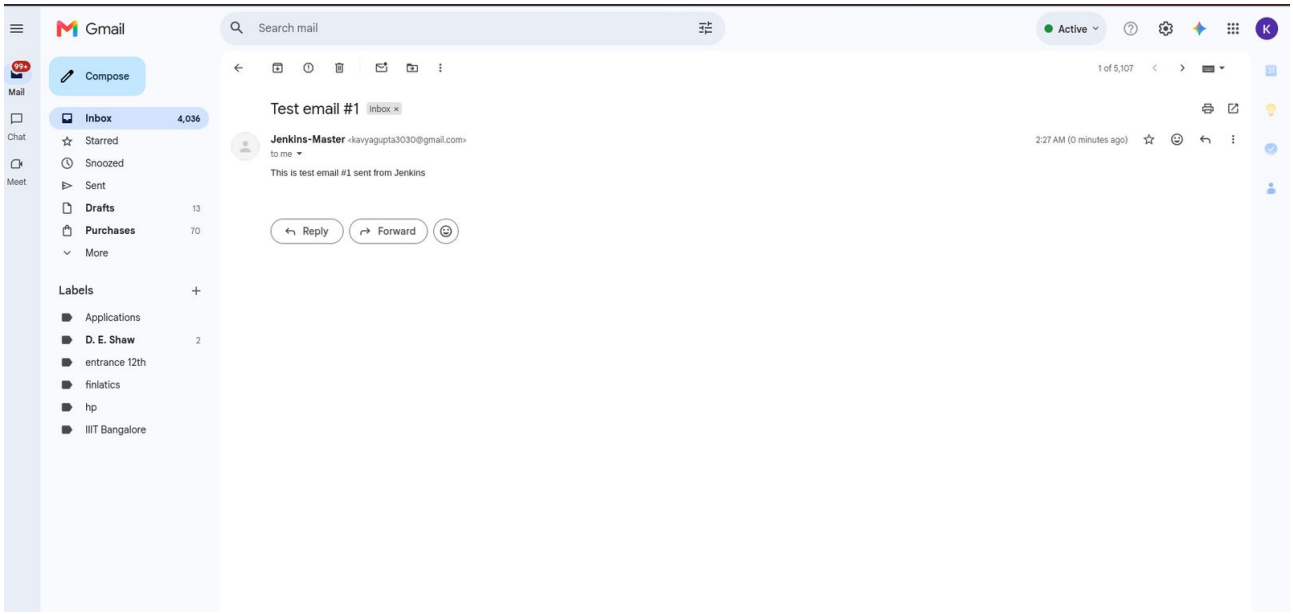


Figure 18: Test email received.

3 Repository and GitHub workflow

3.1 Repository and first commits

The public repository `calculator-devops` was created empty, so that every file entered the history as a deliberate commit. It was cloned into the project folder and the default branch name was set to `main`.

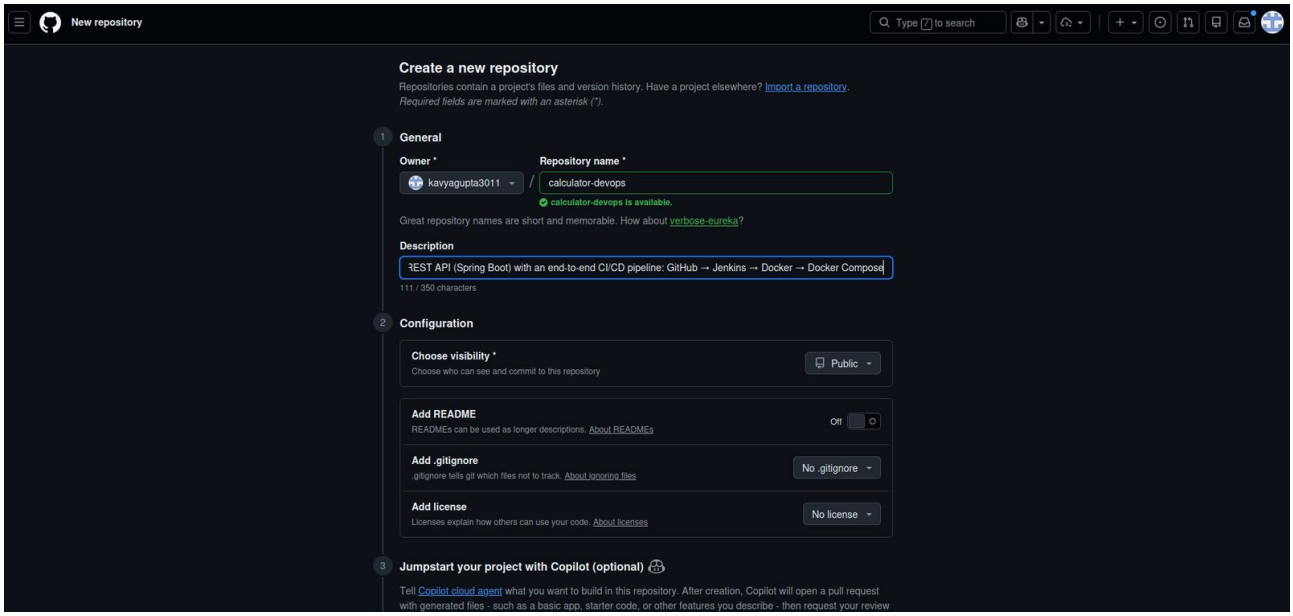


Figure 19: New repository: public, no generated files.

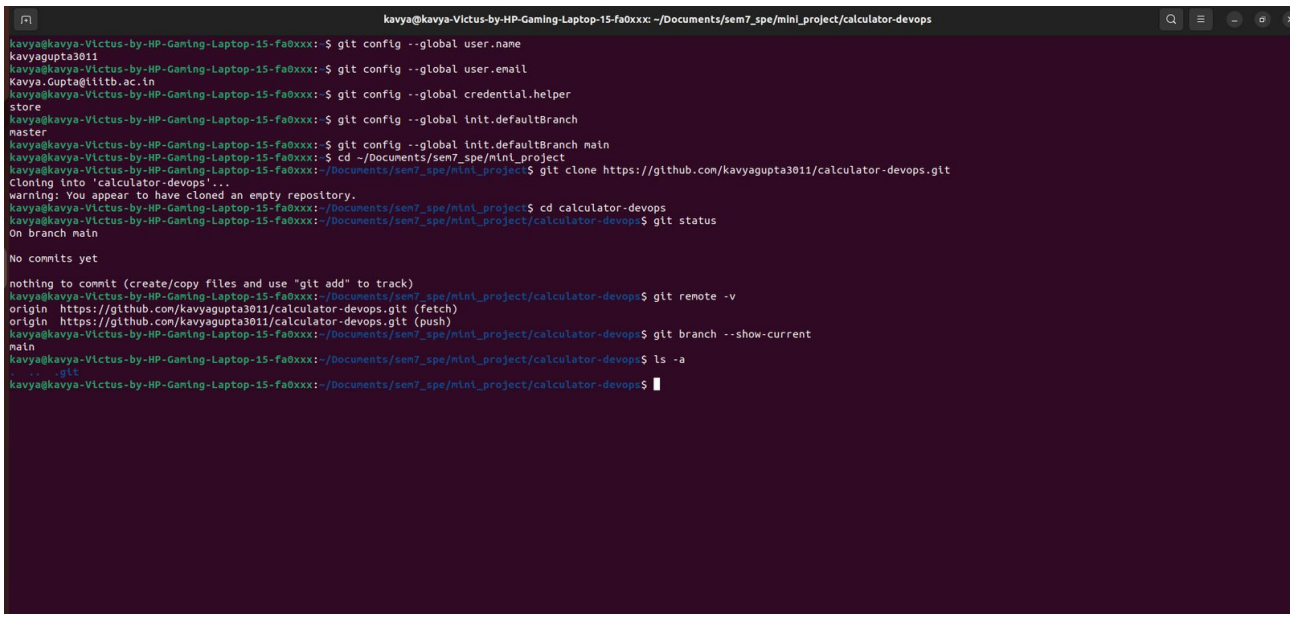


Figure 20: Git identity check, clone of the empty repository and `git remote -v`.

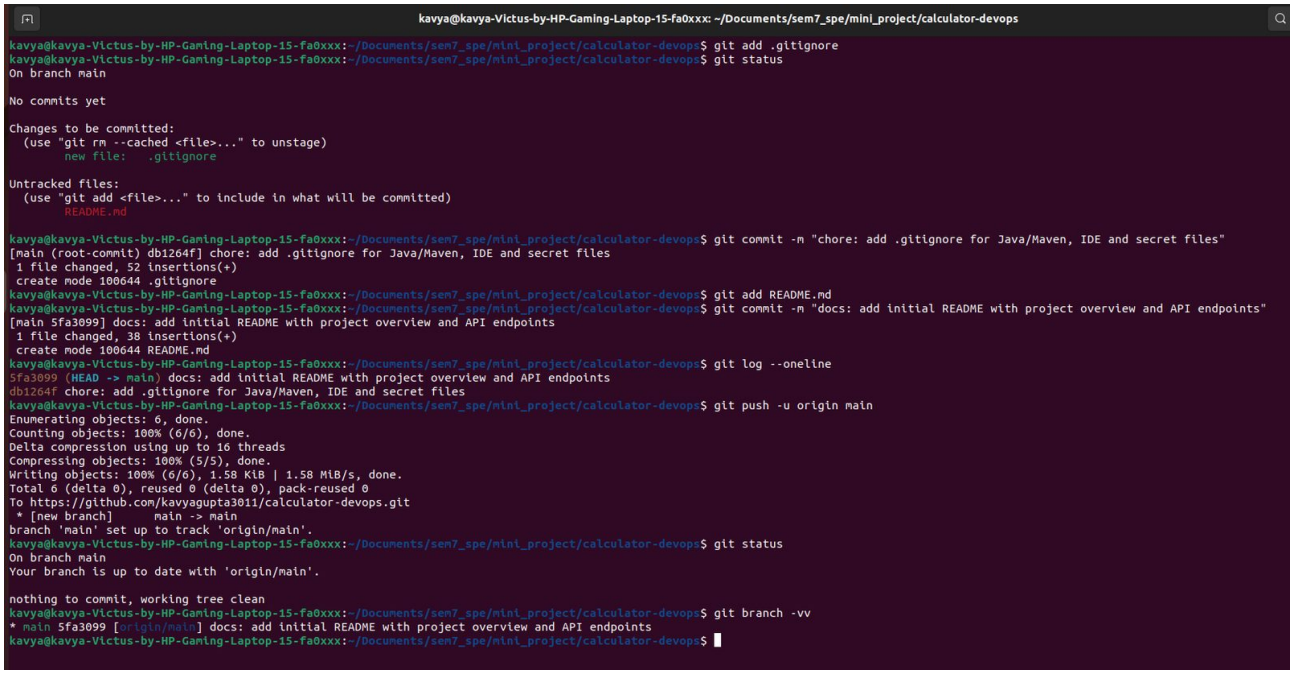


Figure 21: First commits (`.gitignore` as root commit, then README) and the first `git push -u origin main`.

The `.gitignore` excludes Maven output (`target/`), IDE folders, logs and secret files (`.env`, keys, `.git-credentials`) while allowing `.env.example`.

3.2 Templates

A pull request template (summary, `Closes #`, testing, checklist) and two issue templates (feature/task and bug report) were added under `.github/`. Blank issues are disabled so every issue follows a template, and the templates assign the author and apply a label automatically.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .github/

nothing added to commit but untracked files present (use "git add" to track)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add .github/pull_request_template.md
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -m "chore: add pull request template"
[main 78d734c] chore: add pull request template
 1 file changed, 26 insertions(+)
 create mode 100644 .github/pull_request_template.md
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add .github/ISSUE_TEMPLATE/
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -m "chore: add issue templates for features and bugs"
[main 6958639] chore: add issue templates for features and bugs
 3 files changed, 55 insertions(+)
 create mode 100644 .github/ISSUE_TEMPLATE/bug_report.md
 create mode 100644 .github/ISSUE_TEMPLATE/config.yml
 create mode 100644 .github/ISSUE_TEMPLATE/feature_request.md
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline
6958639 (HEAD -> main) chore: add issue templates for features and bugs
78d734c chore: add pull request template
5fa3099 (origin/main) docs: add initial README with project overview and API endpoints
db1264f chore: add .gitignore for Java/Maven, IDE and secret files
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 1.68 KiB | 1.68 MiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/kavyagupta3011/calculator-devops.git
   5fa3099..6958639  main -> main
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 22: Template commits pushed; four commits on `main`.

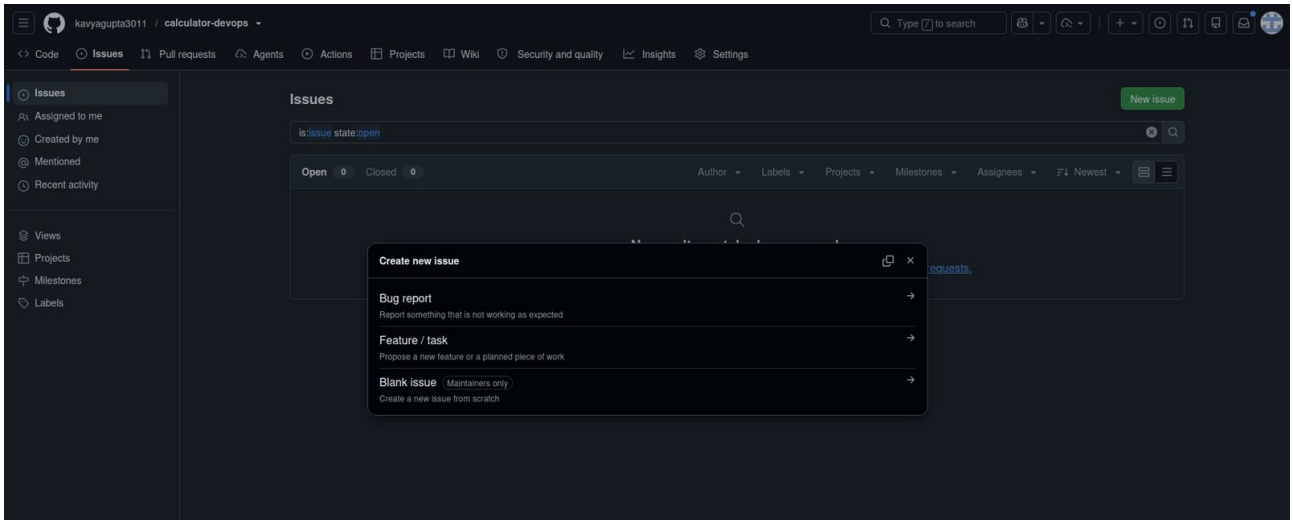


Figure 23: Issue template chooser on GitHub.

3.3 Labels, milestones, issues and project board

Labels group work by type (`feature`, `test`, `docker`, `ci-cd`, `docs`, `setup`, `bug`, `enhancement`); milestones group it by release (`v1.0 -- Requirements`, `v1.1 -- Extras`). One issue per phase lists tasks and acceptance criteria. A Kanban board with automation adds every new issue and pull request to *Todo* and moves closed issues and merged pull requests to *Done*.

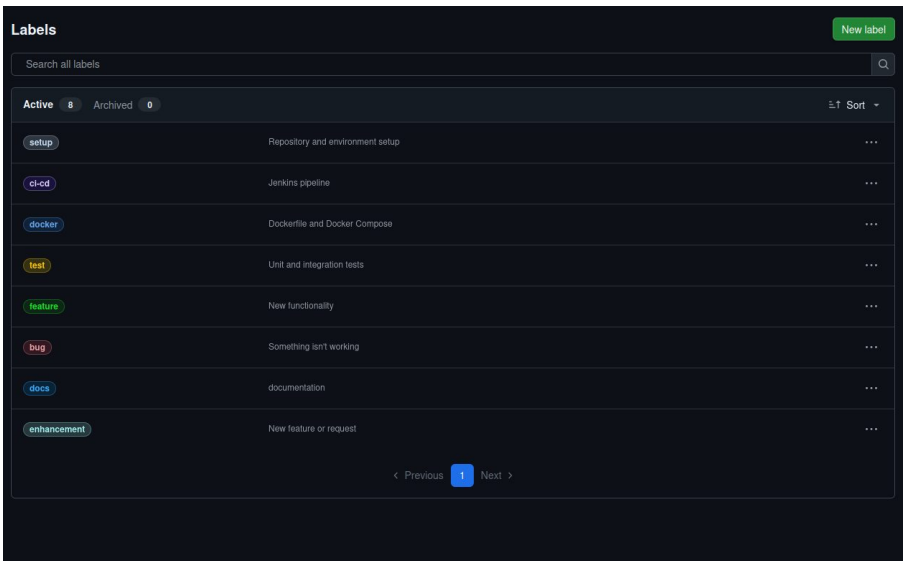


Figure 24: Labels.

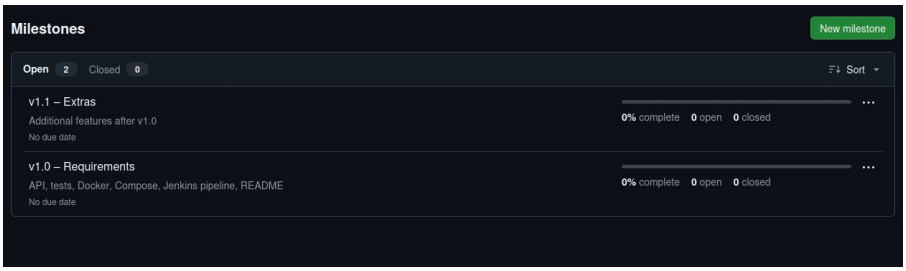


Figure 25: Milestones.

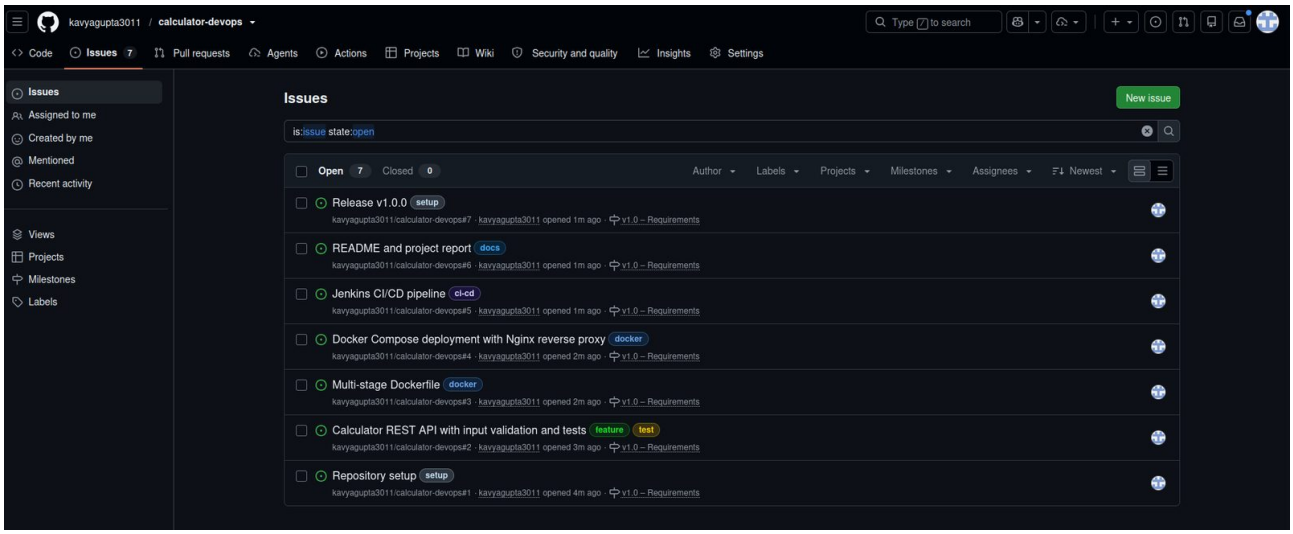


Figure 26: Issues for each phase with labels and milestone.

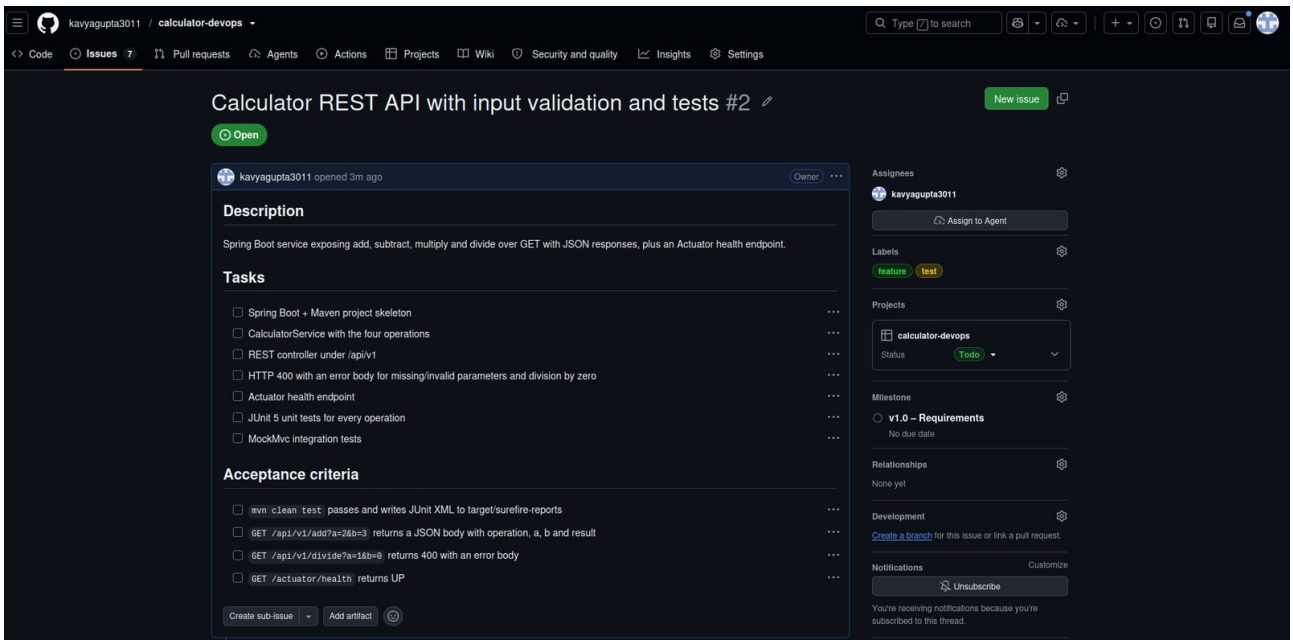


Figure 27: Issue #2 with tasks, acceptance criteria, labels, project and milestone.

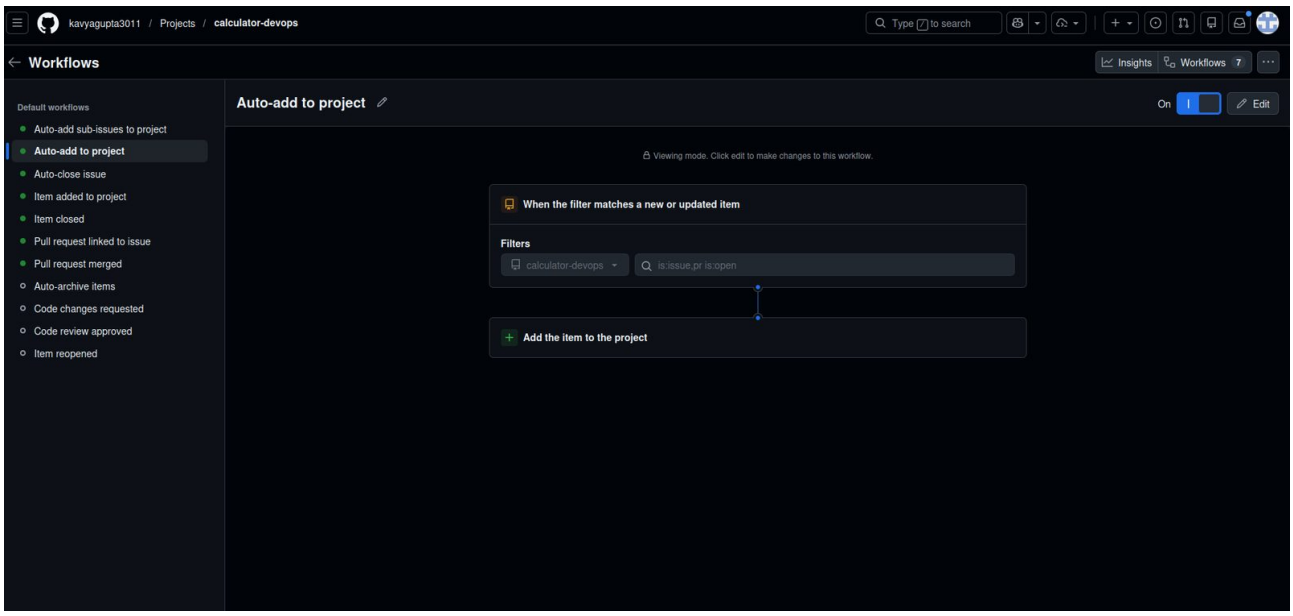


Figure 28: Board automation: auto-add issues and pull requests from the repository.

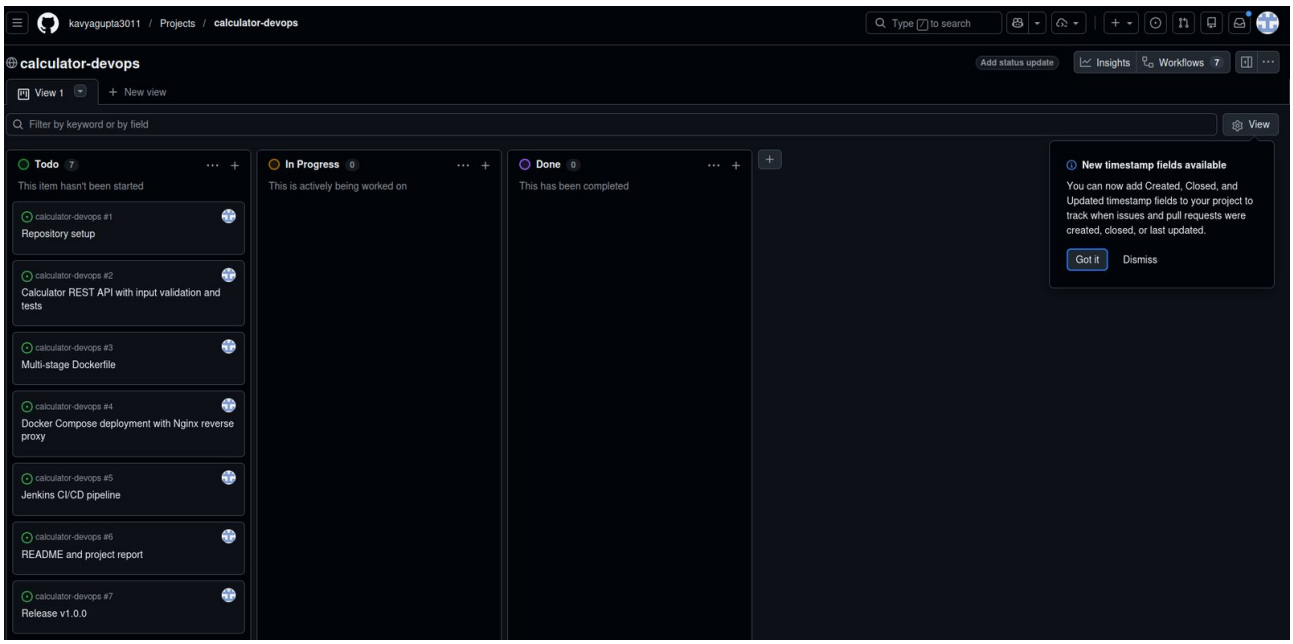


Figure 29: Project board after creating the issues.

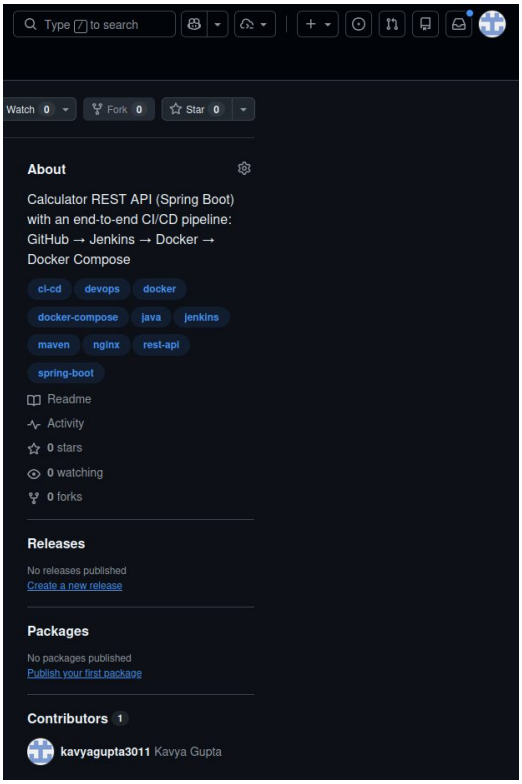


Figure 30: Repository description and topics.

3.4 Branch protection

After the setup commits, `main` was protected: changes require a pull request, review conversations must be resolved, and the rule applies to administrators. Approvals are not required because a single author cannot approve their own pull request. A direct push was attempted to confirm the rule, then the local test commit was discarded with `git reset --hard origin/main` (safe because it had never been pushed).

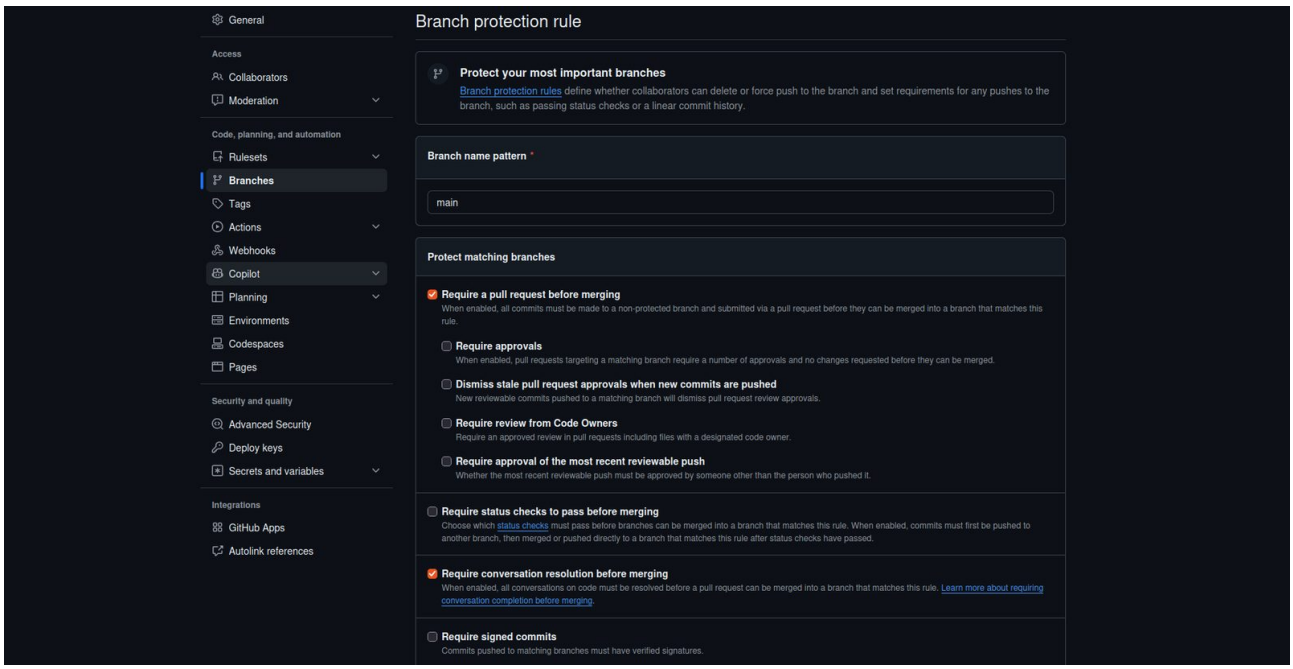


Figure 31: Branch protection rule for `main`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ cd ~/Documents/sem7_spe/mini_project/calculator-devops
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit --allow-empty -m "test: direct push to main"
[main b6fc357] test: direct push to main
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 203 bytes | 203.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
remote: error: GH006: Protected branch update failed for refs/heads/main.
remote:
remote: - Changes must be made through a pull request.
To https://github.com/kavyagupta3011/calculator-devops.git
 ! [remote rejected] main -> main (protected branch hook declined)
error: failed to push some refs to 'https://github.com/kavyagupta3011/calculator-devops.git'
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git reset --hard origin/main
HEAD is now at 1bc4057 docs: simplify README overview
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline
1bc4057 (HEAD -> main, origin/main) docs: simplify README overview
6958639 chore: add issue templates for features and bugs
78d734c chore: add pull request template
5fa3099 docs: add initial README with project overview and API endpoints
db1264f chore: add .gitignore for Java/Maven, IDE and secret files
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch main
Your branch is up to date with 'origin/main'.

kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 32: Direct push rejected: *Changes must be made through a pull request.*

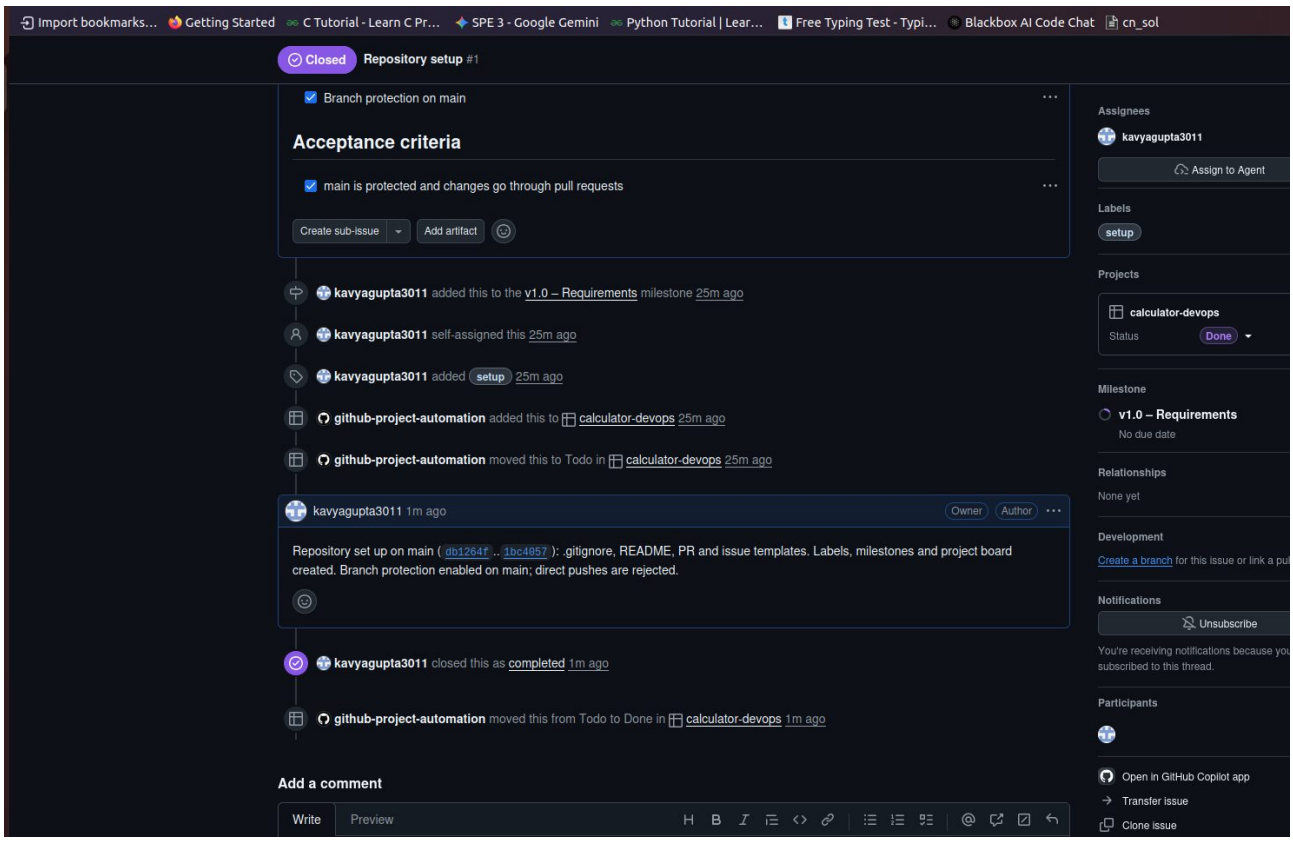


Figure 33: Issue #1 closed with a summary of the setup commits.

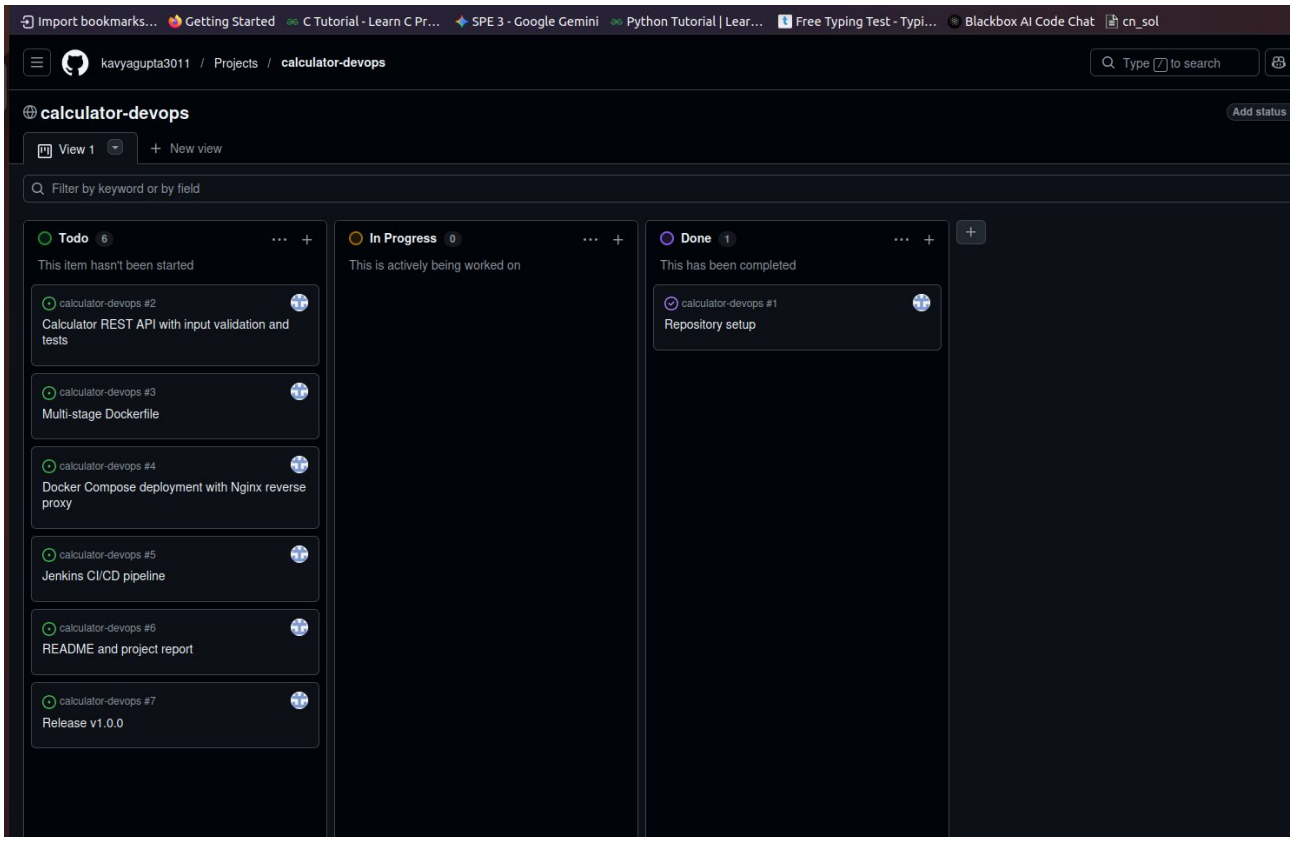


Figure 34: Board automation moved issue #1 to *Done*.

3.5 No secrets in the repository

Tokens exist only in the Jenkins credentials store, and `.gitignore` blocks environment and key files. The full history of every branch and the list of tracked files were searched for token prefixes, passwords, private keys and secret files.

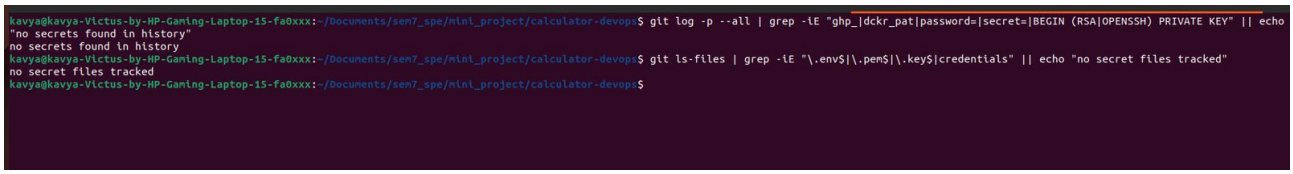


Figure 35: No secrets found in the history of any branch, and no secret files tracked.

4 Application and tests

4.1 Branch and project skeleton

The branch `feature/calculator-api` was created from issue #2, which links the branch to the issue. The Maven project uses the Spring Boot parent 3.5.6, Java 21 and `<finalName>calculator</finalName>`, so the build always produces `target/calculator.jar`. The port is `${SERVER_PORT:8081}` because Jenkins occupies 8080 on the host.

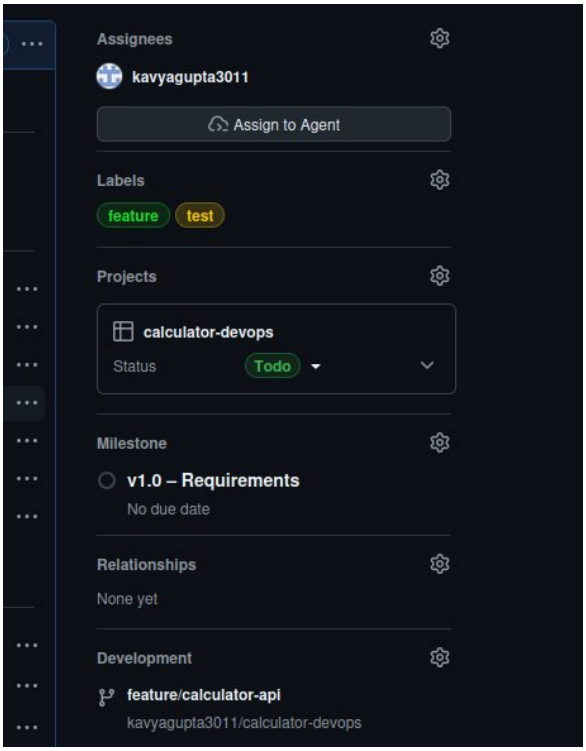


Figure 36: Branch linked to issue #2.

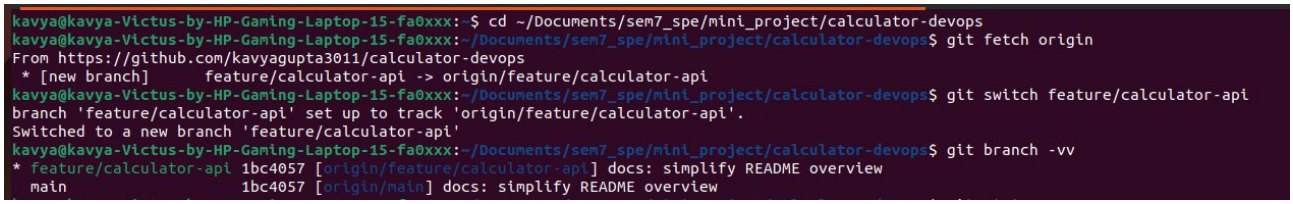


Figure 37: Switching to the new branch.

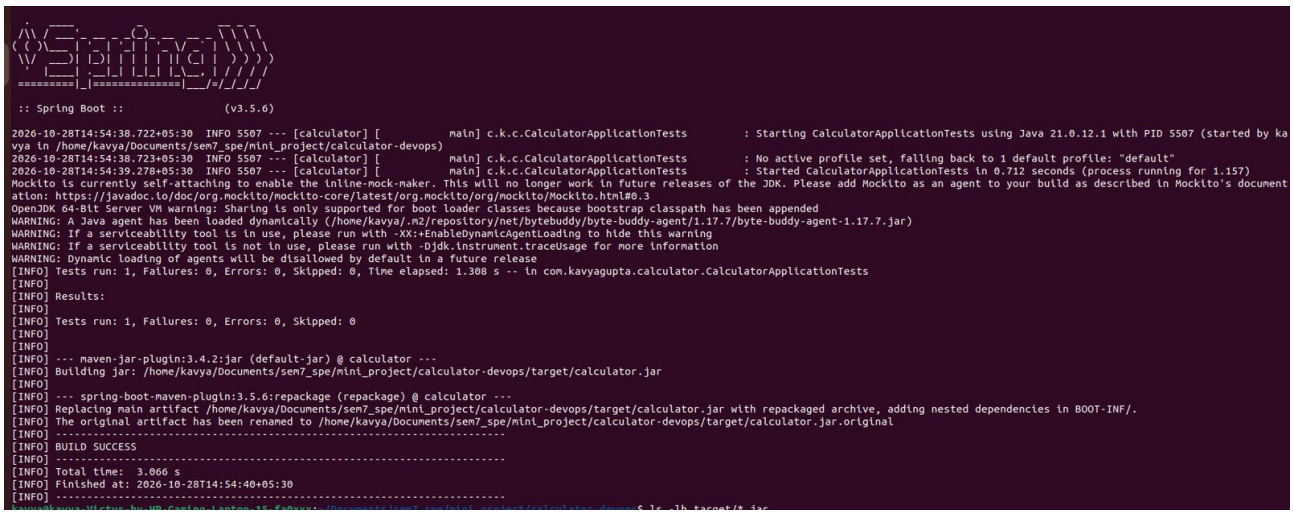


Figure 38: `mvn clean package`: context test passes, executable JAR built.

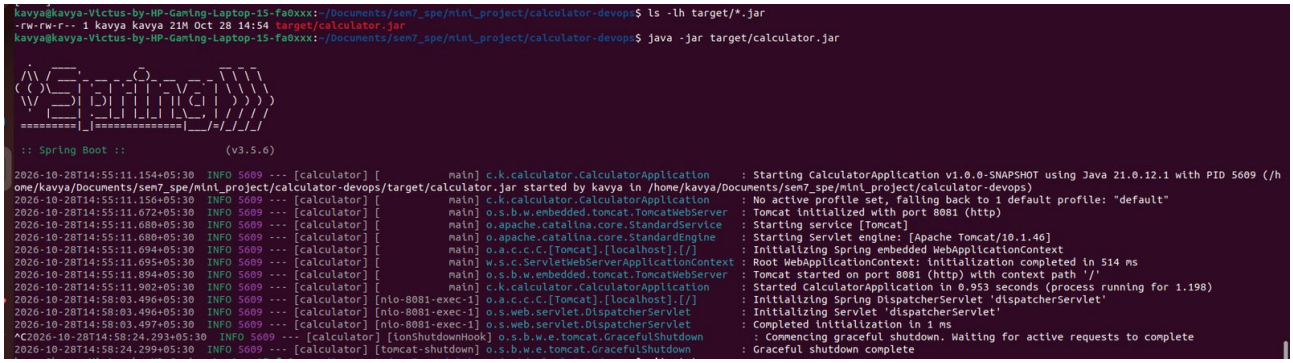


Figure 39: Application started on port 8081.

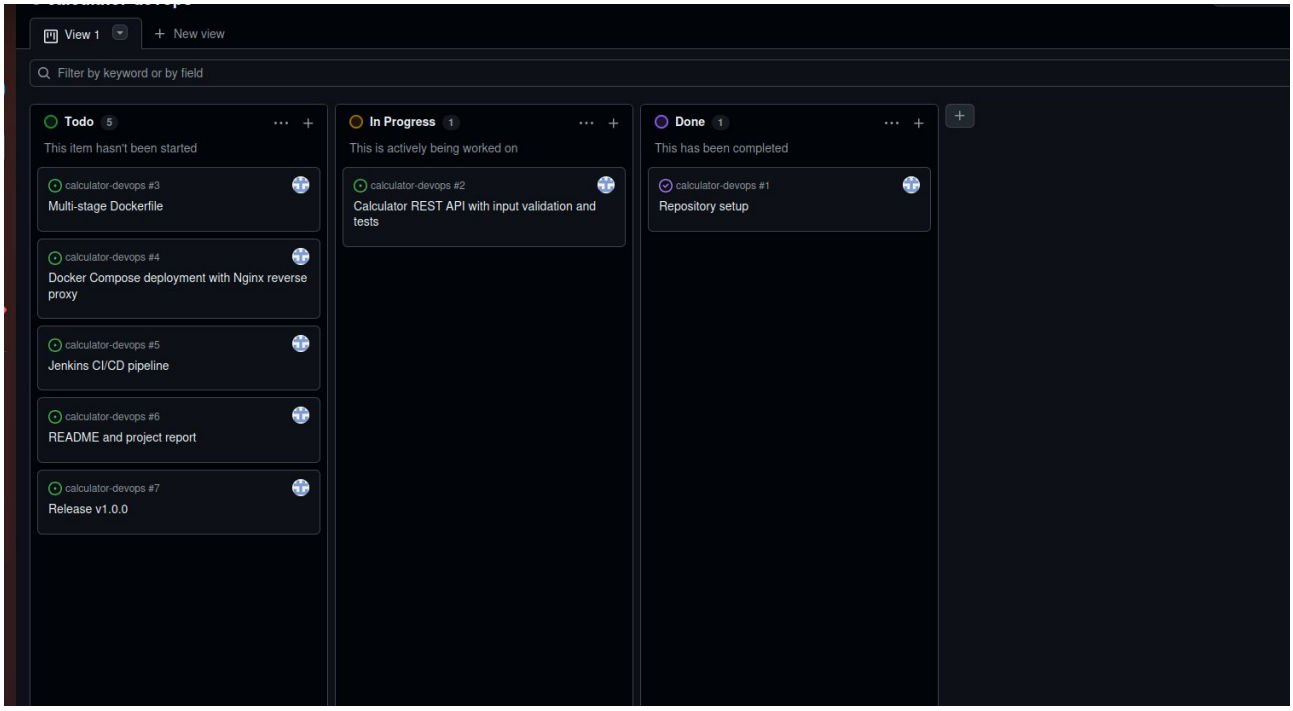


Figure 40: Issue #2 moved to *In Progress*.

4.2 Service and unit tests

`CalculatorService` performs the arithmetic with `BigDecimal` and `MathContext.DECIMAL64` (16 significant digits, so non-terminating divisions such as 1/3 do not throw). Division by zero is detected with `signum()` so that `0`, `0.0` and `0.00` are all rejected, and results are normalised with `stripTrailingZeros()`.

`CalculatorServiceTest` uses `@ParameterizedTest` with `@CsvSource`: 18 input rows across the four operations (negatives, decimals, zero, 1/3) plus division-by-zero and normalisation tests – 21 test cases that run without starting Spring.

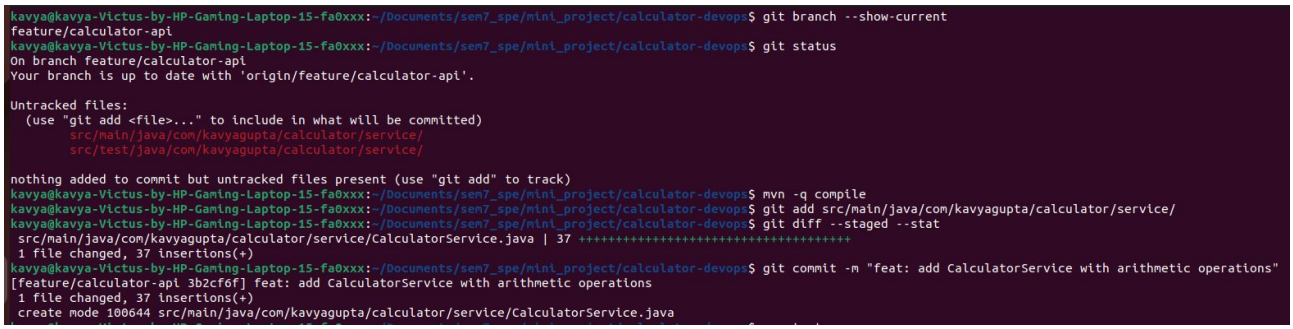


Figure 41: Service compiled and committed (`git diff --staged --stat`).

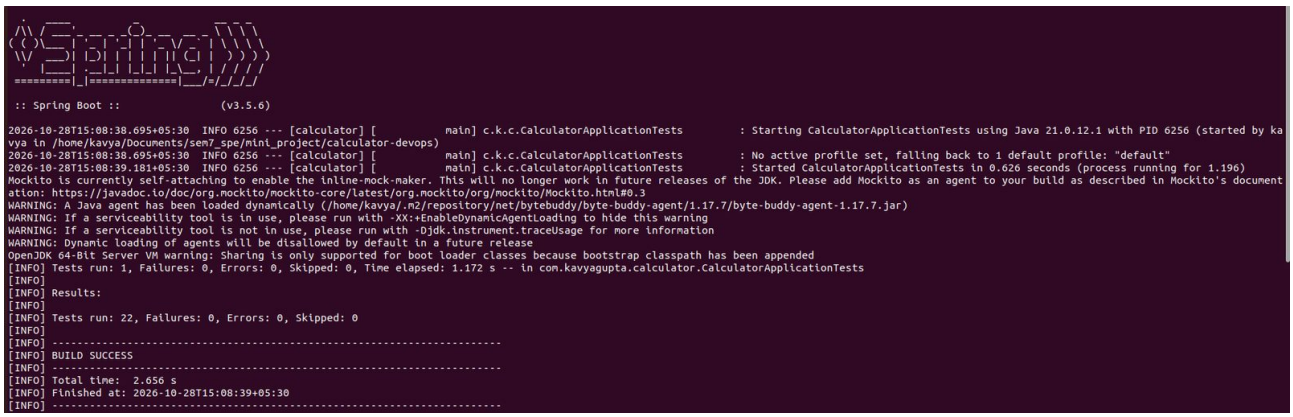


Figure 42: `mvn test`: 22 tests passing.

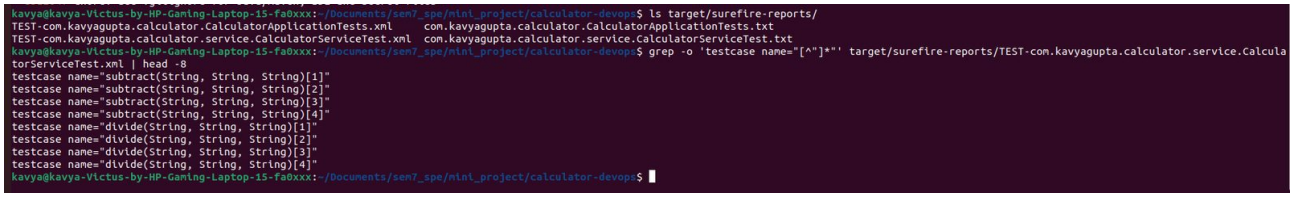


Figure 43: JUnit XML reports with readable parameterized test names.

4.3 Controller and error handling

`CalculatorController` exposes `GET /api/v1/{add,subtract,multiply,divide}` and returns a `CalculationResponse` record serialised as {`"operation","a","b","result"`}. Before error handling was added, division by zero produced HTTP 500 and invalid input produced a 400 without any message.



Figure 44: Successful responses, including exact decimal results.



Figure 45: Before the handler: 500 for division by zero and 400 responses without a message.

`GlobalExceptionHandler` (`@RestControllerAdvice`) maps the three client errors to HTTP 400 with a consistent body:

Exception	Request	Message
`MissingServletRequestParameterException`	`add?a=5`	`Required parameter 'b' is missing`
`MethodArgumentTypeMismatchException`	`add?a=abc&b=3`, `NaN`, `Infinity`	`Parameter 'a' must be a number, but was 'abc'`
`ArithmeticException`	`divide?a=1&b=0`	`Division by zero is not allowed`

Table 4: Error mapping.

While the handler was in progress, `git stash -u` set the new (untracked) files aside and `git stash pop` restored them.

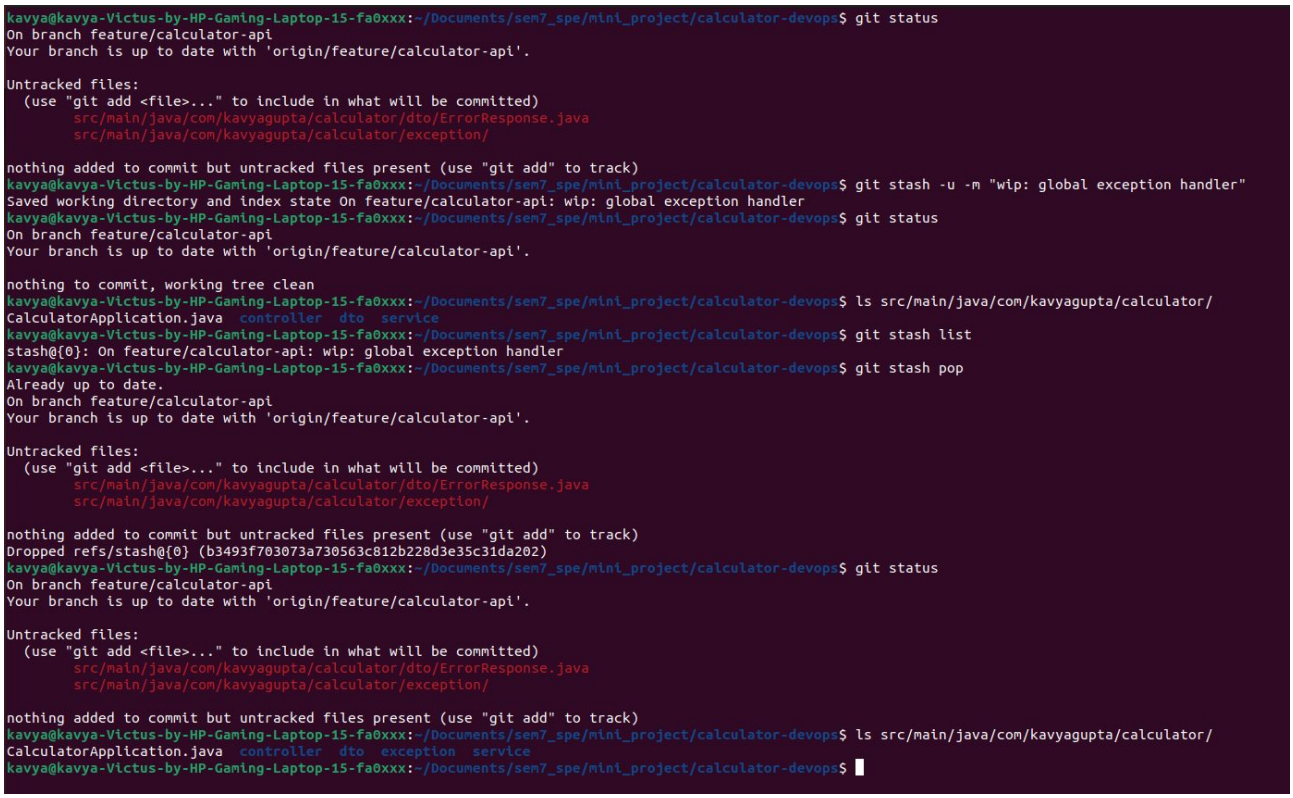


Figure 46: `git stash -u`, clean working tree, `stash list`, `stash pop`.

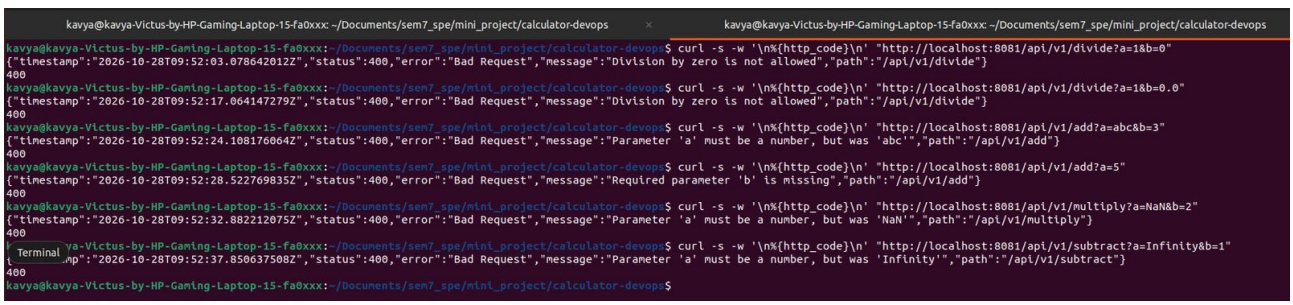


Figure 47: After the handler: every invalid request returns 400 with a specific message.

A parameter that is present but empty (`?a=`) is converted to `null` and reported by Spring as missing, so it is handled by the same rule.



Figure 48: Empty parameters (`?a=` and `?b=`) return 400 through Nginx.

4.4 MockMvc integration tests

`CalculatorControllerTest` uses `@WebMvcTest` with the real service imported, so each request passes through the controller, service, exception handler and JSON serialisation. Ten tests check the status code, content type and JSON fields with `jsonPath`. To confirm the tests guard the API contract, the error message was changed temporarily: the corresponding test failed, and `git restore` reverted the change.

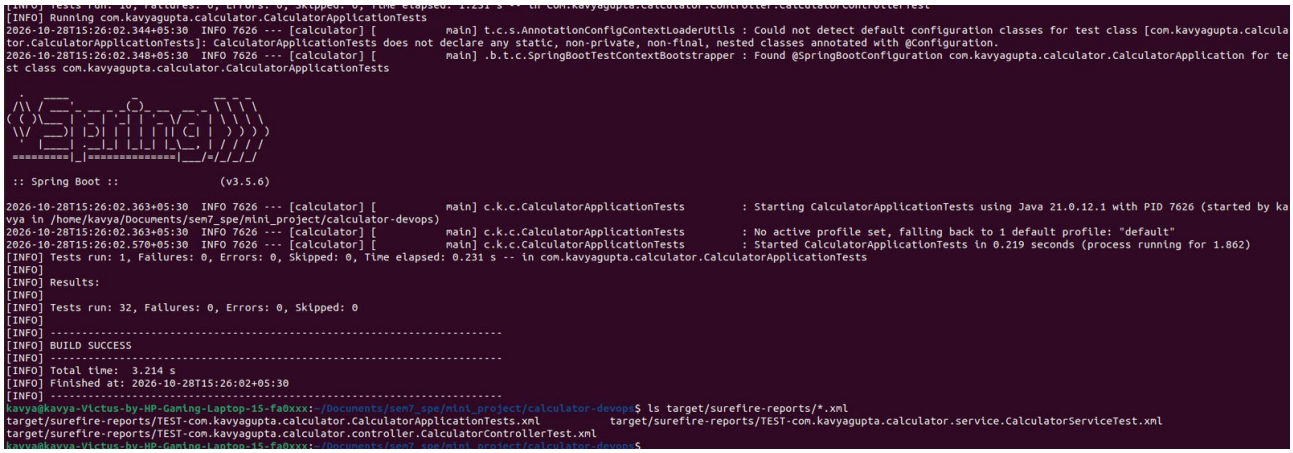


Figure 49: 32 tests passing with three XML reports.

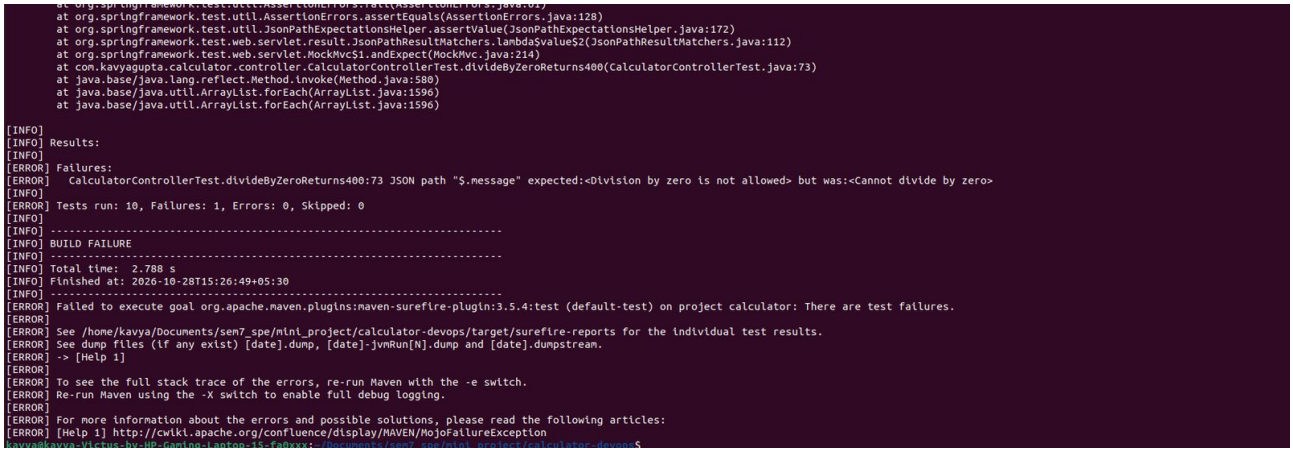


Figure 50: Deliberate change caught: *expected Division by zero is not allowed but was Cannot divide by zero.*



Figure 51: `git restore` discards the change; tests green again.

4.5 Actuator health

Spring Boot Actuator exposes only `health` and `info`; liveness and readiness probes are enabled. `CalculatorApplicationTests` (`@SpringBootTest` with `@AutoConfigureMockMvc`) checks the health, liveness and readiness endpoints and one request through the full context, bringing the total to 36 tests.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git branch --show-current
feature/calculator-api
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch feature/calculator-api
Your branch is up to date with 'origin/feature/calculator-api'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   pom.xml
        modified:   src/main/resources/application.properties
        modified:   src/test/java/com/kavyagupta/calculator/CalculatorApplicationTests.java

no changes added to commit (use "git add" and/or "git commit -a")
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git diff pom.xml src/main/resources/application.properties
diff --git a/pom.xml b/pom.xml
index a4e9494..212f795 100644
--- a/pom.xml
+++ b/pom.xml
@@ -27,6 +27,11 @@
             <artifactId>spring-boot-starter-web</artifactId>
         </dependency>

+        <dependency>
+            <groupId>org.springframework.boot</groupId>
+            <artifactId>spring-boot-starter-actuator</artifactId>
+        </dependency>
+
         <dependency>
             <groupId>org.springframework.boot</groupId>
             <artifactId>spring-boot-starter-test</artifactId>
diff --git a/src/main/resources/application.properties b/src/main/resources/application.properties
index fc80528..b44a816 100644
--- a/src/main/resources/application.properties
+++ b/src/main/resources/application.properties
@@ -5,3 +5,7 @@ server.port=${SERVER_PORT:8081}

 # Write numbers like 10 instead of 1E+1 in JSON responses.
 spring.jackson.generator.write-bigdecimal-as-plain=true
+
+# Actuator: expose only health and info over HTTP.
+management.endpoints.web.exposure.include=health,info
+management.endpoint.health.probes.enabled=true
```

Figure 52: Actuator dependency and configuration.

```
2026-10-28T15:31:01.822+05:30  INFO 8338 --- [calculator] [           main] o.s.t.web.servlet.TestDispatcherServlet  : Completed initialization in 11 ms
2026-10-28T15:31:01.846+05:30  INFO 8338 --- [calculator] [           main] c.k.c.CalculatorApplicationTests         : Started CalculatorApplicationTests in 0.632 sec
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.750 s -- in com.kavyagupta.calculator.CalculatorApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 36, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.4.2:jar (default-jar) @ calculator ---
[INFO] Building jar: /home/kavya/Documents/sem7_spe/mini_project/calculator-devops/target/calculator.jar
[INFO]
[INFO] --- spring-boot-maven-plugin:3.5.6:repackage (repackage) @ calculator ---
[INFO] Replacing main artifact /home/kavya/Documents/sem7_spe/mini_project/calculator-devops/target/calculator.jar with repackaged archive, adding nested dependencies
[INFO] The original artifact has been renamed to /home/kavya/Documents/sem7_spe/mini_project/calculator-devops/target/calculator.jar.original
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.037 s
[INFO] Finished at: 2026-10-28T15:31:02+05:30
[INFO] ------------------------------------------------------------------------
```

Figure 53: 36 tests passing.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8081/actuator/health
{"status":"UP","groups":["liveness","readiness"]}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8081/actuator/health/liveness
{"status":"UP"}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8081/actuator/health/readiness
{"status":"UP"}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8081/actuator/info
{}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8081/actuator/env
{"timestamp":"2026-10-28T10:02:14.422+00:00","status":404,"error":"Not Found","path":"/actuator/env"}
404
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 54: Health, liveness and readiness `UP`; `/actuator/env` is not exposed (404).

4.6 Pull request #8

The branch was documented in the README and merged through pull request #8 with a merge commit, keeping all nine commits and adding a merge commit with two parents. Review comments were added on specific lines and resolved before merging.

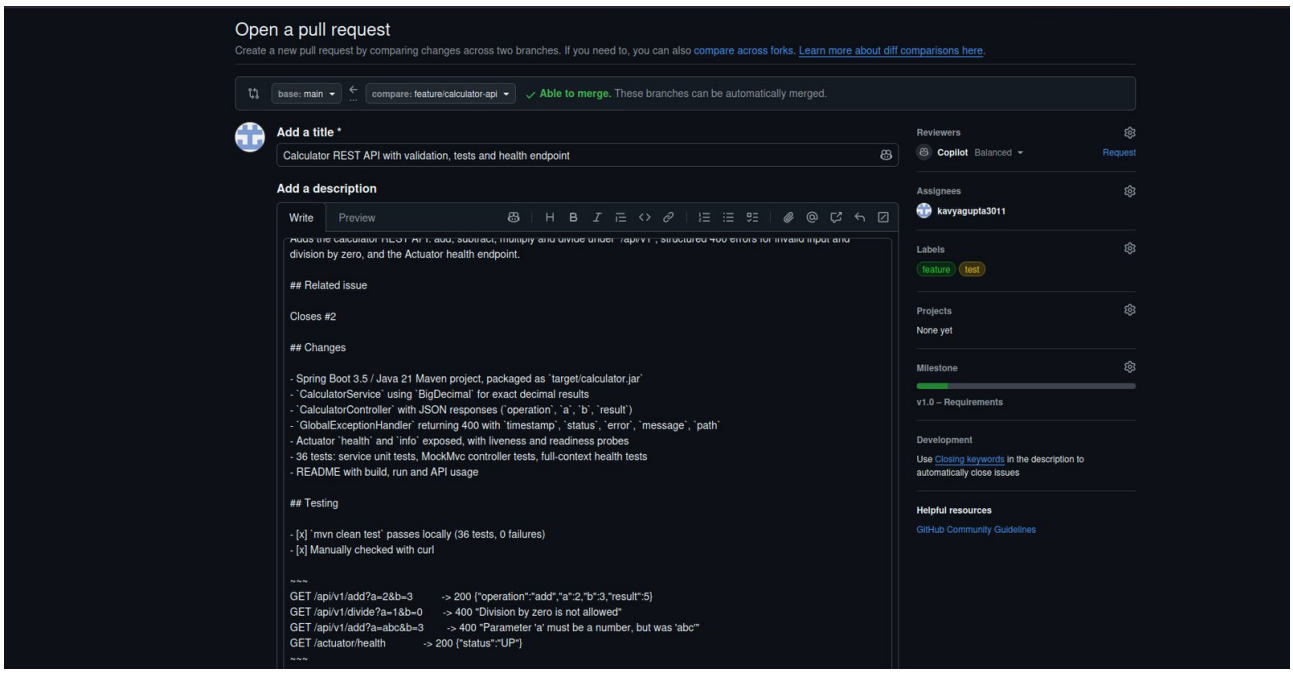


Figure 55: Pull request #8 created from the template, closing issue #2.

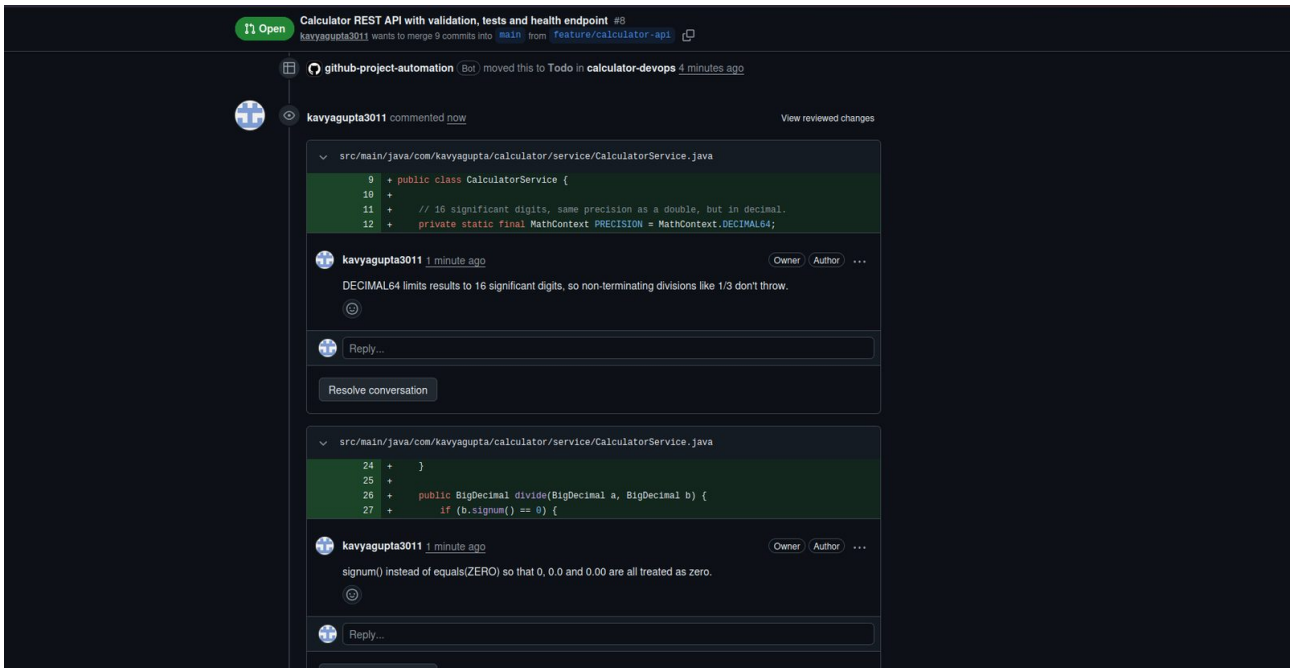


Figure 56: Line review comments, resolved before merge.

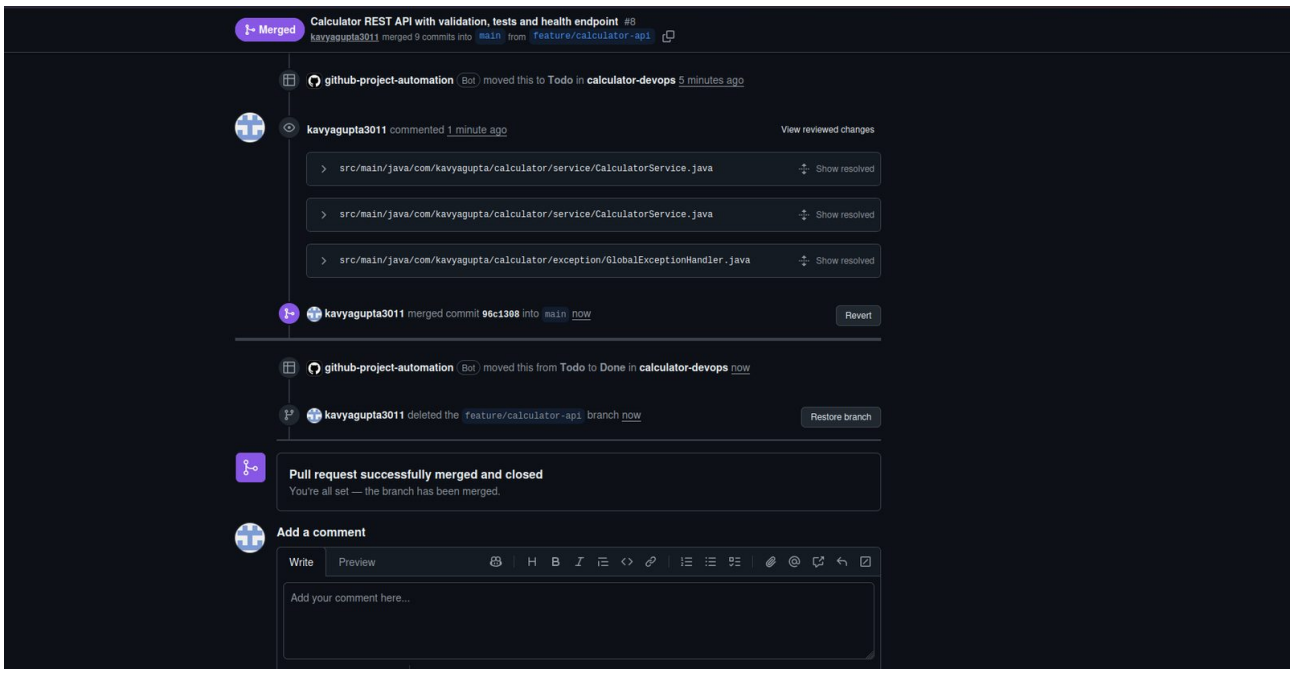


Figure 57: Merged with a merge commit; branch deleted.

```
 pom.xml                                                                 |  51 +++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 src/main/java/com/kavyagupta/calculator/CalculatorApplication.java      |  12 ++++++++++++
 src/main/java/com/kavyagupta/calculator/controller/CalculatorController.java |  42 ++++++++++++++++++++++++++++++++++++++++++
 src/main/java/com/kavyagupta/calculator/dto/CalculationResponse.java    |   6 ++++++
 src/main/java/com/kavyagupta/calculator/dto/ErrorResponse.java          |   6 ++++++
 src/main/java/com/kavyagupta/calculator/exception/GlobalExceptionHandler.java |  50 ++++++++++++++++++++++++++++++++++++++++++++++++++
 src/main/java/com/kavyagupta/calculator/service/CalculatorService.java  |  37 +++++++++++++++++++++++++++++++++++++
 src/main/resources/application.properties                               |  11 +++++++++++
 src/test/java/com/kavyagupta/calculator/CalculatorApplicationTests.java |  51 +++++++++++++++++++++++++++++++++++++++++++++++++++
 src/test/java/com/kavyagupta/calculator/controller/CalculatorControllerTest.java | 105 +++++++++++++++++++++++++++++++++++++++
```

`.dockerignore` excludes `target/`, `.git/`, IDE files, CI and deployment files and secrets.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ cd ~/Documents/sem7_spe/mini_project/calculator-devops
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch main
Already on 'main'
Your branch is up to date with 'origin/main'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git pull
Already up to date.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .dockerignore
        Dockerfile

nothing added to commit but untracked files present (use "git add" to track)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch -c feature/docker
Switched to a new branch 'feature/docker'
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push -u origin feature/docker
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/docker' on GitHub by visiting:
remote:      https://github.com/kavyagupta3011/calculator-devops/pull/new/feature/docker
remote:
To https://github.com/kavyagupta3011/calculator-devops.git
 * [new branch]      feature/docker -> feature/docker
branch 'feature/docker' set up to track 'origin/feature/docker'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add .dockerignore
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -m "build: add .dockerignore to keep the build context small"
[feature/docker c6ddf60] build: add .dockerignore to keep the build context small
 1 file changed, 23 insertions(+)
 create mode 100644 .dockerignore
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add Dockerfile
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -m "build: add multi-stage Dockerfile with non-root runtime user"
[feature/docker 89fbaa4] build: add multi-stage Dockerfile with non-root runtime user
 1 file changed, 33 insertions(+)
 create mode 100644 Dockerfile
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push

Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.33 KiB | 1.33 MiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/kavyagupta3011/calculator-devops.git
   96c1308..89fbaa4  feature/docker -> feature/docker
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline --graph -4
* 89fbaa4 (HEAD -> feature/docker, origin/feature/docker) build: add multi-stage Dockerfile with non-root runtime user
* c6ddf60 build: add .dockerignore to keep the build context small
*   96c1308 (origin/main, main) Merge pull request #8 from kavyagupta3011/feature/calculator-api
|\
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 59: Branch `feature/docker` and the two build commits.

```
Step 8/16 : LABEL org.opencontainers.image.title="calculator"       org.opencontainers.image.description="Calculator REST API"       org.opencontainers.image.source="https://github.c
ulator-devops"
 ---> Running in 991821b8c3f6
Removing intermediate container 991821b8c3f6
 ---> c89b650e4090
Step 9/16 : RUN addgroup -S app && adduser -S app -G app
 ---> Running in d2de28cffa8f
Removing intermediate container d2de28cffa8f
 ---> a97fe0640b51
Step 10/16 : WORKDIR /app
 ---> Running in a77021df0b02
Removing intermediate container a77021df0b02
 ---> 4319140faecc
Step 11/16 : COPY --from=build --chown=app:app /build/target/calculator.jar app.jar
 ---> 3f78a52ec6d4
Step 12/16 : USER app
 ---> Running in 8b7b1856b44d
Removing intermediate container 8b7b1856b44d
 ---> 3fb91ec23f97
Step 13/16 : ENV SERVER_PORT=8081
 ---> Running in 9030303928d9
Removing intermediate container 9030303928d9
 ---> 6cd997e220f3
Step 14/16 : EXPOSE 8081
 ---> Running in 18421de76832
Removing intermediate container 18421de76832
 ---> 6de9351a6bc0
Step 15/16 : HEALTHCHECK --interval=15s --timeout=3s --start-period=30s --retries=3   CMD wget -qO- "http://localhost:${SERVER_PORT}/actuator/health" || exit 1
 ---> Running in d5f94dfa4ee2
Removing intermediate container d5f94dfa4ee2
 ---> 83da2ed25542
Step 16/16 : ENTRYPOINT ["java", "-jar", "/app/app.jar"]
 ---> Running in c9dc2f48052f
Removing intermediate container c9dc2f48052f
 ---> e68686030fb1
Successfully built e68686030fb1
Successfully tagged kavyag3011/calculator-devops:dev
```

Figure 60: Image built: 16 steps, only the JAR copied into the runtime stage.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker images | grep -E "REPOSITORY|calculator-devops|maven|temurin"
REPOSITORY                   TAG                       IMAGE ID       CREATED              SIZE
kavyag3011/calculator-devops   dev                       e68686030fb1   About a minute ago   231MB
eclipse-temurin              21-jre-alpine             bb1b8368a51b   6 days ago           208MB
maven                        3.9.11-eclipse-temurin-21   b9611ad9b2f6   11 months ago        531MB
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker history kavyag3011/calculator-devops:dev
IMAGE          CREATED              CREATED BY                                      SIZE      COMMENT
e68686030fb1   About a minute ago   /bin/sh -c #(nop)  ENTRYPOINT ["java" "-jar"…   0B
83da2ed25542   About a minute ago   /bin/sh -c #(nop)  HEALTHCHECK &{["CMD-SHELL…   0B
6de9351a6bc0   About a minute ago   /bin/sh -c #(nop)  EXPOSE 8081                  0B
6cd997e220f3   About a minute ago   /bin/sh -c #(nop)  ENV SERVER_PORT=8081         0B
3fb91ec23f97   About a minute ago   /bin/sh -c #(nop)  USER app                     0B
3f78a52ec6d4   About a minute ago   /bin/sh -c #(nop) COPY --chown=app:appfile:4…   23.6MB
4319140faecc   About a minute ago   /bin/sh -c #(nop) WORKDIR /app                  0B
a97fe0640b51   About a minute ago   /bin/sh -c addgroup -S app && adduser -S app…   3.03kB
c89b650e4090   About a minute ago   /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
bb1b8368a51b   6 days ago           ENTRYPOINT ["/__cacert_entrypoint.sh"]          0B        buildkit.dockerfile.v0
<missing>      6 days ago           COPY --chmod=755 entrypoint.sh /__cacert_ent…   5.97kB    buildkit.dockerfile.v0
<missing>      6 days ago           RUN /bin/sh -c set -eux;     echo "Verifying…   0B        buildkit.dockerfile.v0
<missing>      6 days ago           RUN /bin/sh -c set -eux;     ARCH="$(apk --p…   164MB     buildkit.dockerfile.v0
<missing>      6 days ago           ENV JAVA_VERSION=jdk-21.0.12.1+1                0B        buildkit.dockerfile.v0
<missing>      6 days ago           RUN /bin/sh -c set -eux;     apk add --no-ca…   34.9MB    buildkit.dockerfile.v0
<missing>      6 days ago           ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_AL…   0B        buildkit.dockerfile.v0
<missing>      6 days ago           ENV PATH=/opt/java/openjdk/bin:/usr/local/sb…   0B        buildkit.dockerfile.v0
<missing>      6 days ago           ENV JAVA_HOME=/opt/java/openjdk                 0B        buildkit.dockerfile.v0
<missing>      6 days ago           CMD ["/bin/sh"]                                 0B        buildkit.dockerfile.v0
<missing>      6 days ago           ADD alpine-minirootfs-3.24.3-x86_64.tar.gz /…   8.43MB    buildkit.dockerfile.v0
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker inspect --format '{{.Config.User}} | {{json .Config.ExposedPorts}} | {{json .Config.Healthcheck
.Test}}' kavyag3011/calculator-devops:dev
app | {"8081/tcp":{}} | ["CMD-SHELL","wget -qO- \"http://localhost:${SERVER_PORT}/actuator/health\" || exit 1"]
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 61: Runtime image 231 MB vs 531 MB build image; layers; user `app`, port 8081 and health check.

5.2 Running the container

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker run -d --name calc-test -p 8081:8081 kavyag3011/calculator-devops:dev
18288b7fe878892cae6663f57716af17a77b016d48063e53be5bf38b496cefec
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                            PORTS                                                NAMES
18288b7fe878   kavyag3011/calculator-devops:dev   "java -jar /app/app.…"   8 seconds ago    Up 7 seconds (health: starting)   0.0.0.0:8081->8081/tcp, :::8081->8081/tcp   calc-test
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                    PORTS                                                NAMES
18288b7fe878   kavyag3011/calculator-devops:dev   "java -jar /app/app.…"   54 seconds ago   Up 54 seconds (healthy)   0.0.0.0:8081->8081/tcp, :::8081->8081/tcp   calc-test
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker inspect --format '{{.State.Health.Status}}' calc-test
healthy
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker logs calc-test | tail -5
2026-10-28T13:20:21.844Z  INFO 1 --- [calculator] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port 8081 (http) with context path '/'
2026-10-28T13:20:21.865Z  INFO 1 --- [calculator] [           main] c.k.calculator.CalculatorApplication     : Started CalculatorApplication in 1.673 seconds (process running for 1.917)
2026-10-28T13:20:35.045Z  INFO 1 --- [calculator] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-10-28T13:20:35.045Z  INFO 1 --- [calculator] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet        : Initializing Servlet 'dispatcherServlet'
2026-10-28T13:20:35.046Z  INFO 1 --- [calculator] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet        : Completed initialization in 1 ms
```

Figure 62: Single `docker run`: *health: starting*, then *healthy*.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' "http://localhost:8081/api/v1/add?a=2&b=3"
{"operation":"add","a":2,"b":3,"result":5}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' "http://localhost:8081/api/v1/divide?a=1&b=0"
{"timestamp":"2026-10-28T13:21:51.3107635262","status":400,"error":"Bad Request","message":"Division by zero is not allowed","path":"/api/v1/divide"}
400
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8081/actuator/health
{"status":"UP","groups":["liveness","readiness"]}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 63: API responding from the container.

```
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker exec calc-test whoami
app
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker exec calc-test id
uid=100(app) gid=101(app) groups=101(app)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker exec calc-test ps -o pid,user,args
PID   USER     COMMAND
    1 app      java -jar /app/app.jar
  154 app      ps -o pid,user,args
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 64: `whoami` returns `app`; PID 1 is the JVM running as `app`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker run -d --name calc-9090 -e SERVER_PORT=9090 -p 9090:9090 kavyag3011/calculator-devops:dev
f7b257b936d674ee85bd55d7c9aecbfda14b983b599d2a9ce7dfbb0a051c1664
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                    PORTS                                                NAMES
f7b257b936d6   kavyag3011/calculator-devops:dev   "java -jar /app/app.…"   57 seconds ago   Up 56 seconds (healthy)   8081/tcp, 0.0.0.0:9090->9090/tcp, :::9090->9090/tcp   calc-9090
18288b7fe878   kavyag3011/calculator-devops:dev   "java -jar /app/app.…"   5 minutes ago    Up 5 minutes (healthy)    0.0.0.0:8081->8081/tcp, :::8081->8081/tcp            calc-test
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n' "http://localhost:9090/api/v1/multiply?a=6&b=7"
{"operation":"multiply","a":6,"b":7,"result":42}
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 65: The same image on port 9090 by setting `SERVER_PORT`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ time docker stop calc-test
calc-test

real    0m0.322s
user    0m0.012s
sys     0m0.012s
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker stop calc-9090
calc-9090
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker rm calc-test calc-9090
calc-test
calc-9090
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker ps -a | grep calc || echo "no calculator containers left"
no calculator containers left
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 66: `docker stop` completes in 0.3 s because the JVM receives SIGTERM directly.

5.3 Layer caching

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ time docker build -t kavyag3011/calculator-devops:dev .
Sending build context to Docker daemon   42.5kB
Step 1/16 : FROM maven:3.9.11-eclipse-temurin-21 AS build
 ---> b9611ad9b2f6
Step 2/16 : WORKDIR /build
 ---> Using cache
 ---> b65b85fa57ca
Step 3/16 : COPY pom.xml .
 ---> Using cache
 ---> b2f693ae096c
Step 4/16 : RUN mvn -B -q dependency:go-offline
 ---> Using cache
 ---> f089c00a53bc
Step 5/16 : COPY src ./src
 ---> Using cache
 ---> cc3b15adcdd5
Step 6/16 : RUN mvn -B -q package -DskipTests
 ---> Using cache
 ---> 08a0d0373a2d
Step 7/16 : FROM eclipse-temurin:21-jre-alpine
 ---> bb1b8368a51b
Step 8/16 : LABEL org.opencontainers.image.title="calculator"       org.opencontainers.image.description="Calculator REST API"       org.opencontainers.image.source="https://github.com/kavyagupta3011/calc
ulator-devops"
 ---> Using cache
 ---> c89b650e4090
Step 9/16 : RUN addgroup -S app && adduser -S app -G app
 ---> Using cache
 ---> a97fe0640b51
Step 10/16 : WORKDIR /app
 ---> Using cache
 ---> 4319140faecc
Step 11/16 : COPY --from=build --chown=app:app /build/target/calculator.jar app.jar
 ---> Using cache
 ---> 3f78a52ec6d4
Step 12/16 : USER app
 ---> Using cache
 ---> 3fb91ec23f97
Step 13/16 : ENV SERVER_PORT=8081
 ---> Using cache
 ---> 6cd997e220f3
Step 14/16 : EXPOSE 8081
 ---> Using cache
 ---> 6de9351a6bc0
Step 15/16 : HEALTHCHECK --interval=15s --timeout=3s --start-period=30s --retries=3   CMD wget -qO- "http://localhost:${SERVER_PORT}/actuator/health" || exit 1
 ---> Using cache
 ---> 83da2ed25542
Step 16/16 : ENTRYPOINT ["java", "-jar", "/app/app.jar"]
 ---> Using cache
 ---> e68686030fb1
Successfully built e68686030fb1
Successfully tagged kavyag3011/calculator-devops:dev

real    0m0.050s
user    0m0.006s
sys     0m0.017s
```

Figure 67: Rebuild without changes: every step cached.

```
 ---> Using cache
 ---> b65b85fa57ca
Step 3/16 : COPY pom.xml .
 ---> Using cache
 ---> b2f693ae096c
Step 4/16 : RUN mvn -B -q dependency:go-offline
 ---> Using cache
 ---> f089c00a53bc
Step 5/16 : COPY src ./src
 ---> 53ff1f82f6ca
Step 6/16 : RUN mvn -B -q package -DskipTests
 ---> Running in b32e6bf1114d
Removing intermediate container b32e6bf1114d
 ---> 423604065df7
Step 7/16 : FROM eclipse-temurin:21-jre-alpine
 ---> bb1b8368a51b
Step 8/16 : LABEL org.opencontainers.image.title="calculator"       org.opencontainers.image.description="Calculator REST API"       org.opencontainers.image.source="https://github.com/kavyagupta3011/calc
ulator-devops"
 ---> Using cache
 ---> c89b650e4090
Step 9/16 : RUN addgroup -S app && adduser -S app -G app
 ---> Using cache
 ---> a97fe0640b51
Step 10/16 : WORKDIR /app
 ---> Using cache
 ---> 4319140faecc
Step 11/16 : COPY --from=build --chown=app:app /build/target/calculator.jar app.jar
 ---> 20634c0968d9
Step 12/16 : USER app
 ---> Running in 6b2d5a72df27
Removing intermediate container 6b2d5a72df27
 ---> 44d76c7f6e16
Step 13/16 : ENV SERVER_PORT=8081
 ---> Running in dee516a4e246
Removing intermediate container dee516a4e246
 ---> 539ef6ae61b0
Step 14/16 : EXPOSE 8081
 ---> Running in f4dbf8edbc9a
Removing intermediate container f4dbf8edbc9a
 ---> 4a608bb917bc
Step 15/16 : HEALTHCHECK --interval=15s --timeout=3s --start-period=30s --retries=3   CMD wget -qO- "http://localhost:${SERVER_PORT}/actuator/health" || exit 1
 ---> Running in 61e2fafb90d4
Removing intermediate container 61e2fafb90d4
 ---> 0175885bf577
Step 16/16 : ENTRYPOINT ["java", "-jar", "/app/app.jar"]
 ---> Running in fe3ac93e66c4
Removing intermediate container fe3ac93e66c4
 ---> b989da071209
Successfully built b989da071209
Successfully tagged kavyag3011/calculator-devops:dev

real    0m6.952s
user    0m0.009s
sys     0m0.018s
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 68: After changing one Java file: dependency layers cached, rebuild starts at `COPY src` (6.9 s).

5.4 Docker Hub

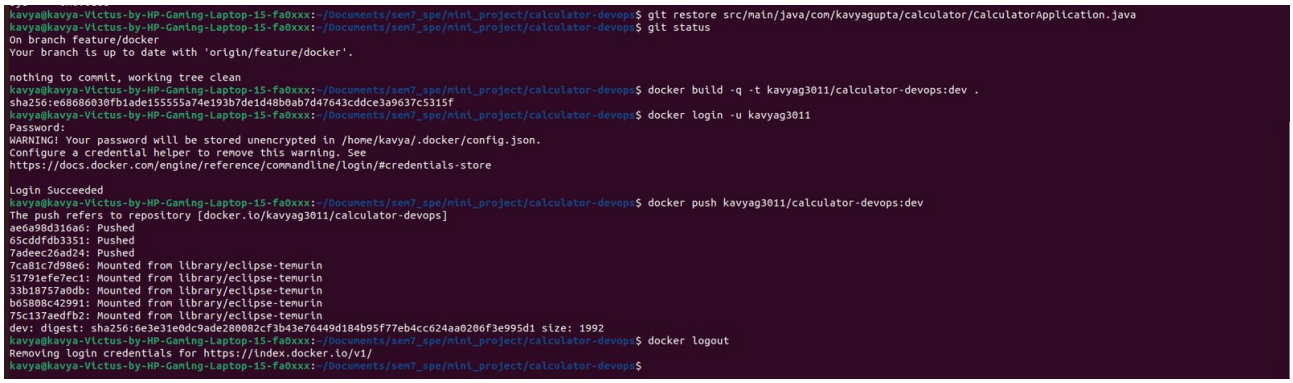


Figure 69: Manual push: application layer pushed, base layers mounted from `eclipse-temurin`.

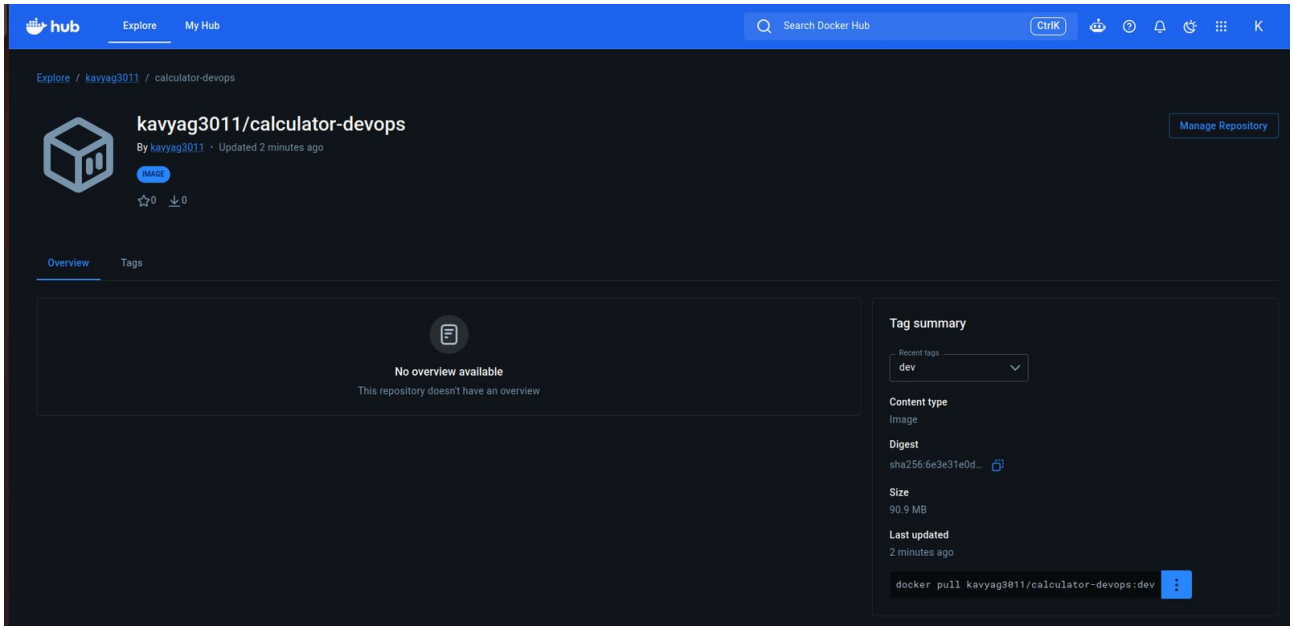


Figure 70: Public Docker Hub repository.

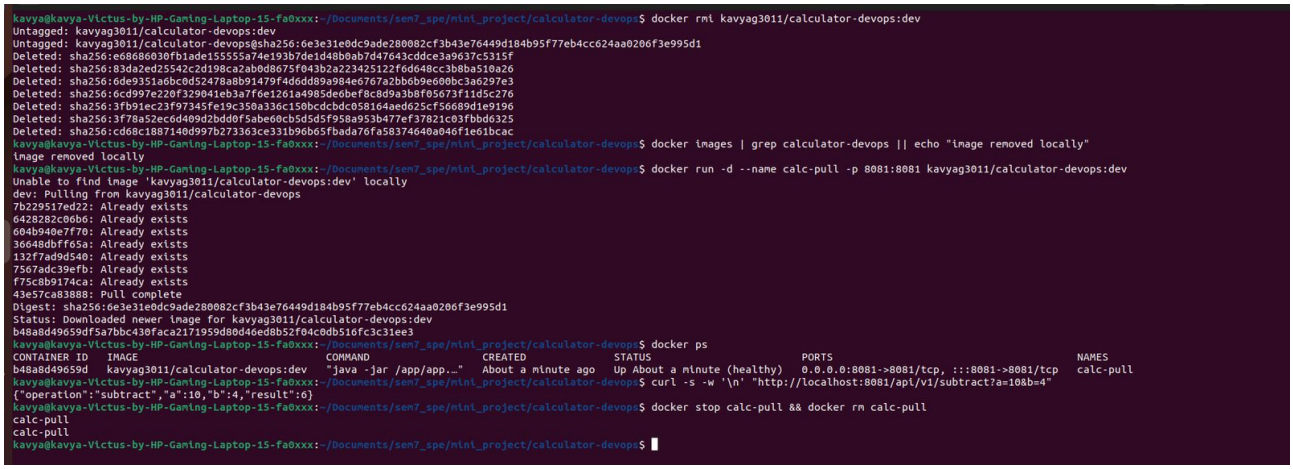


Figure 71: Local image removed, pulled back from Docker Hub and run.

5.5 Rebase and squash merge

A small documentation pull request (#9) was merged first, so `main` moved ahead of the point where `feature/docker` started. The feature branch was rebased onto `main`: its two commits were replayed with new

IDs on top of the new tip, giving a straight history. Because the commits were rewritten, the branch was pushed with `--force-with-lease`, which refuses to overwrite remote work it has not seen. Pull request #10 was then merged with *squash and merge*, adding a single commit to `main`.

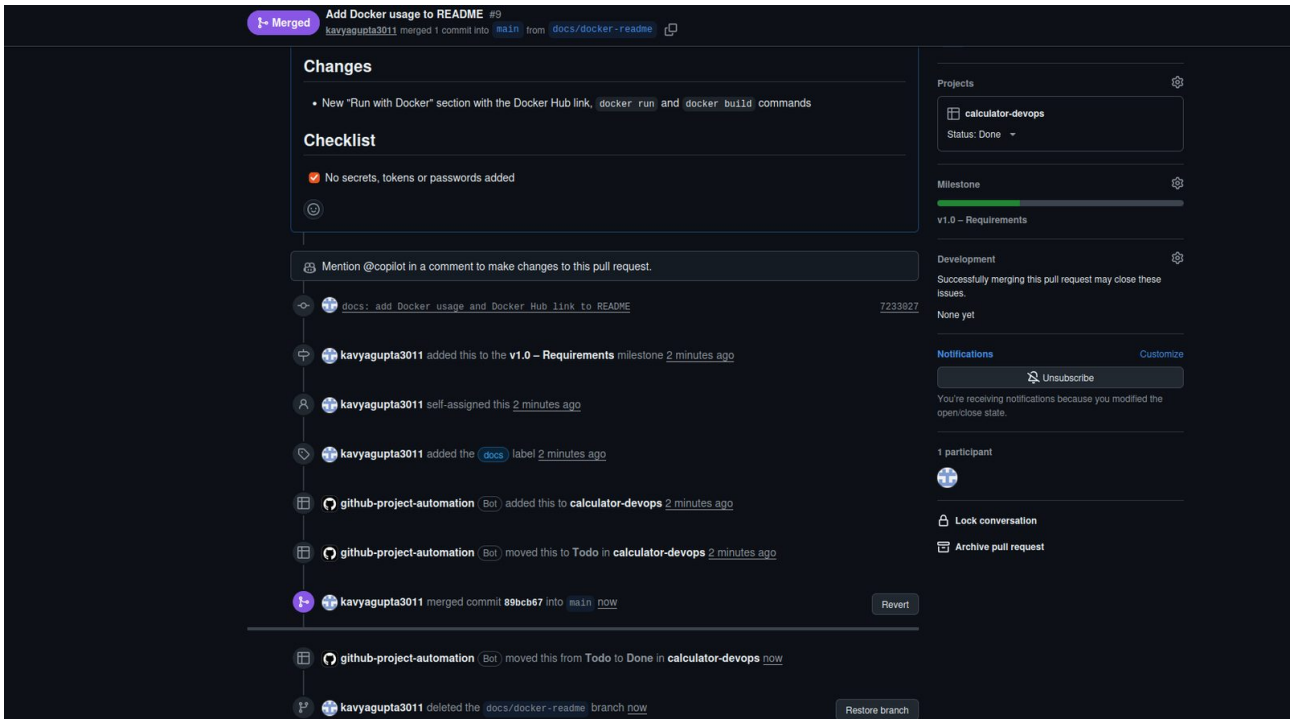


Figure 72: Documentation pull request #9 merged.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch main
Already on 'main'
Your branch is up to date with 'origin/main'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 2), reused 2 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.24 KiB | 1.24 MiB/s, done.
From https://github.com/kavyagupta3011/calculator-devops
   96c1308..89bcb67  main       -> origin/main
Updating 96c1308..89bcb67
Fast-forward
 README.md | 17 ++++++++++++++++
 1 file changed, 17 insertions(+)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git branch -d docs/docker-readme
warning: deleting branch 'docs/docker-readme' that has been merged to
         'refs/remotes/origin/docs/docker-readme', but not yet merged to HEAD.
Deleted branch docs/docker-readme (was 7233027).
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git fetch --prune
From https://github.com/kavyagupta3011/calculator-devops
 - [deleted]         (none)     -> origin/docs/docker-readme
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline --graph --all -8
* 89bcb67 (HEAD -> main, origin/main) docs: add Docker usage and Docker Hub link to README (#9)
| * 89fbaa4 (origin/feature/docker, feature/docker) build: add multi-stage Dockerfile with non-root runtime user
| * c6ddf60 build: add .dockerignore to keep the build context small
|/
*   96c1308 Merge pull request #8 from kavyagupta3011/feature/calculator-api
|\
| * 1d38198 docs: document build, run and API usage in README
| * bd2a881 test: verify actuator health endpoints in full application context
| * 58e0329 feat: expose actuator health endpoint with liveness and readiness probes
| * 7b86484 test: add MockMvc tests for calculator endpoints
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 73: Before the rebase: `feature/docker` still based on the old `main`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch feature/docker
Switched to branch 'feature/docker'
Your branch is up to date with 'origin/feature/docker'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline -3
89fbaa4 (HEAD -> feature/docker, origin/feature/docker) build: add multi-stage Dockerfile with non-root runtime user
c6ddf60 build: add .dockerignore to keep the build context small
96c1308 Merge pull request #8 from kavyagupta3011/feature/calculator-api
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git rebase main
Successfully rebased and updated refs/heads/feature/docker.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline --graph --all -8
* 746ae17 (HEAD -> feature/docker) build: add multi-stage Dockerfile with non-root runtime user
* d97099b build: add .dockerignore to keep the build context small
* 89bcb67 (origin/main, main) docs: add Docker usage and Docker Hub link to README (#9)
| * 89fbaa4 (origin/feature/docker) build: add multi-stage Dockerfile with non-root runtime user
| * c6ddf60 build: add .dockerignore to keep the build context small
|/
*   96c1308 Merge pull request #8 from kavyagupta3011/feature/calculator-api
|\
| * 1d38198 docs: document build, run and API usage in README
| * bd2a881 test: verify actuator health endpoints in full application context
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch feature/docker
Your branch and 'origin/feature/docker' have diverged,
and have 3 and 2 different commits each, respectively.
  (use "git pull" to merge the remote branch into yours)

nothing to commit, working tree clean
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push --force-with-lease
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.35 KiB | 1.35 MiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/kavyagupta3011/calculator-devops.git
 + 89fbaa4...746ae17 feature/docker -> feature/docker (forced update)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
On branch feature/docker
Your branch is up to date with 'origin/feature/docker'.

nothing to commit, working tree clean
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 74: `git rebase main`, the branch diverges from its remote copy, `git push --force-with-lease`.

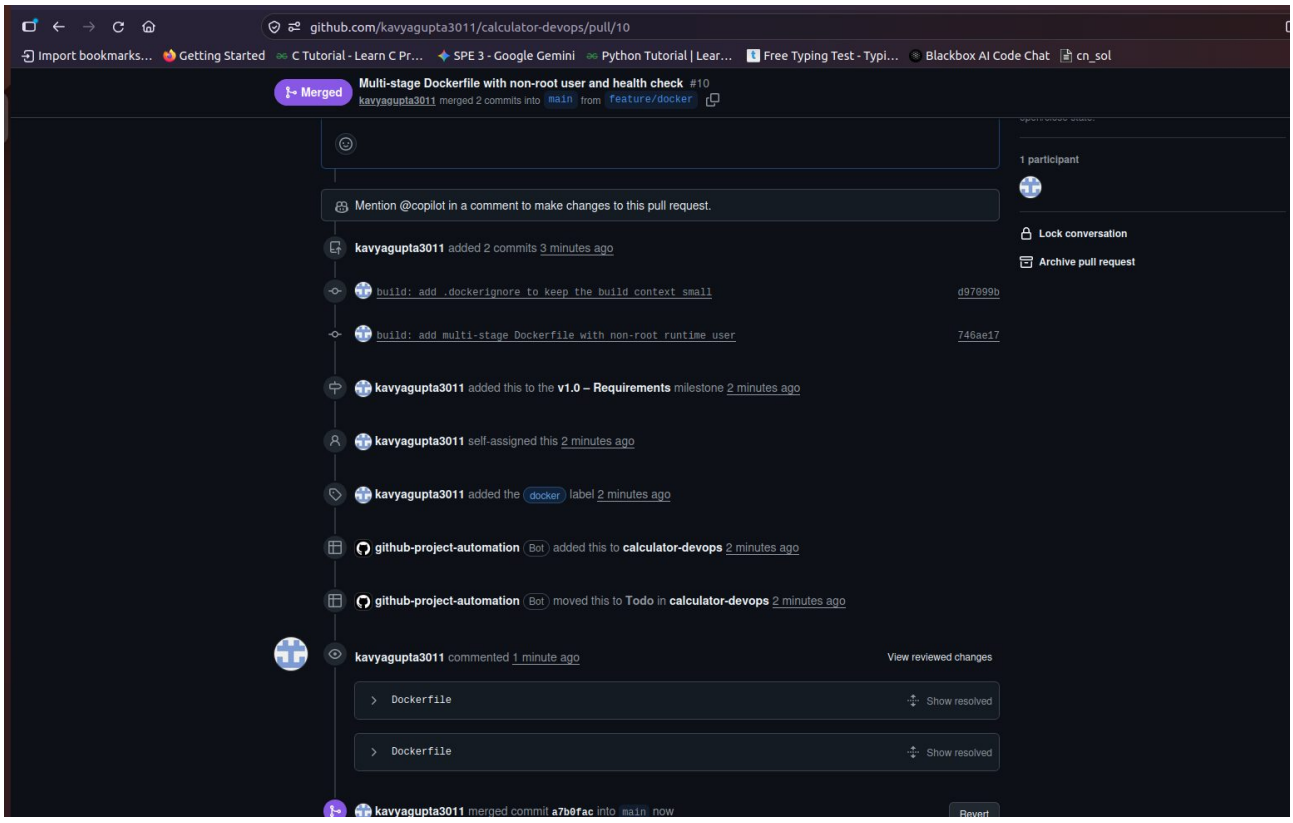


Figure 75: Pull request #10 squashed and merged.

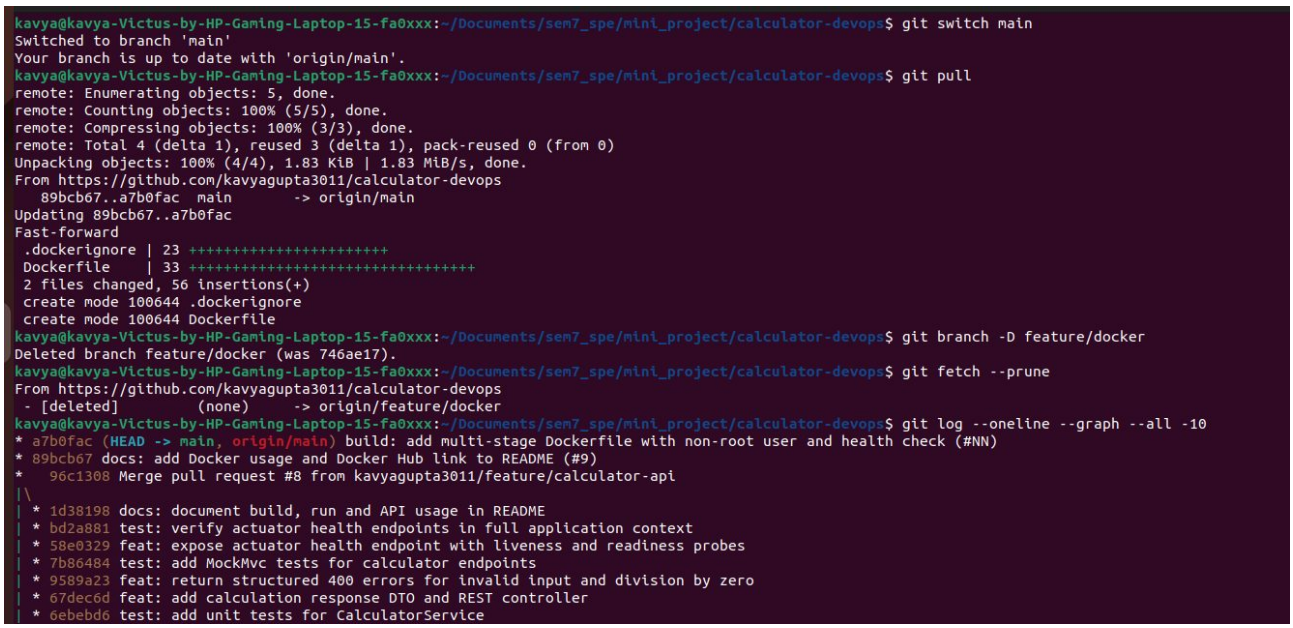


Figure 76: `main` after the squash: one commit per pull request.

6 Docker Compose deployment

6.1 Compose file and Nginx

The stack (Appendix A.3) has two services under the fixed project name `calculator`:

- **calculator** – image `${IMAGE_NAME}:${IMAGE_TAG}` (defaults `kavyag3011/calculator-devops:dev`), environment `SERVER_PORT` and `JAVA_TOOL_OPTIONS=-XX:MaxRAMPercentage=75`, `expose: 8081` only, health check on `/actuator/health`, `restart: unless-stopped`, attached only to the `backend` network.
- **nginx** – `nginx:1.27-alpine`, published on `${NGINX_PORT:-8000}`, configuration mounted read-only, starts only when `calculator` is healthy (`depends_on: condition: service_healthy`), its own health check, attached to `frontend` and `backend`.
- **Networks** – `backend` is `internal`, so the API cannot be reached from the host or reach the internet.

Nginx forwards only `/api/` and `/actuator/health`; other paths return a JSON 404. It resolves the service name through Docker's DNS (`resolver 127.0.0.11 valid=10s`) and a variable in `proxy_pass`, so a container recreated with a new IP is picked up without restarting Nginx. `.env.example` documents the variables; a real `.env` is ignored by Git.

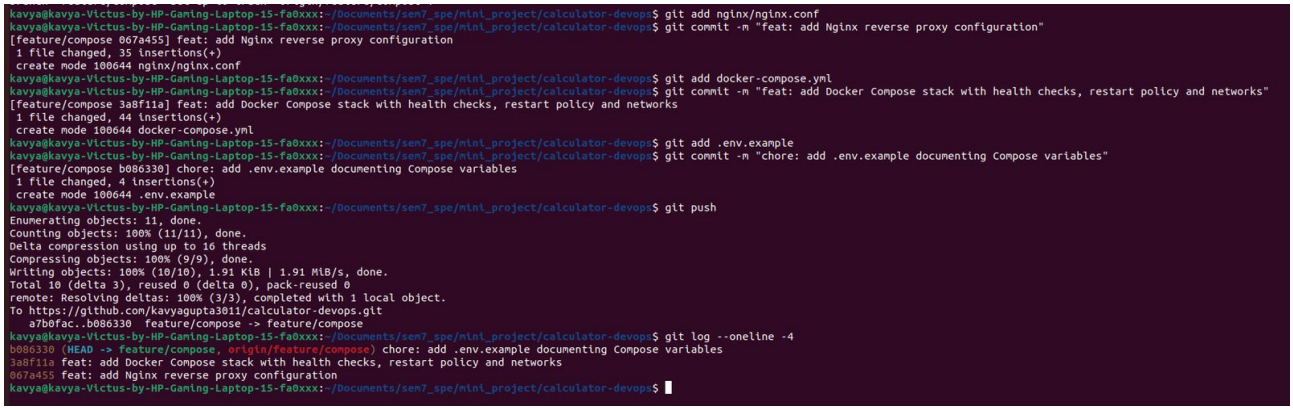


Figure 77: Branch `feature/compose` with three commits.

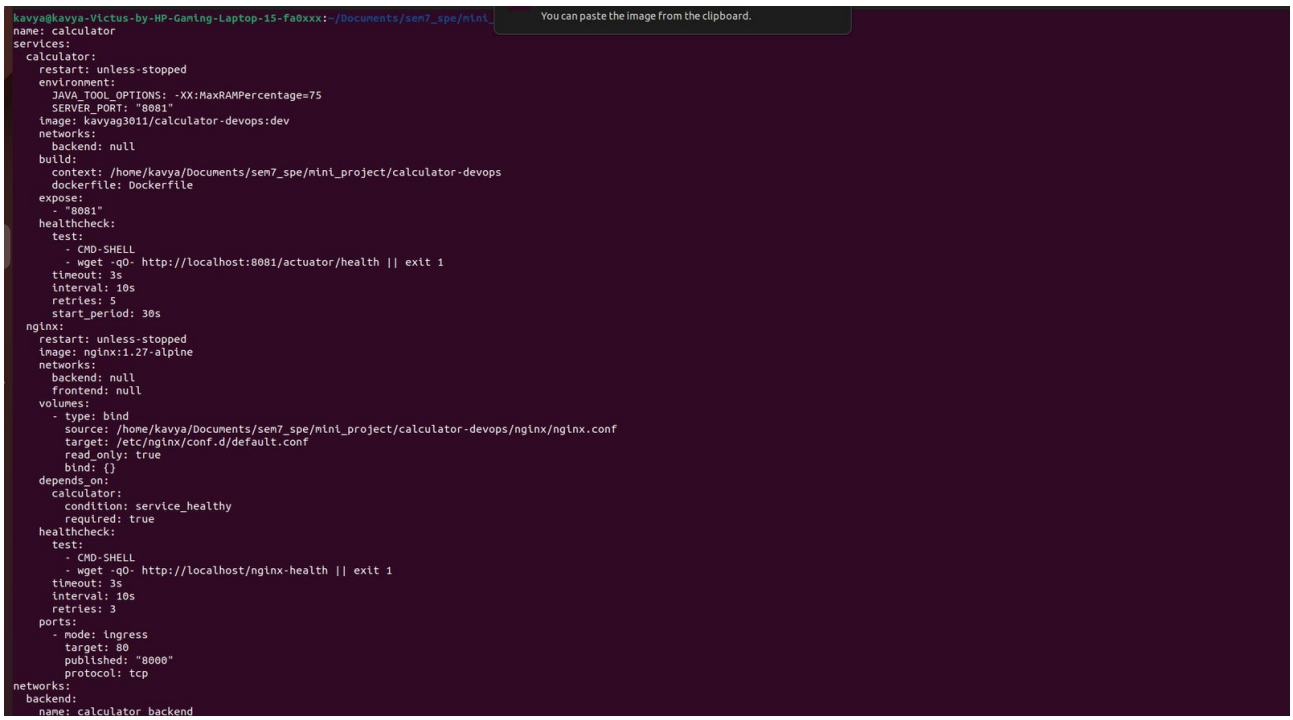


Figure 78: `docker compose config` with variables resolved.

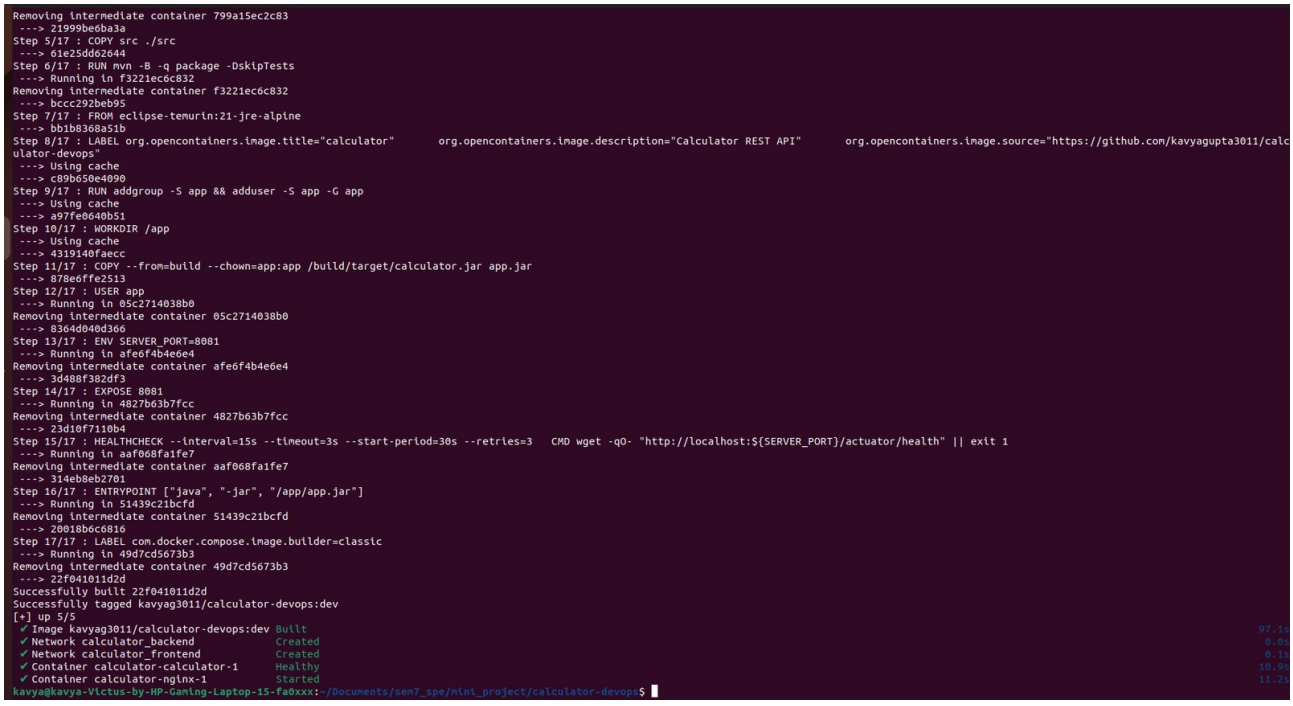


Figure 79: `docker compose up -d --build`: networks created, `calculator` healthy before `nginx` starts.



Figure 80: Both services healthy; only Nginx publishes a host port.



Figure 81: Requests through Nginx; port 8081 is not reachable from the host.

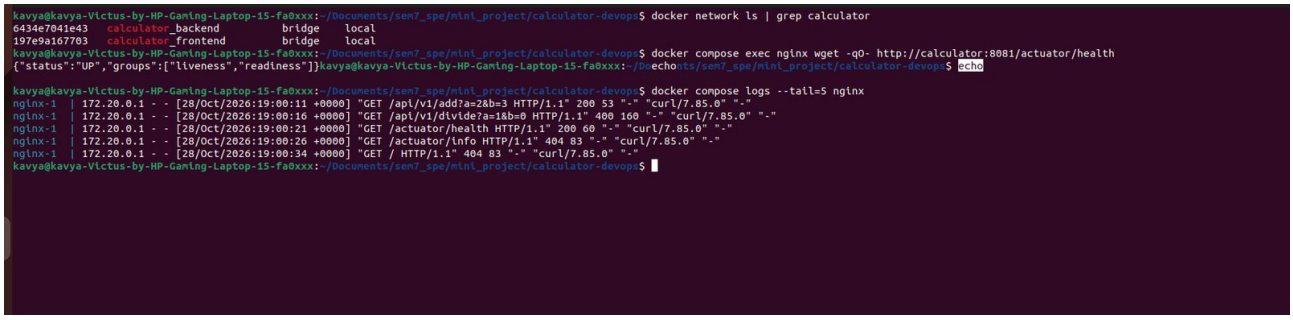


Figure 82: Compose networks, service-name resolution from inside Nginx, and the Nginx access log.

The first `compose up` failed because pulling `nginx:1.27-alpine` from Docker Hub timed out; a separate `docker pull` succeeded on retry.

6.2 Behaviour checks

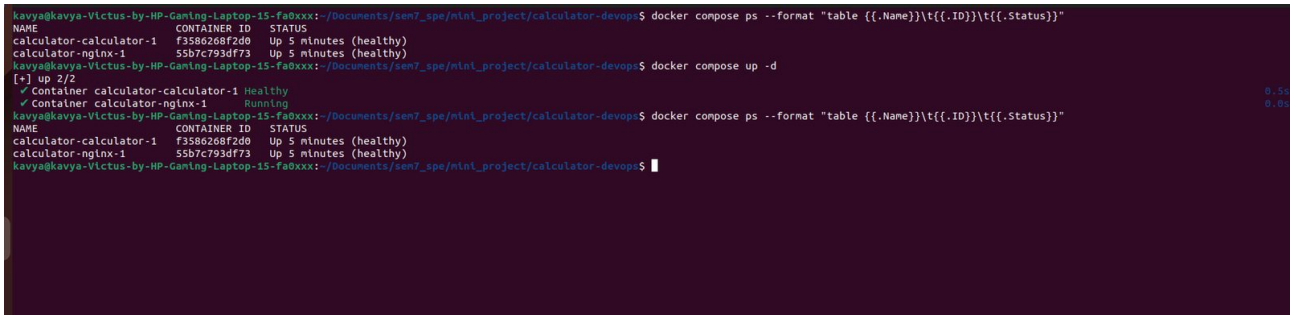


Figure 83: Running `docker compose up -d` again: same container IDs, nothing recreated.

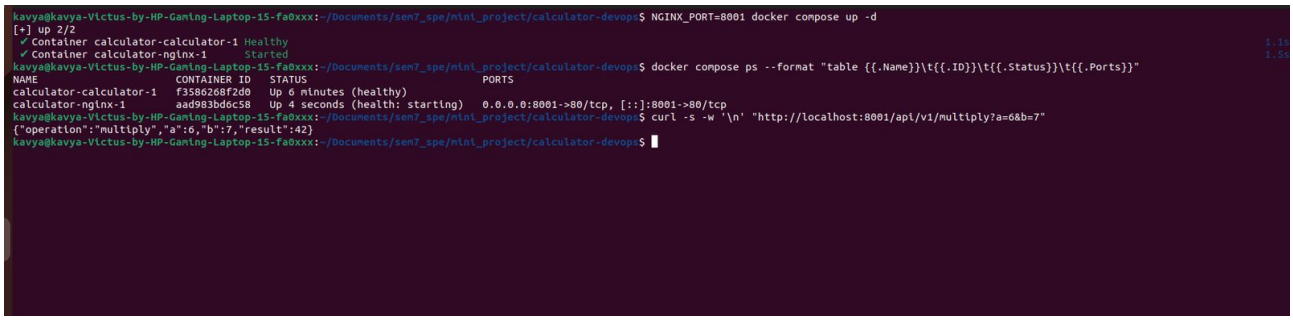


Figure 84: Changing `NGINX_PORT`: only Nginx is recreated.

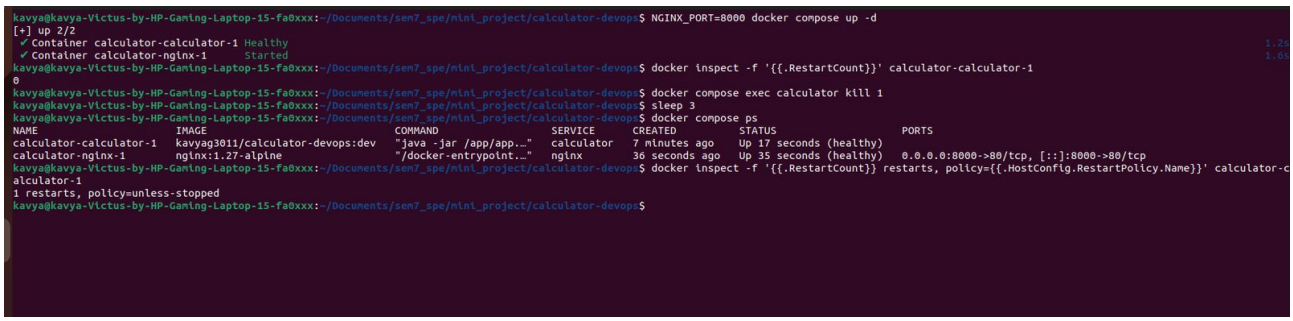


Figure 85: Killing the JVM: Docker restarts the container (`1 restarts, policy=unless-stopped`).

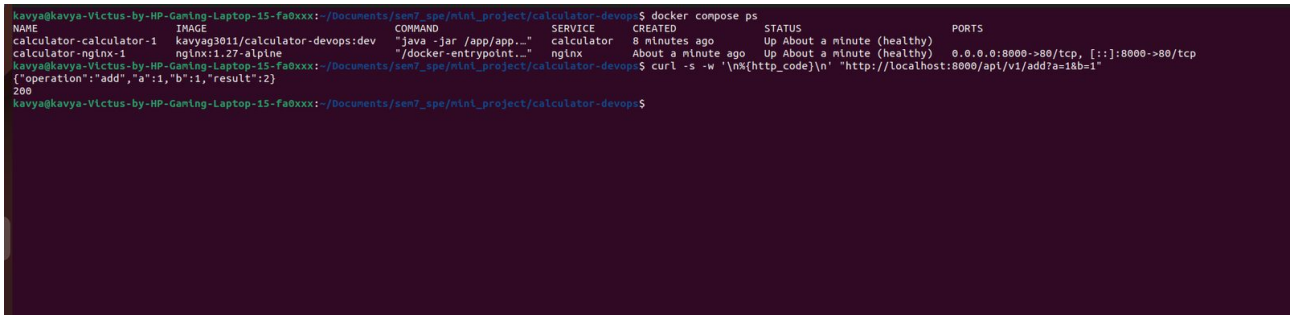


Figure 86: Container healthy again and serving through Nginx.

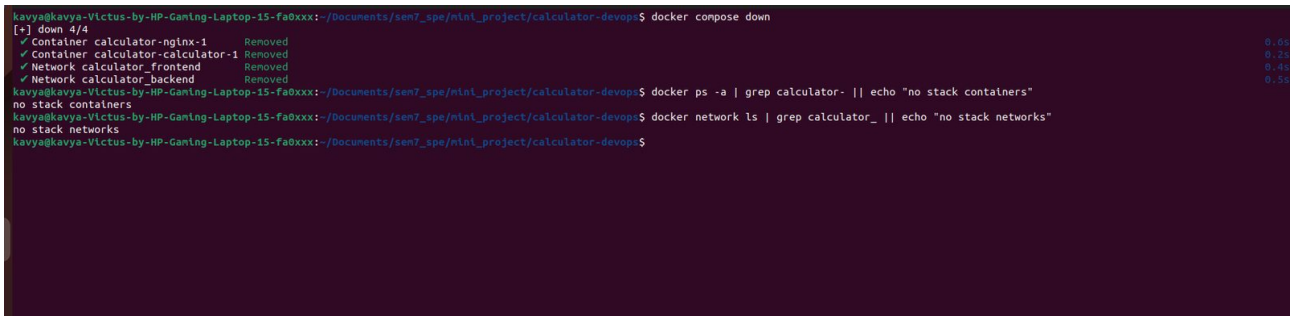


Figure 87: `docker compose down` removes both containers and networks.

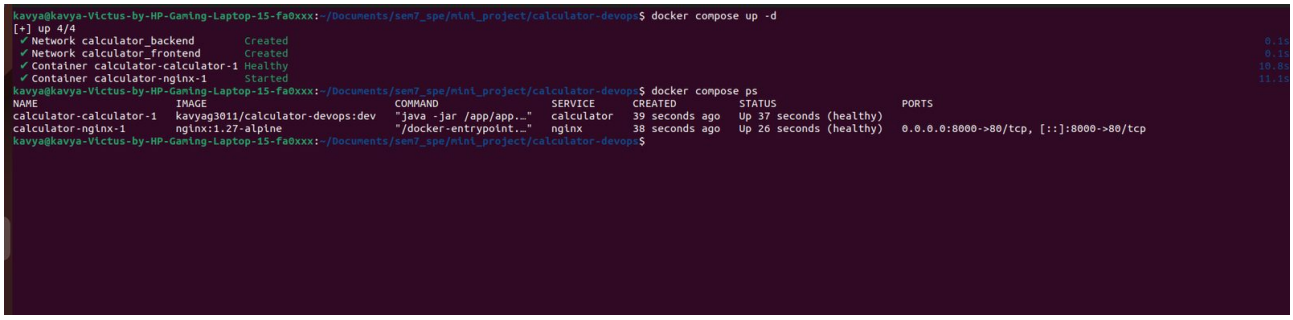


Figure 88: A single `docker compose up -d` recreates the full stack.

6.3 Merge conflict and rebase-and-merge

While the Compose branch was open, another pull request (#11) changed the same README line on `main`. Pull request #12 then reported a conflict. It was resolved during `git rebase origin/main`: Git stopped at the README commit, the conflict markers were replaced by a line keeping both changes, the file was staged and the rebase continued. In a rebase, `HEAD` is the branch being rebased onto (`main`) and the other side is the commit being replayed. Pull request #12 was merged with *rebase and merge*, which adds the four commits to `main` individually without a merge commit.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git pull
Already up to date.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch -c docs/readme-webhook
Switched to a new branch 'docs/readme-webhook'
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ sed -i 's/^GitHub -> Jenkins/GitHub (webhook) -> Jenkins/' README.md
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git diff
diff --git a/README.md b/README.md
index f71fc60..b5c6c9c 100644
--- a/README.md
+++ b/README.md
@@ -3,7 +3,7 @@
 Calculator REST API built with Java 21, Spring Boot and Maven, deployed through a CI/CD pipeline using GitHub, Jenkins, Docker and Docker Compose.

 ```
-GitHub -> Jenkins -> Maven build and test -> Docker image -> Docker Hub -> Docker Compose
+GitHub (webhook) -> Jenkins -> Maven build and test -> Docker image -> Docker Hub -> Docker Compose
 ```

 ## Tech stack
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -am "docs: mention webhook trigger in pipeline overview"
[docs/readme-webhook 84a9c48] docs: mention webhook trigger in pipeline overview
 1 file changed, 1 insertion(+), 1 deletion(-)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push -u origin docs/readme-webhook
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 335 bytes | 335.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'docs/readme-webhook' on GitHub by visiting:
remote: https://github.com/kavyagupta3011/calculator-devops/pull/new/docs/readme-webhook
remote:
To https://github.com/kavyagupta3011/calculator-devops.git
 * [new branch]   docs/readme-webhook -> docs/readme-webhook
branch 'docs/readme-webhook' set up to track 'origin/docs/readme-webhook'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 89: Change to the same README line on a separate branch.

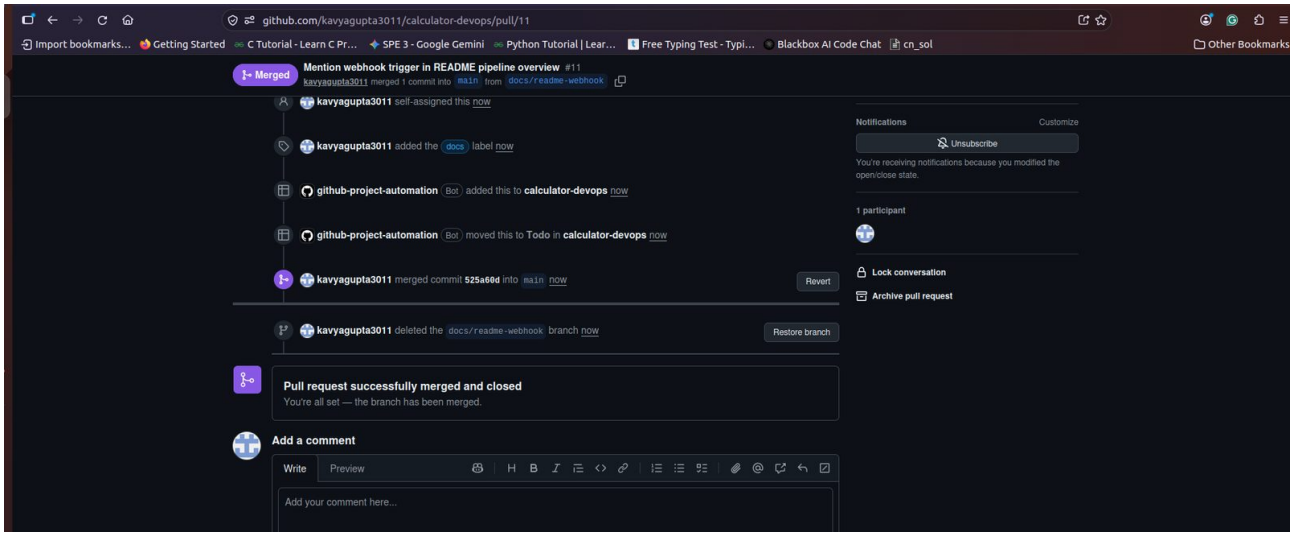


Figure 90: Pull request #11 merged into `main`.

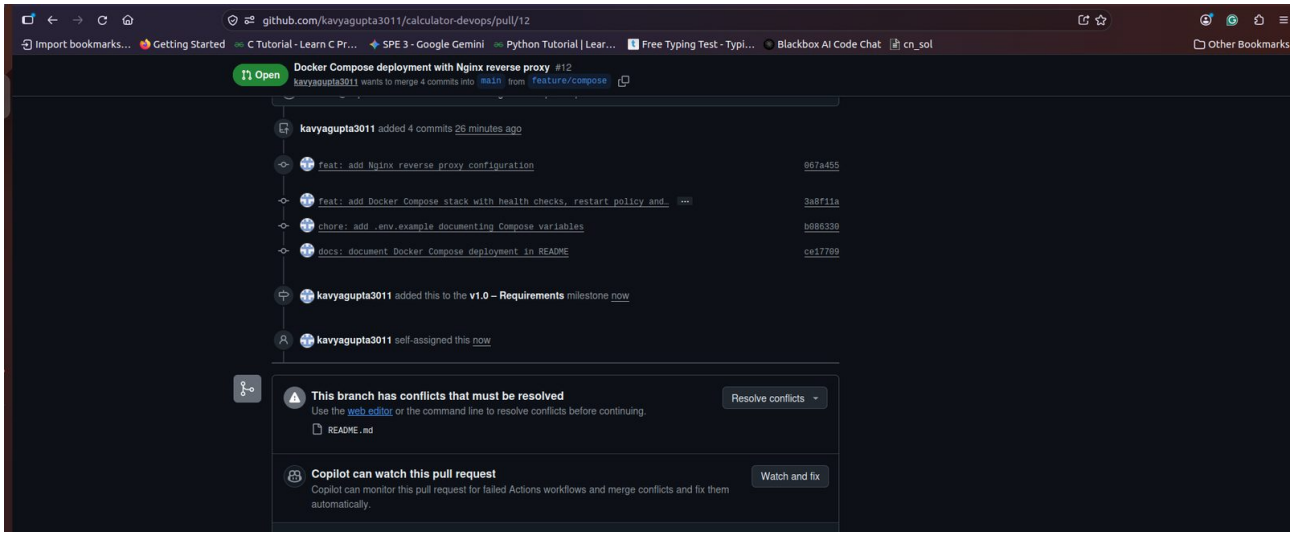


Figure 91: Pull request #12: *This branch has conflicts that must be resolved.*

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git fetch origin
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git rebase origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply ce17709... docs: document Docker Compose deployment in README
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
Could not apply ce17709... docs: document Docker Compose deployment in README
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
interactive rebase in progress; onto 525a60d
Last commands done (4 commands done):
   pick b086330 chore: add .env.example documenting Compose variables
   pick ce17709 docs: document Docker Compose deployment in README
  (see more in file .git/rebase-merge/done)
No commands remaining.
You are currently rebasing branch 'feature/compose' on '525a60d'.
  (fix conflicts and then run "git rebase --continue")
  (use "git rebase --skip" to skip this patch)
  (use "git rebase --abort" to check out the original branch)

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ grep -n -A6 "<<<<<<<" README.md
6:<<<<<<< HEAD
7-GitHub (webhook) -> Jenkins -> Maven build and test -> Docker image -> Docker Hub -> Docker Compose
8-=======
9-GitHub -> Jenkins -> Maven build and test -> Docker image -> Docker Hub -> Docker Compose (Nginx + API)
10->>>>>>> ce17709 (docs: document Docker Compose deployment in README)
11-```
12-
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 92: Rebase stops at the conflicting commit; conflict markers in `README.md`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ nano README.md
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ grep -c -E "^(<<<<<<<|=======|>>>>>>>)" README.md
0
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ sed -n '1,8p' README.md
# calculator-devops

Calculator REST API built with Java 21, Spring Boot and Maven, deployed through a CI/CD pipeline using GitHub, Jenkins, Docker and Docker Compose.

```
 GitHub (webhook) -> Jenkins -> Maven build and test -> Docker image -> Docker Hub -> Docker Compose (Nginx + API)
```

kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add README.md
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git status
interactive rebase in progress; onto 525a60d
Last commands done (4 commands done):
   pick b086330 chore: add .env.example documenting Compose variables
   pick ce17709 docs: document Docker Compose deployment in README
  (see more in file .git/rebase-merge/done)
No commands remaining.
You are currently rebasing branch 'feature/compose' on '525a60d'.
  (all conflicts fixed: run "git rebase --continue")

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified: README.md

kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ GIT_EDITOR=true git rebase --continue
[detached HEAD 008db0a] docs: document Docker Compose deployment in README
 1 file changed, 30 insertions(+), 1 deletion(-)
Successfully rebased and updated refs/heads/feature/compose.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline --graph -7
* 008db0a (HEAD -> feature/compose) docs: document Docker Compose deployment in README
* fde1552 chore: add .env.example documenting Compose variables
* 23008ab feat: add Docker Compose stack with health checks, restart policy and networks
* 086e260 feat: add Nginx reverse proxy configuration
* 525a60d (origin/main, main) docs: mention webhook trigger in pipeline overview (#11)
* a7b8fac build: add multi-stage Dockerfile with non-root user and health check (#NN)
* 89bcb67 docs: add Docker usage and Docker Hub link to README (#9)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 93: Resolution keeps both changes; `git rebase --continue`.

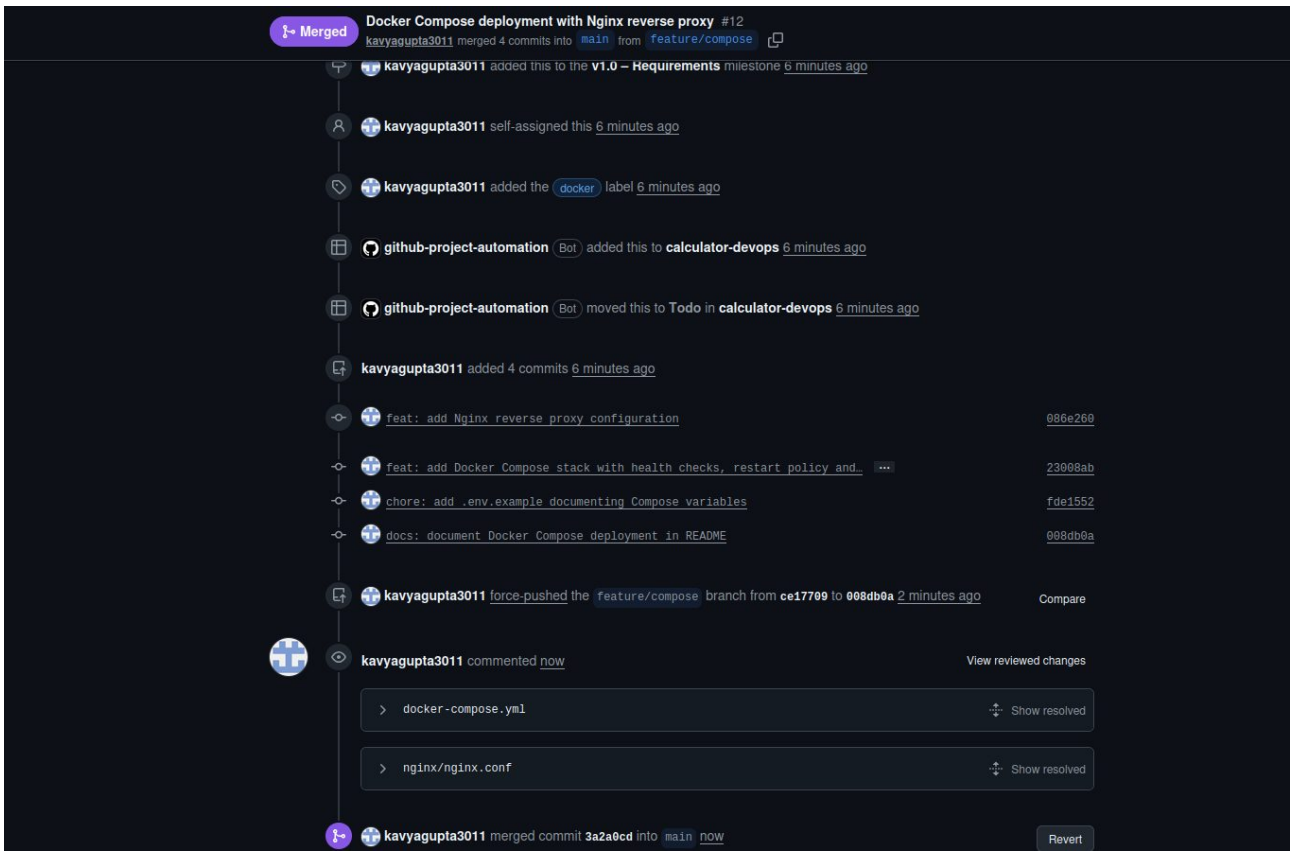


Figure 94: Pull request #12 merged with rebase and merge.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git pull
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 13 (delta 4), reused 9 (delta 4), pack-reused 0 (from 0)
Unpacking objects: 100% (13/13), 3.14 KiB | 357.00 KiB/s, done.
From https://github.com/kavyagupta3011/calculator-devops
   525a60d..3a2a0cd  main       -> origin/main
Updating 525a60d..3a2a0cd
Fast-forward
 .env.example       |  4 ++++
 README.md          | 31 +++++++++++++++++++++++++++++-
 docker-compose.yml | 44 ++++++++++++++++++++++++++++++++++++++++++++
 nginx/nginx.conf   | 35 +++++++++++++++++++++++++++++++++++
 4 files changed, 113 insertions(+), 1 deletion(-)
 create mode 100644 .env.example
 create mode 100644 docker-compose.yml
 create mode 100644 nginx/nginx.conf
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git branch -D feature/compose
Deleted branch feature/compose (was 008db0a).
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git fetch --prune
From https://github.com/kavyagupta3011/calculator-devops
 - [deleted]         (none)     -> origin/feature/compose
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline --graph --all -12
* 3a2a0cd (HEAD -> main, origin/main) docs: document Docker Compose deployment in README
* 9203ea9 chore: add .env.example documenting Compose variables
* 9a0e568 feat: add Docker Compose stack with health checks, restart policy and networks
* e2ec88c feat: add Nginx reverse proxy configuration
* 525a60d docs: mention webhook trigger in pipeline overview (#11)
* a7b0fac build: add multi-stage Dockerfile with non-root user and health check (#NN)
* 89bcb67 docs: add Docker usage and Docker Hub link to README (#9)
*   96c1308 Merge pull request #8 from kavyagupta3011/feature/calculator-api
|\
| * 1d38198 docs: document build, run and API usage in README
| * bd2a881 test: verify actuator health endpoints in full application context
| * 58e0329 feat: expose actuator health endpoint with liveness and readiness probes
| * 7b86484 test: add MockMvc tests for calculator endpoints
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 95: History on `main`: the four Compose commits added individually.

7 Jenkins CI/CD pipeline

7.1 Job and webhook

A Multibranch Pipeline job builds every branch and pull request that contains a `Jenkinsfile` and reports a commit status for each. GitHub reaches the local Jenkins through an ngrok tunnel; the repository webhook sends push and pull request events to `/github-webhook/`. Discovery of pull requests from forks was removed so that external code is never run on the build machine, and a repository scan every hour catches missed events.

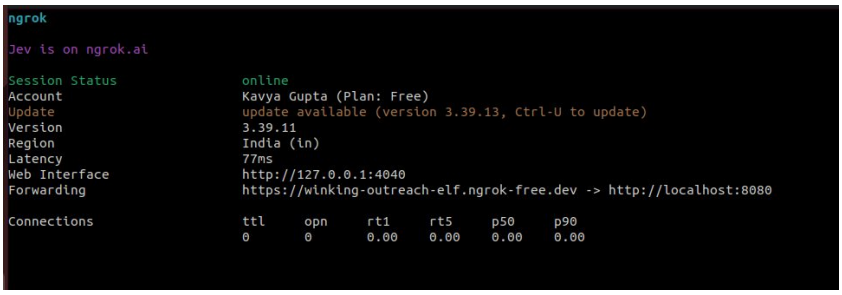


Figure 96: ngrok forwarding a public HTTPS URL to Jenkins on port 8080.

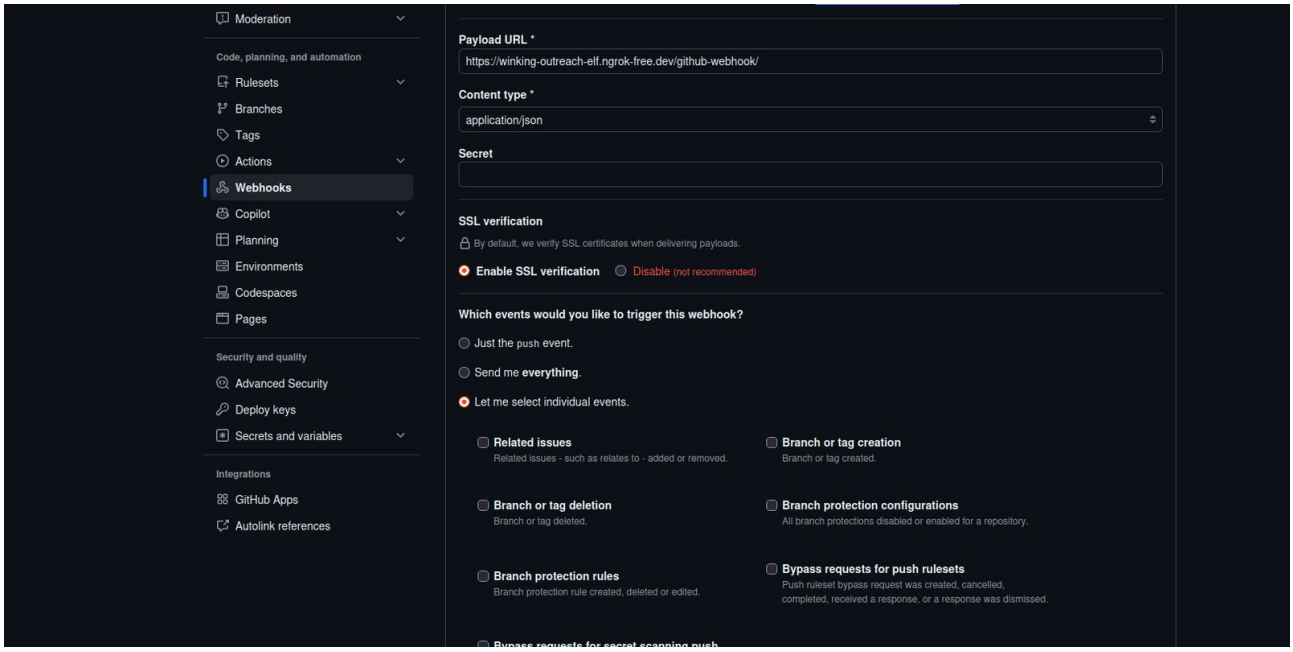


Figure 97: Repository webhook for push and pull request events.

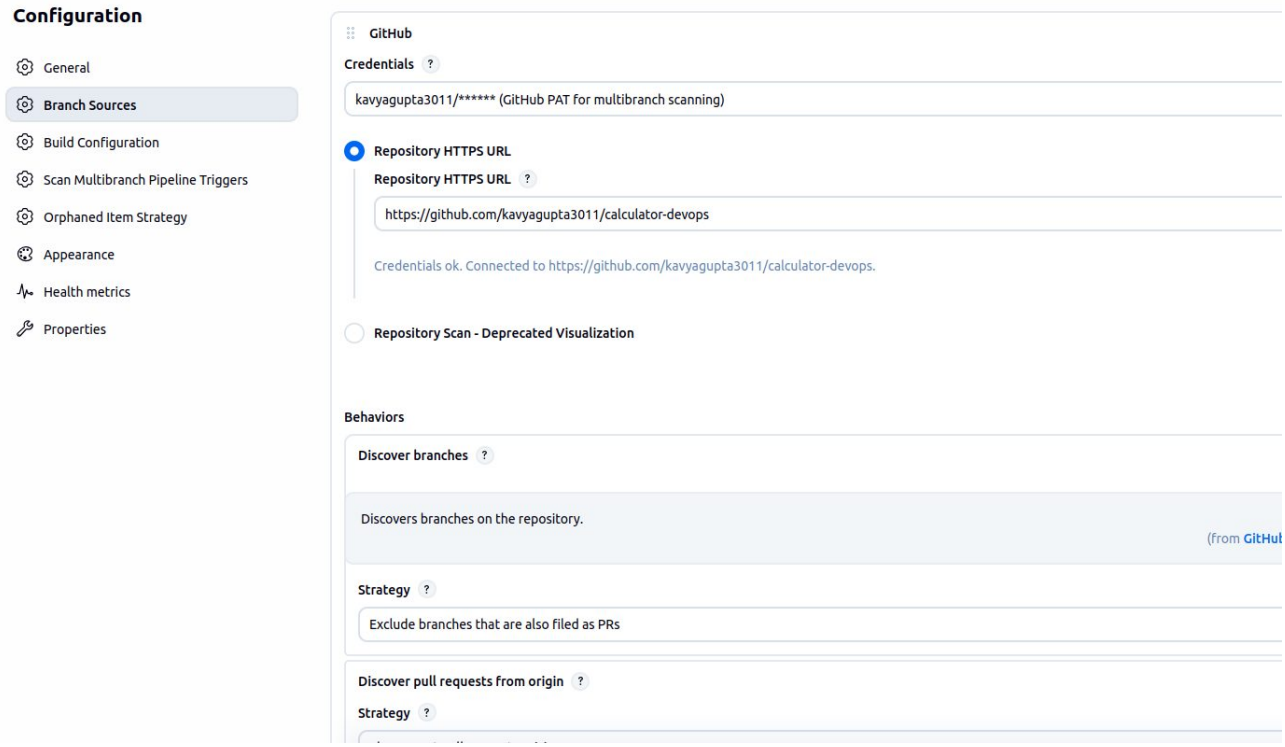


Figure 98: Multibranch branch source with credentials verified.

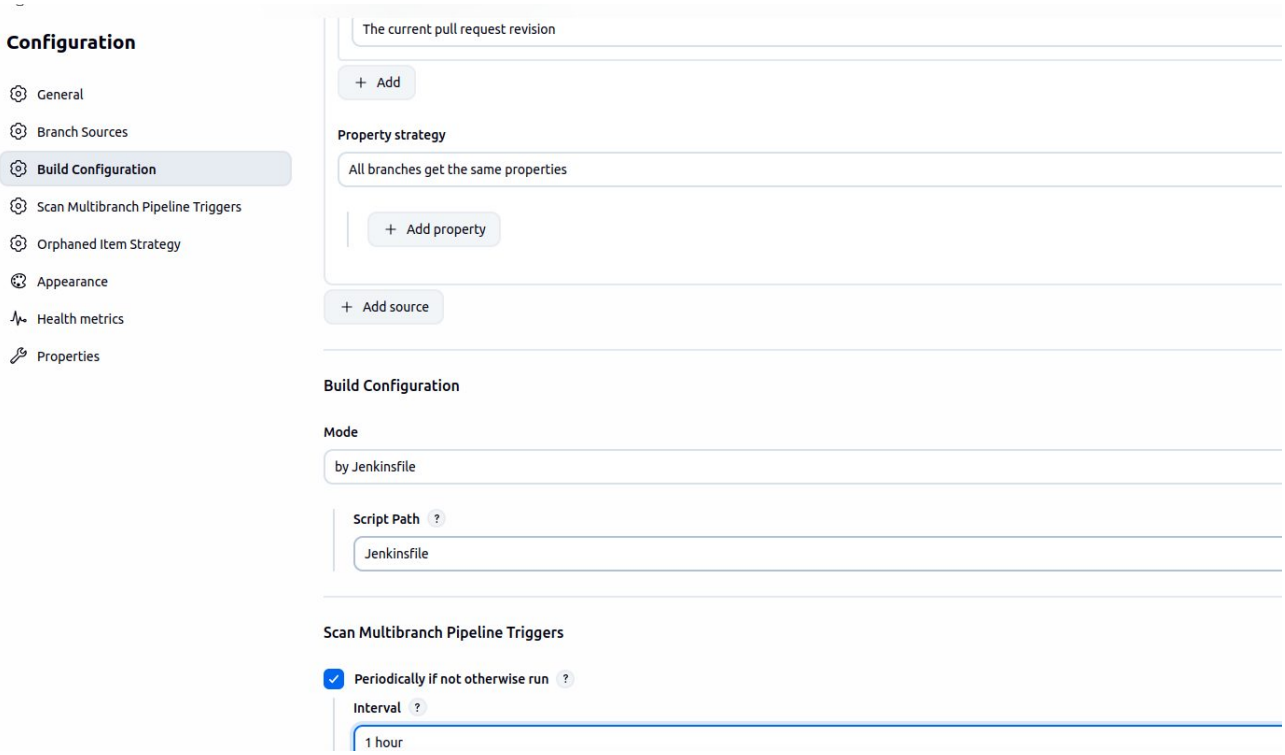


Figure 99: Build configuration (`Jenkinsfile`) and hourly scan trigger.

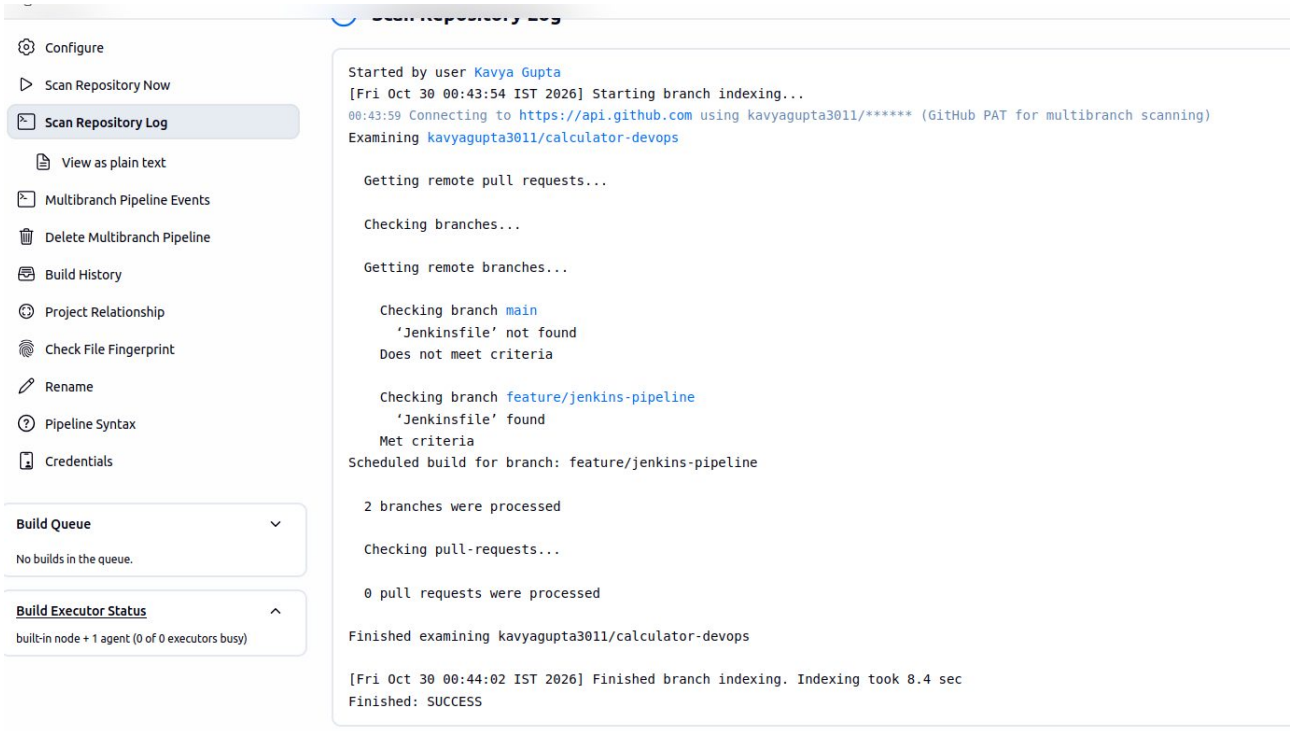


Figure 100: First scan: `main` has no `Jenkinsfile` yet; the feature branch does and is scheduled.

7.2 Building the pipeline stage by stage

The `Jenkinsfile` was developed on `feature/jenkins-pipeline`, one stage per commit, and every push triggered a build through the webhook.

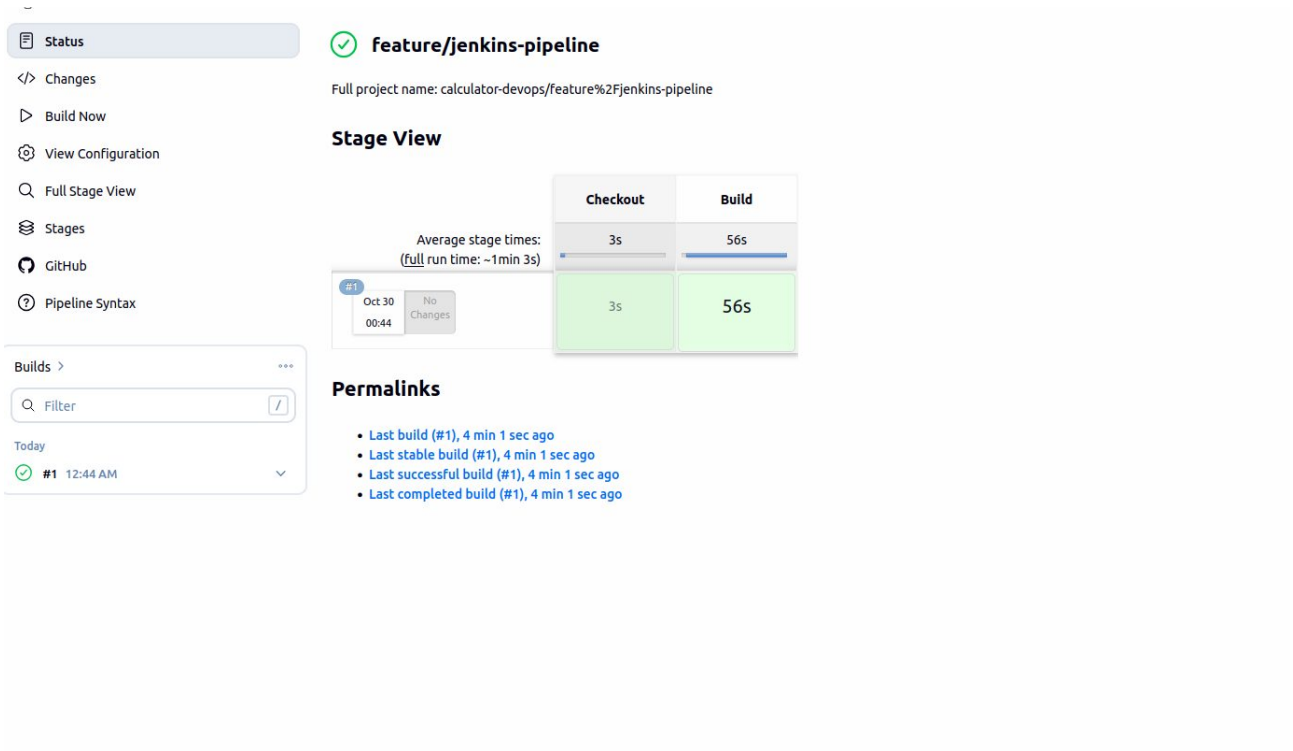


Figure 101: Build #1: Checkout and Build (first run downloads Maven dependencies).

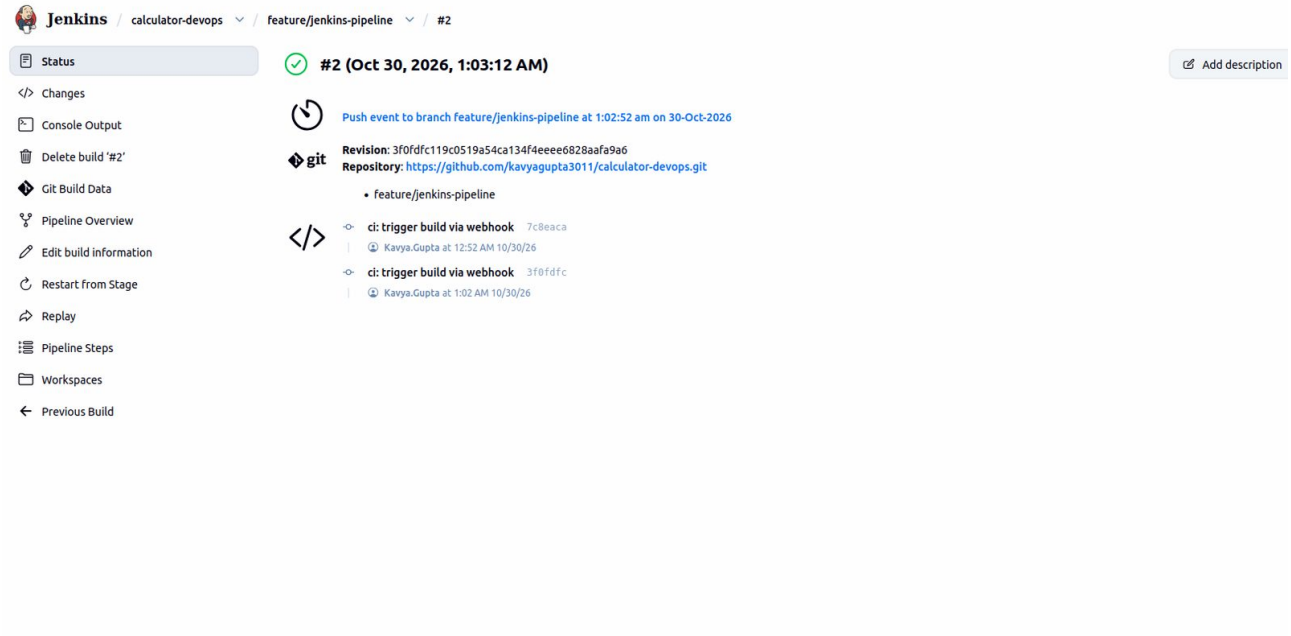


Figure 102: Build triggered by a push event, not by a user.

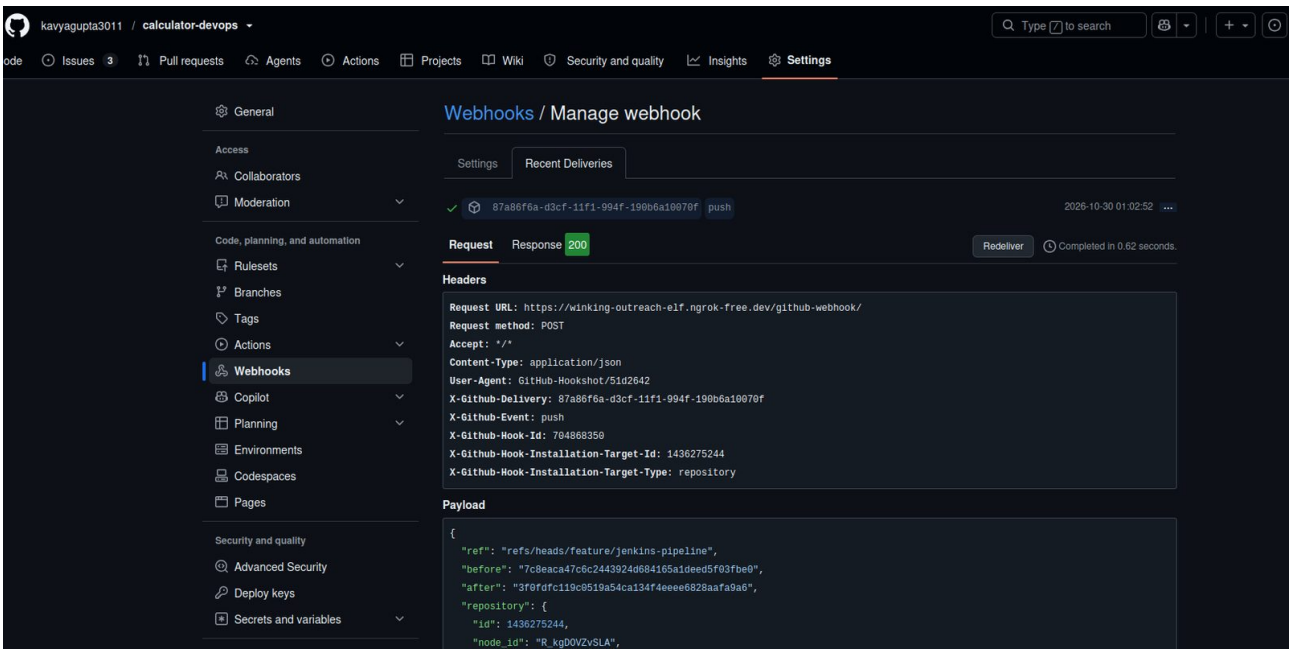


Figure 103: Webhook delivery accepted by Jenkins (HTTP 200).

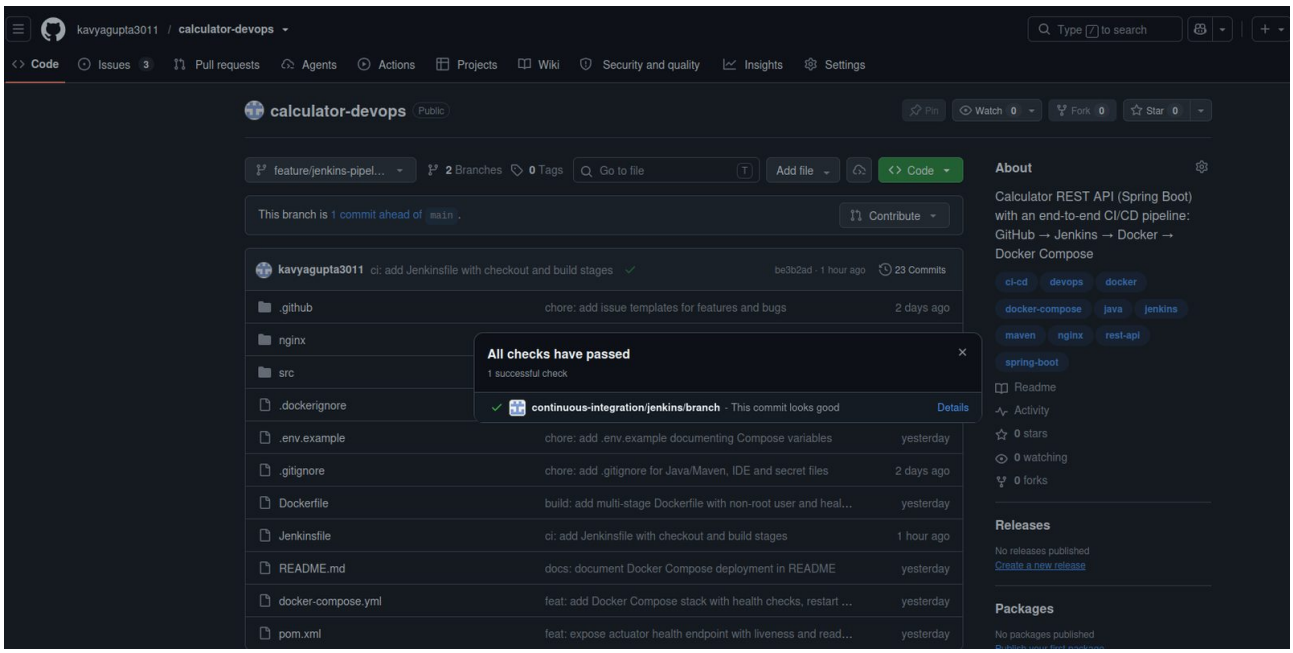


Figure 104: Commit status reported to GitHub.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git branch --show-current
feature/jenkins-pipeline
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git diff
diff --git a/Jenkinsfile b/Jenkinsfile
index 5e50a9b..17f747a 100644
--- a/Jenkinsfile
+++ b/Jenkinsfile
@@ -25,5 +25,16 @@ pipeline {
                 sh 'mvn -B clean package -DskipTests'
             }
         }
+
+        stage('Test') {
+            steps {
+                sh 'mvn -B test'
+            }
+            post {
+                always {
+                    junit testResults: 'target/surefire-reports/*.xml', allowEmptyResults: false
+                }
+            }
+        }
     }
 }
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add Jenkinsfile
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -m "ci: run tests and publish JUnit results"
[feature/jenkins-pipeline e9017d2] ci: run tests and publish JUnit results
 1 file changed, 11 insertions(+)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 438 bytes | 438.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/kavyagupta3011/calculator-devops.git
   3f0fdfc..e9017d2  feature/jenkins-pipeline -> feature/jenkins-pipeline
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 105: Test stage added: `mvn test`, then `junit` in `post { always }` so reports are published even when tests fail.

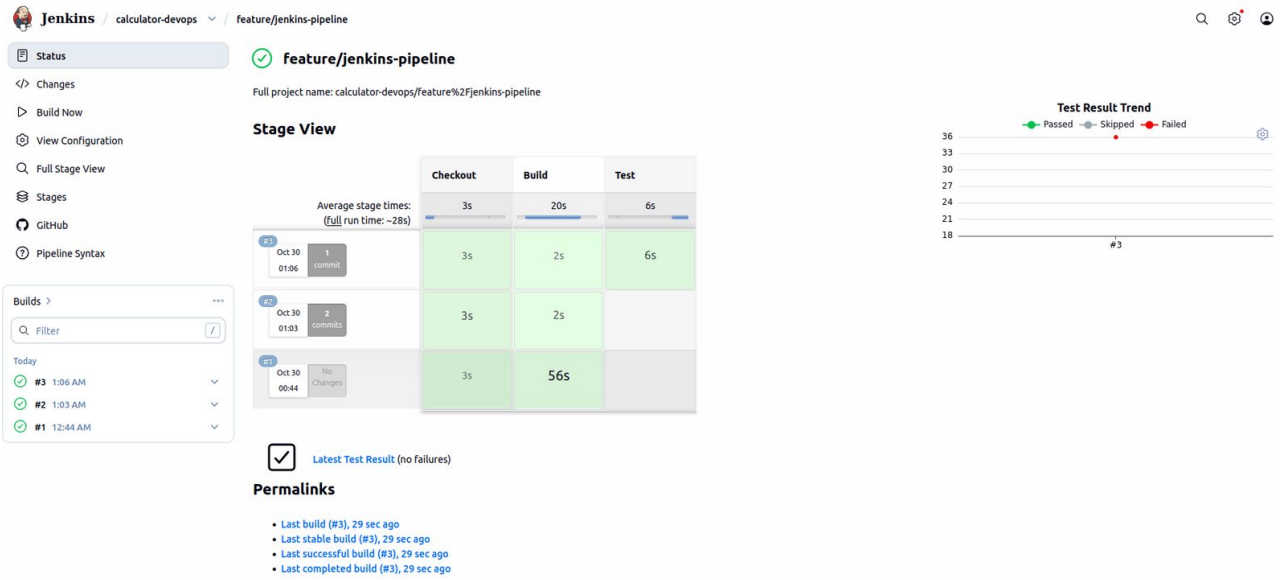


Figure 106: Checkout, Build and Test.

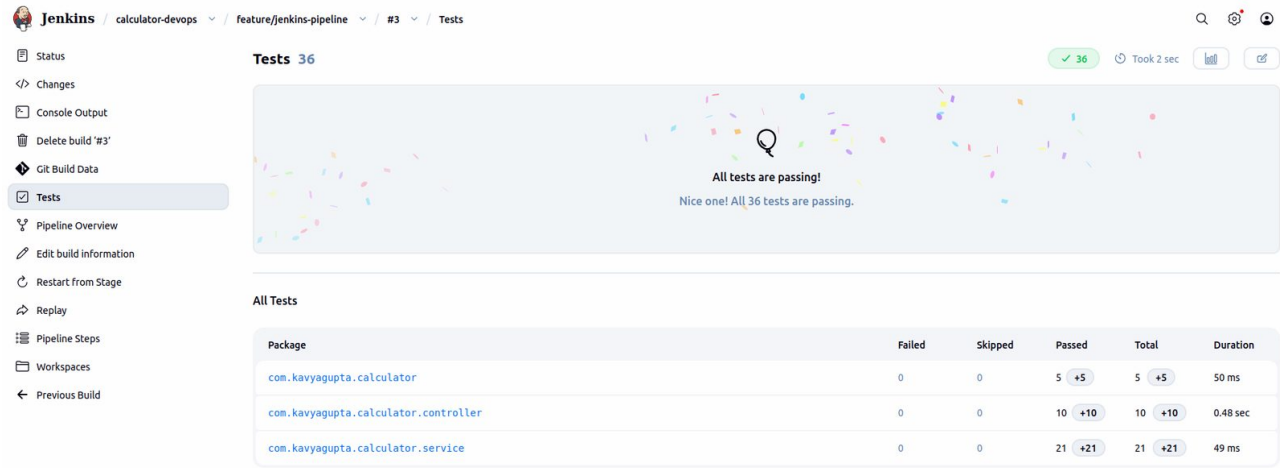


Figure 107: Published test results: 36 tests in three packages.

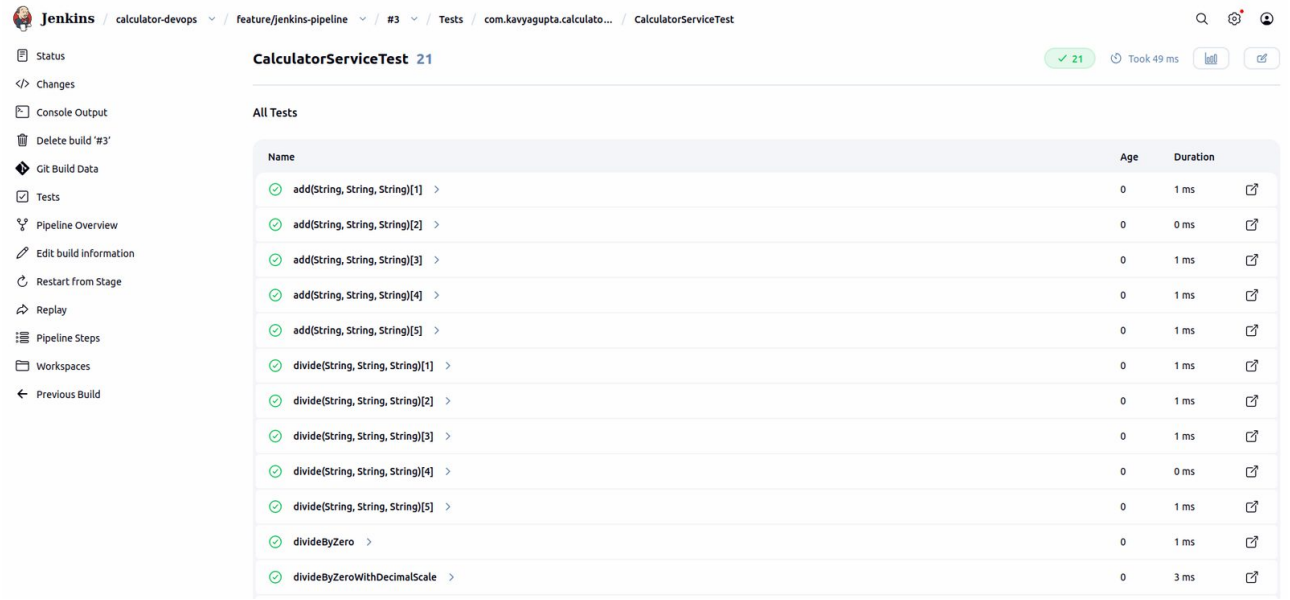


Figure 108: Individual test cases, including parameterized ones.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git branch --show-current
feature/jenkins-pipeline
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git diff
diff --git a/Jenkinsfile b/Jenkinsfile
index 17f747a..08dafb7 100644
--- a/Jenkinsfile
+++ b/Jenkinsfile
@@ -1,3 +1,5 @@
+def appImage
+
 pipeline {
     agent any

@@ -10,6 +12,8 @@ pipeline {

     environment {
         IMAGE_NAME = 'kavyag3011/calculator-devops'
+        // main builds are tagged with the plain build number, other branches get a prefix
+        IMAGE_TAG = "${env.BRANCH_NAME == 'main' ? env.BUILD_NUMBER : env.BRANCH_NAME.replaceAll('[^A-Za-z0-9_.-]', '-') + '-' + env.BUILD_NUMBER}"
     }

     stages {
@@ -36,5 +40,14 @@ pipeline {
                 }
             }
         }
+
+        stage('Docker Build') {
+            steps {
+                script {
+                    appImage = docker.build("${IMAGE_NAME}:${IMAGE_TAG}")
+                }
+                sh 'docker images "$IMAGE_NAME" --format "{{.Repository}}:{{.Tag}}  {{.Size}}" | head -5'
+            }
+        }
     }
 }
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git add Jenkinsfile
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git commit -m "ci: build Docker image tagged with build number"
[feature/jenkins-pipeline e2c94e6] ci: build Docker image tagged with build number
 1 file changed, 13 insertions(+)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 657 bytes | 657.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/kavyagupta3011/calculator-devops.git
   e9017d2..e2c94e6  feature/jenkins-pipeline -> feature/jenkins-pipeline
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 109: Docker Build stage and the image tag rule.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker images kavyag3011/calculator-devops
REPOSITORY                    TAG                        IMAGE ID       CREATED         SIZE
kavyag3011/calculator-devops  feature-jenkins-pipeline-4  8b5339d1b876   3 minutes ago   231MB
kavyag3011/calculator-devops  dev                        22f041011d2d   25 hours ago    231MB
kavyag3011/calculator-devops  <none>                     e68686030fb1   31 hours ago    231MB
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 110: Image built by Jenkins as `feature-jenkins-pipeline-4`.

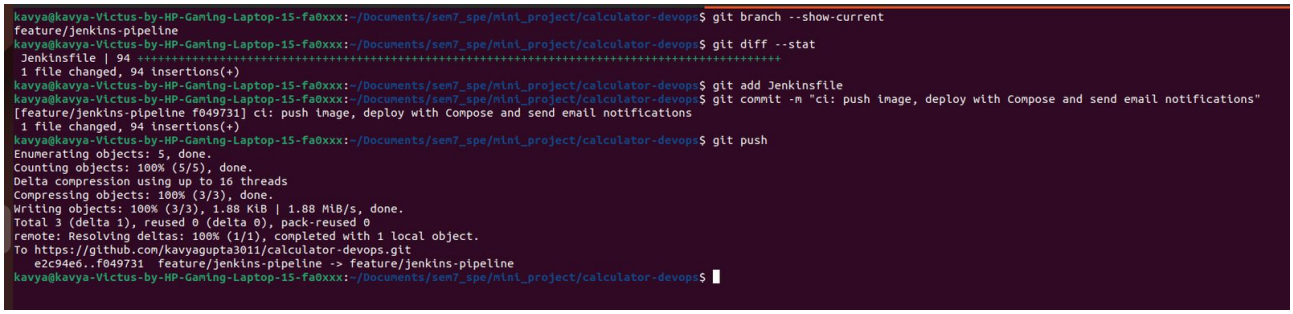
The image tag is the build number on `main`; on other branches the branch name is prefixed, because each branch in a Multibranch job numbers its builds from 1 and tags would otherwise collide.

7.3 Push, deploy and notifications

The last commit added the Push and Deploy stages, both guarded by `when { branch 'main' }`, and the post actions:

- **Push** – `docker.withRegistry('', 'DockerHubCred')` logs in with a temporary configuration for the build and pushes the build-number tag and `latest`.
- **Deploy** – copies `docker-compose.yml` and `nginx.conf` to `/var/lib/jenkins/deploy/calculator` and runs Compose there. The workspace is deleted after each build, and Nginx bind-mounts its configuration, so running Compose from the workspace would leave Nginx pointing at a deleted file. `docker compose up -d --no-build --remove-orphans` uses the image just built (`IMAGE_TAG` is exported by Jenkins), then the stage polls `/actuator/health` through Nginx up to 30 times, confirms the running image is the new tag and calls the API. A failed check prints the container status and logs and fails the build.
- **Post** – `mail` on success and on failure with job, branch, commit, image and links; `cleanWs()` always.

`disableConcurrentBuilds()` prevents two builds from deploying at once, and the fixed Compose project name means each run updates the same stack.


kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git branch --show-current
feature/jenkins-pipeline
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git diff --stat
Jenkinsfile | 94 ++++++

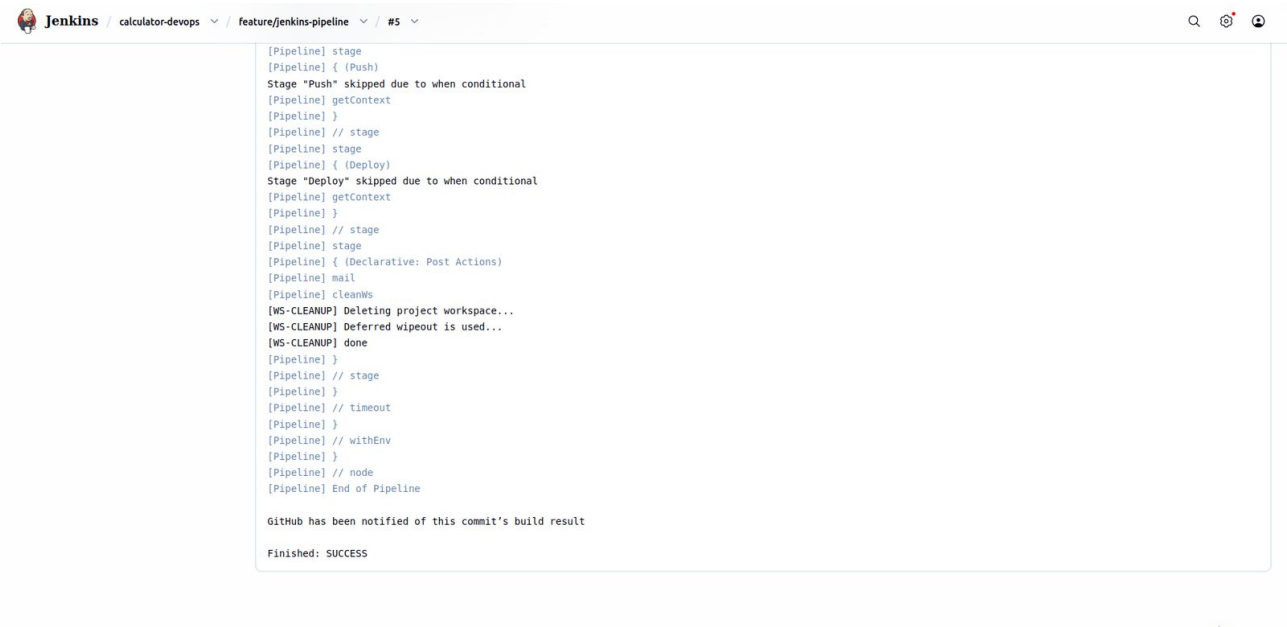


Figure 113: Console: stages skipped, email sent, workspace deleted, GitHub notified.

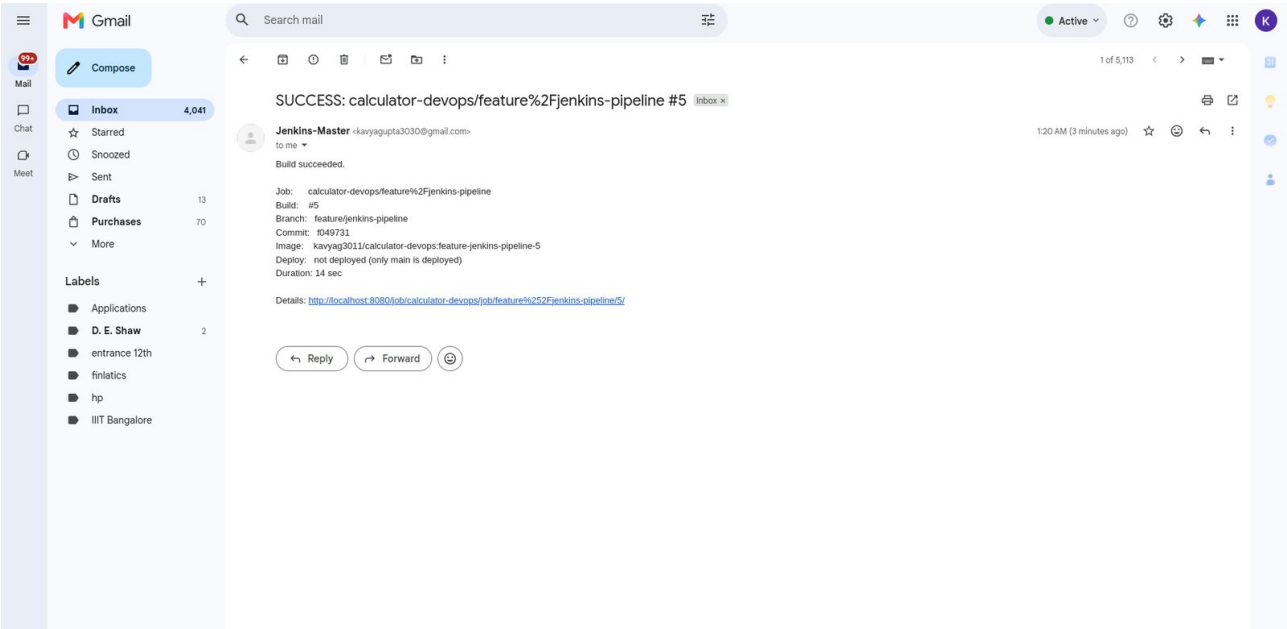


Figure 114: Success email for a branch build (not deployed).

7.4 Pull request #13 and required check

For pull requests Jenkins builds the pull request head and reports `continuous-integration/jenkins/pr-head`. This check was made required in the branch protection rule together with *Require branches to be up to date before merging*, so a pull request cannot be merged until Jenkins has built the latest combination of the branch and `main` successfully.

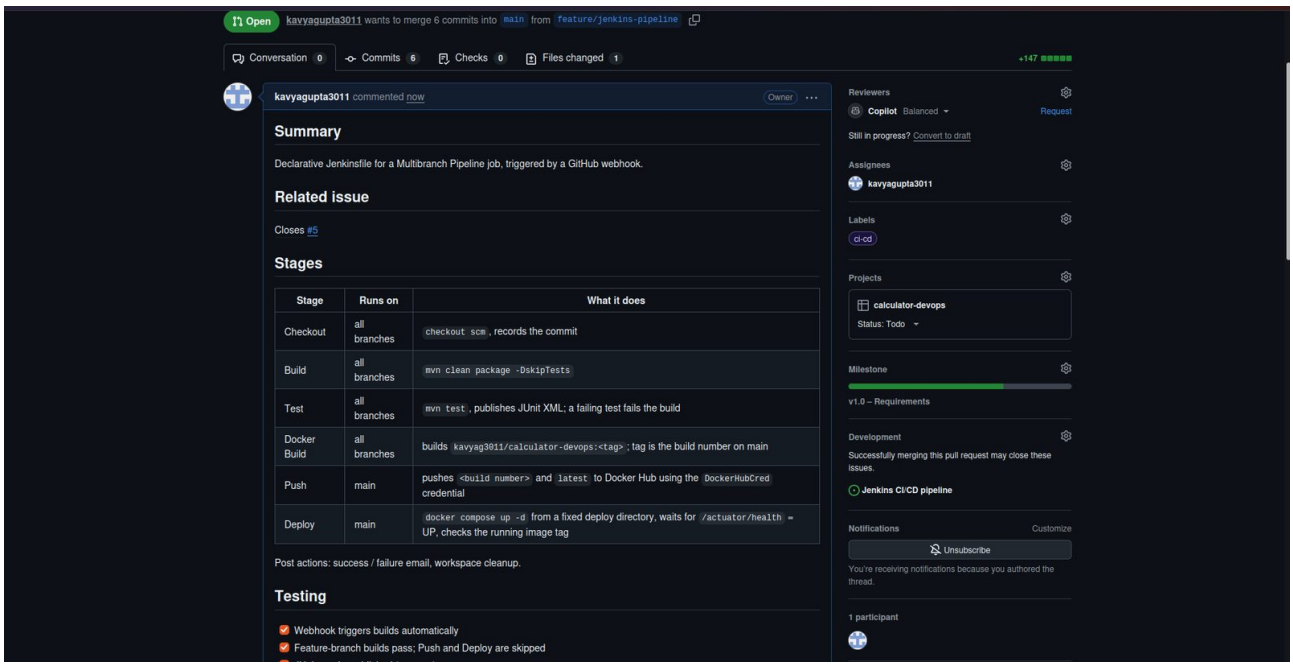


Figure 115: Pull request #13 describing the stages.

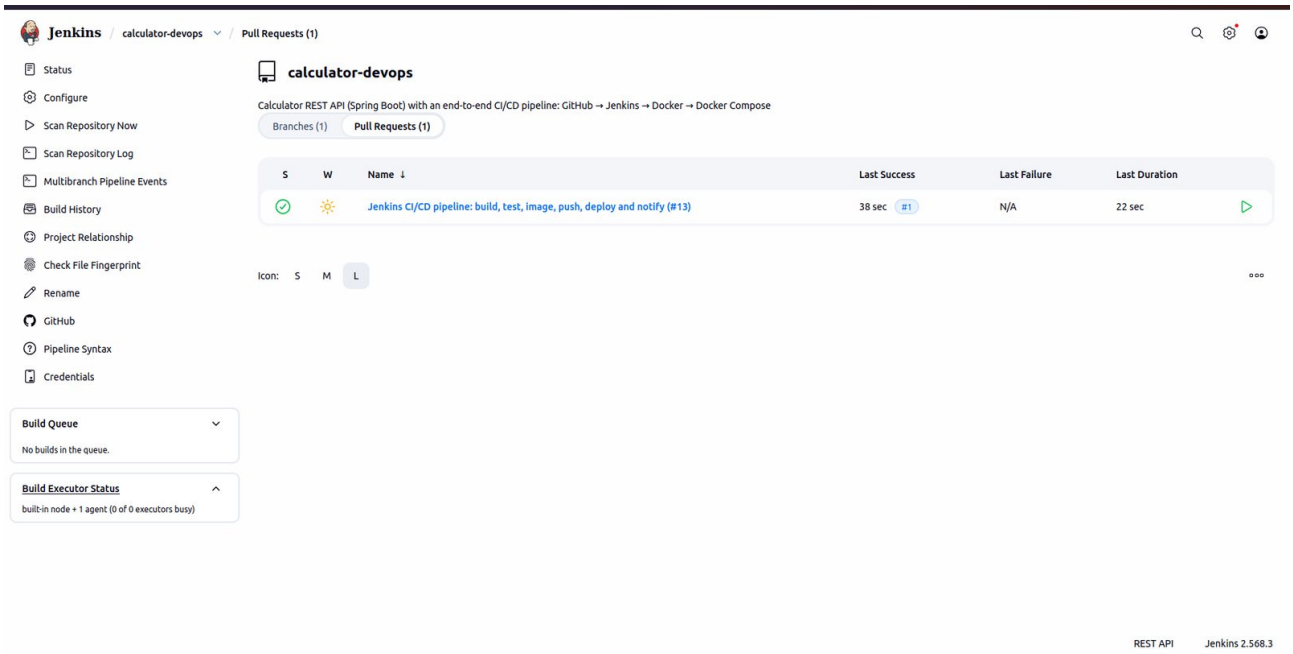


Figure 116: Jenkins building the pull request.

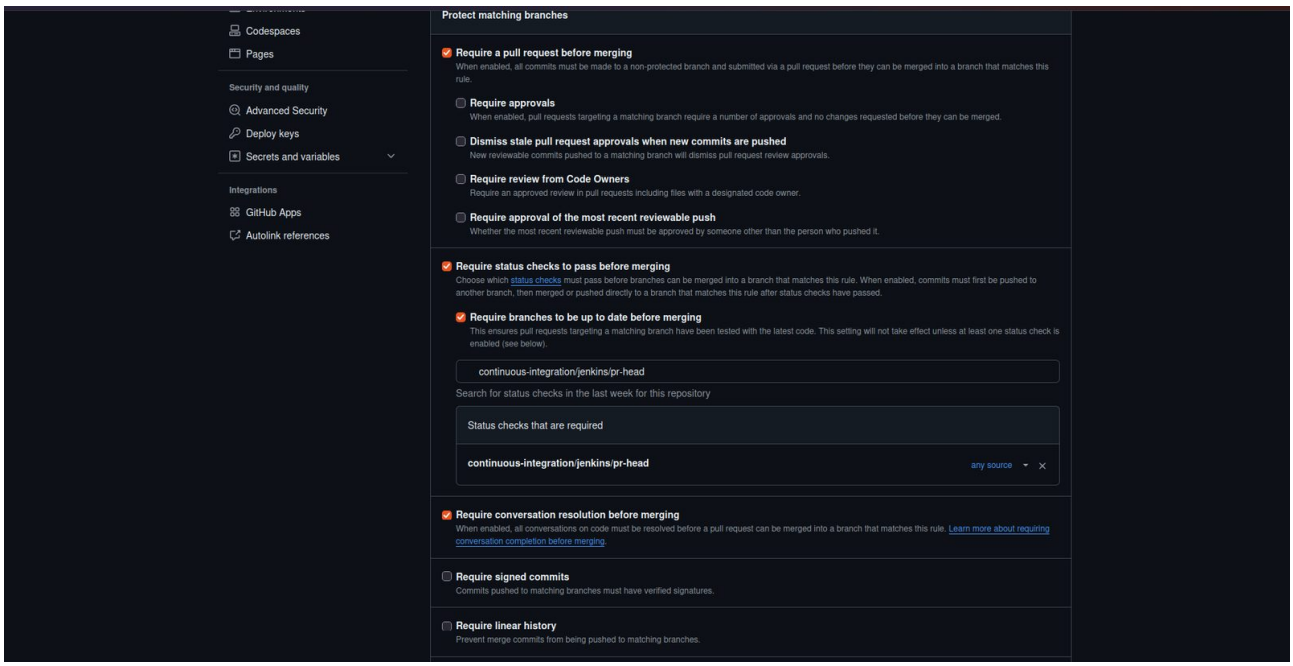


Figure 117: Required status check added to branch protection.

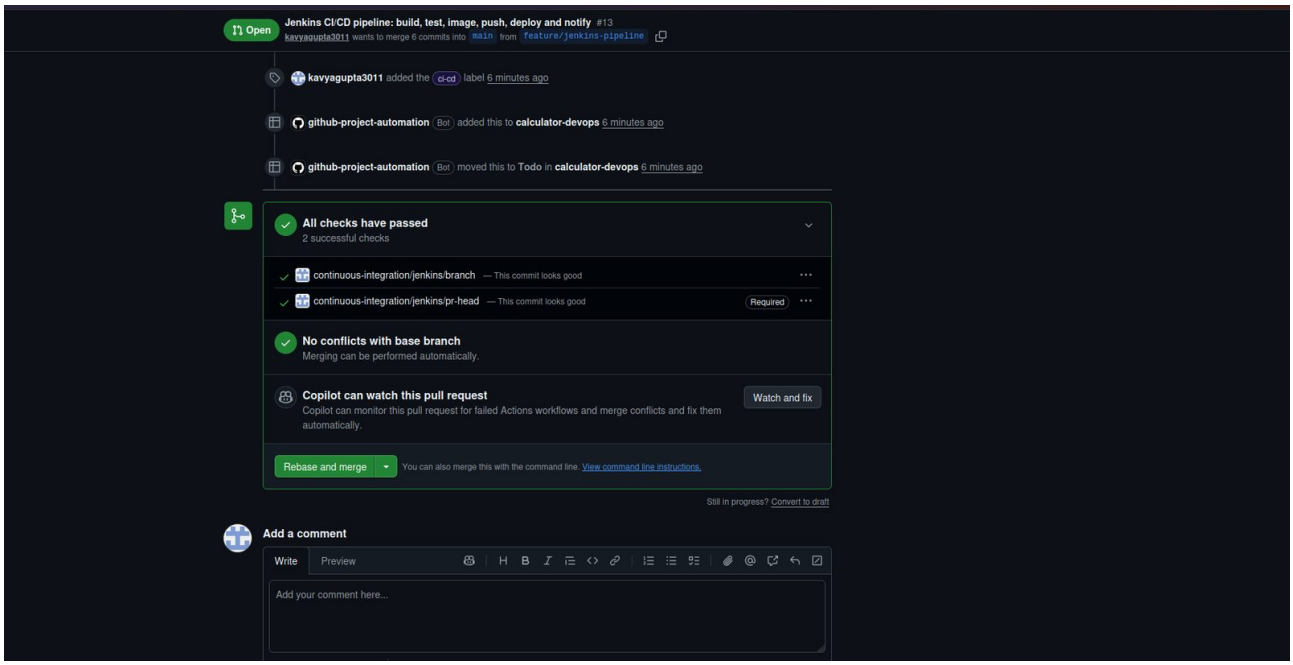


Figure 118: The pull request check marked *Required*.

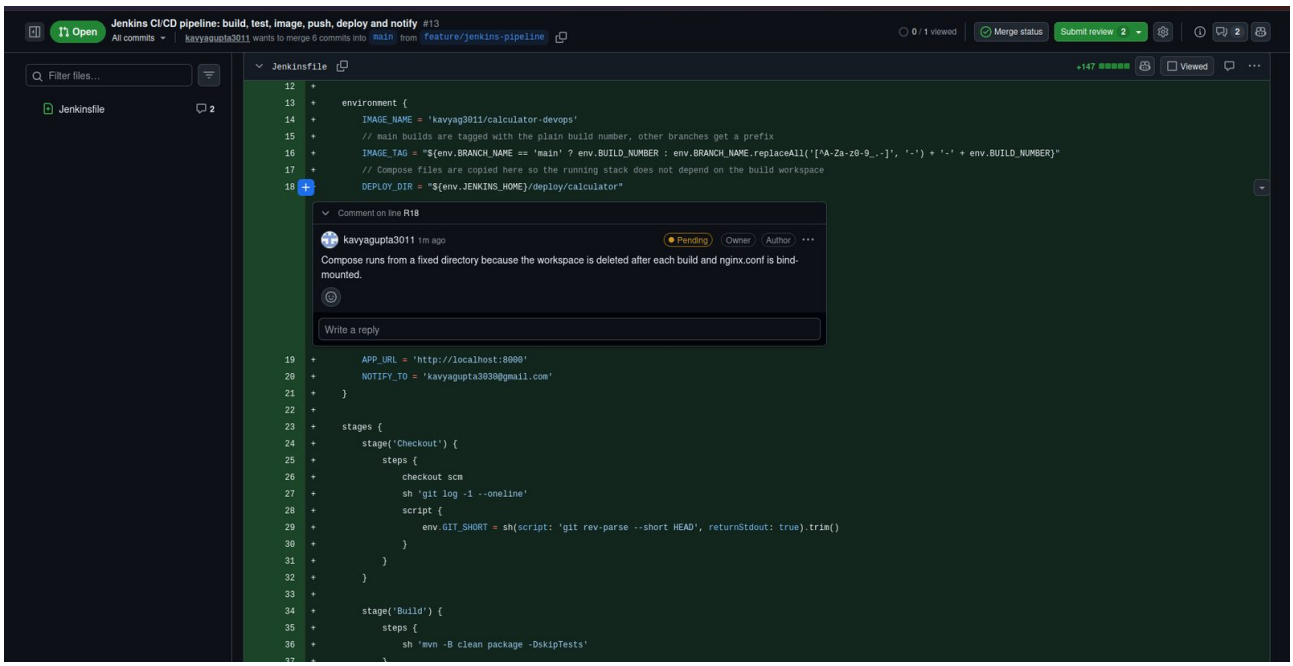


Figure 119: Review comment on the deploy directory.

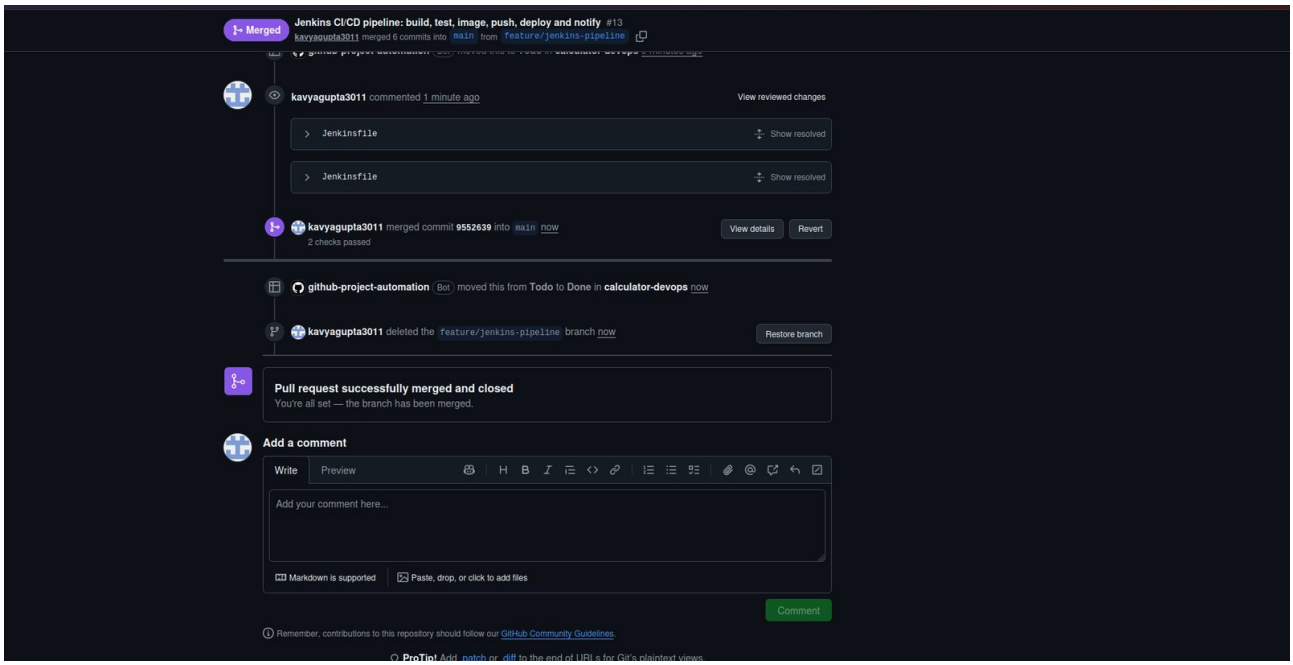


Figure 120: Pull request #13 merged with a merge commit after both checks passed.

7.5 First deployment from main

Merging created the first `Jenkinsfile` on `main`; the webhook started `main` build #1, which ran all six stages.

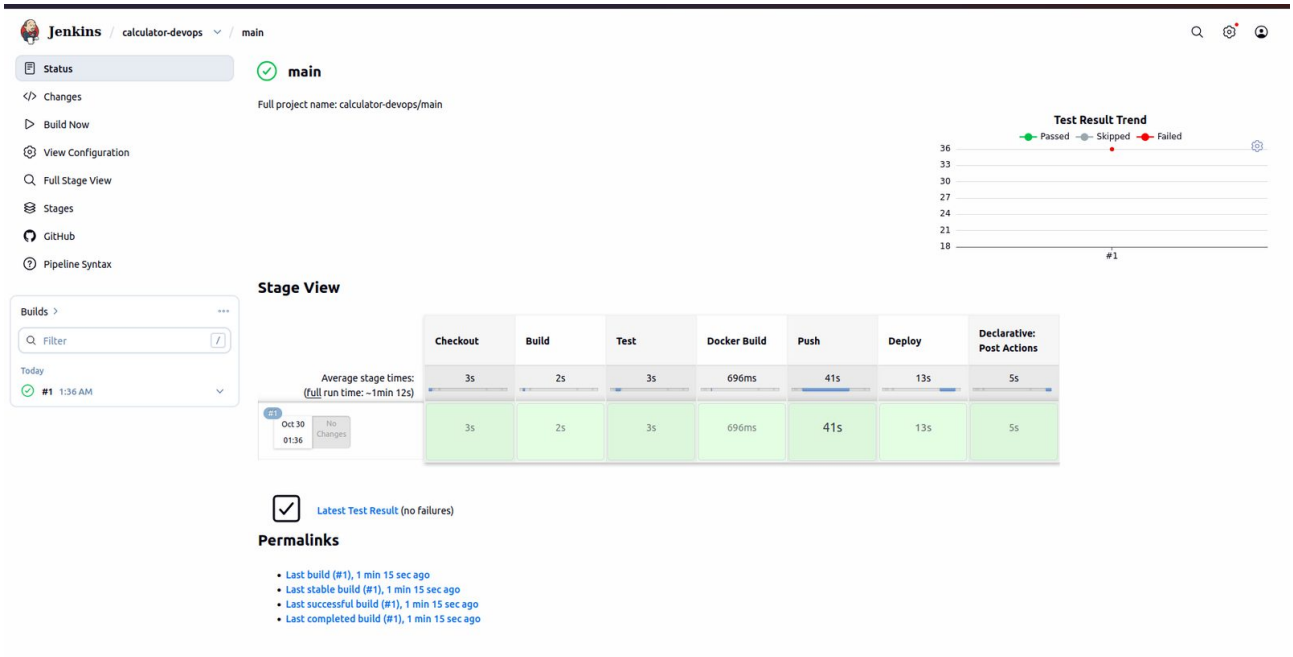


Figure 121: `main` build #1: all stages green.

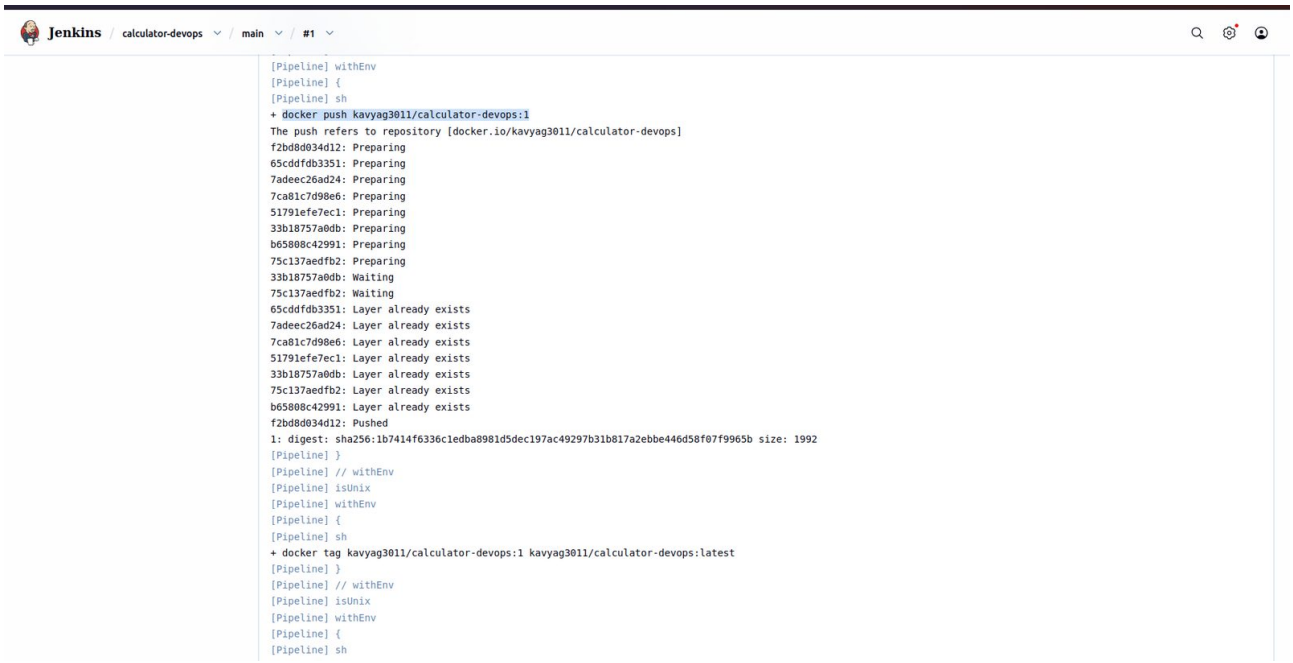


Figure 122: Push: only the application layer is uploaded; tag `1` and `latest`.

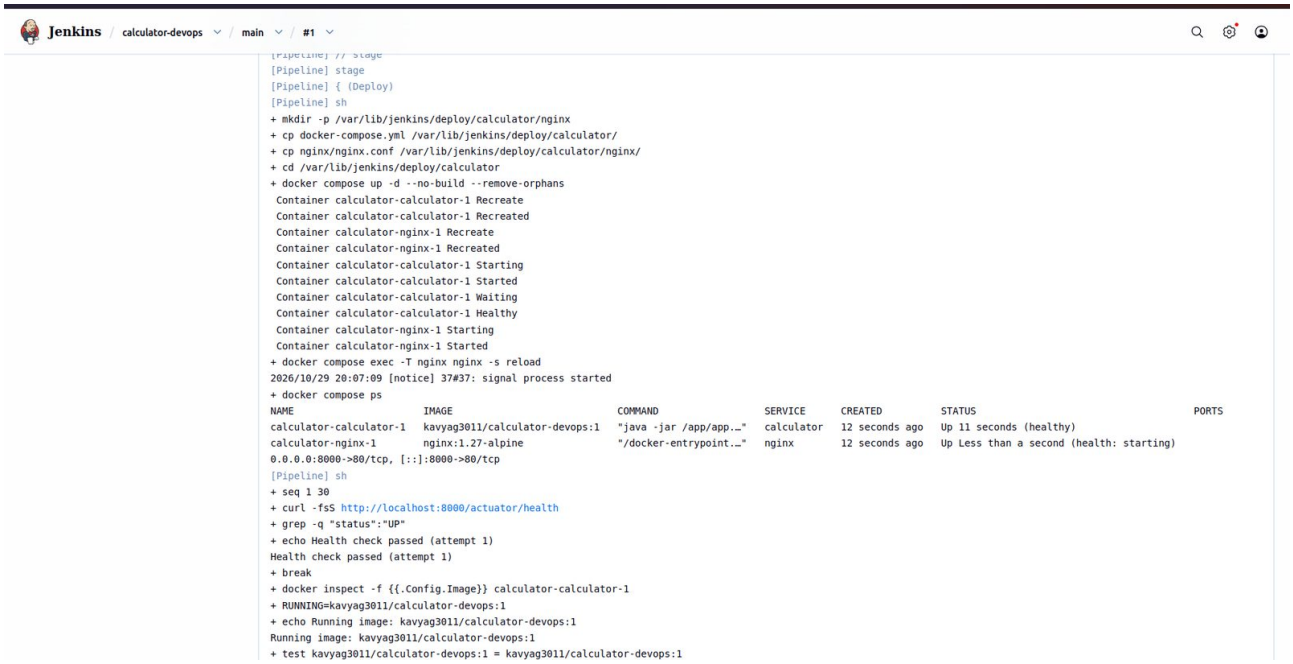


Figure 123: Deploy: containers recreated, API healthy before Nginx starts, health check passed, running image `calculator-devops:1`.

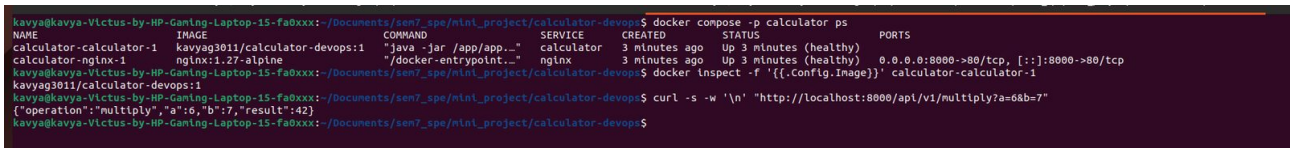


Figure 124: Deployed stack running image tag `1`.

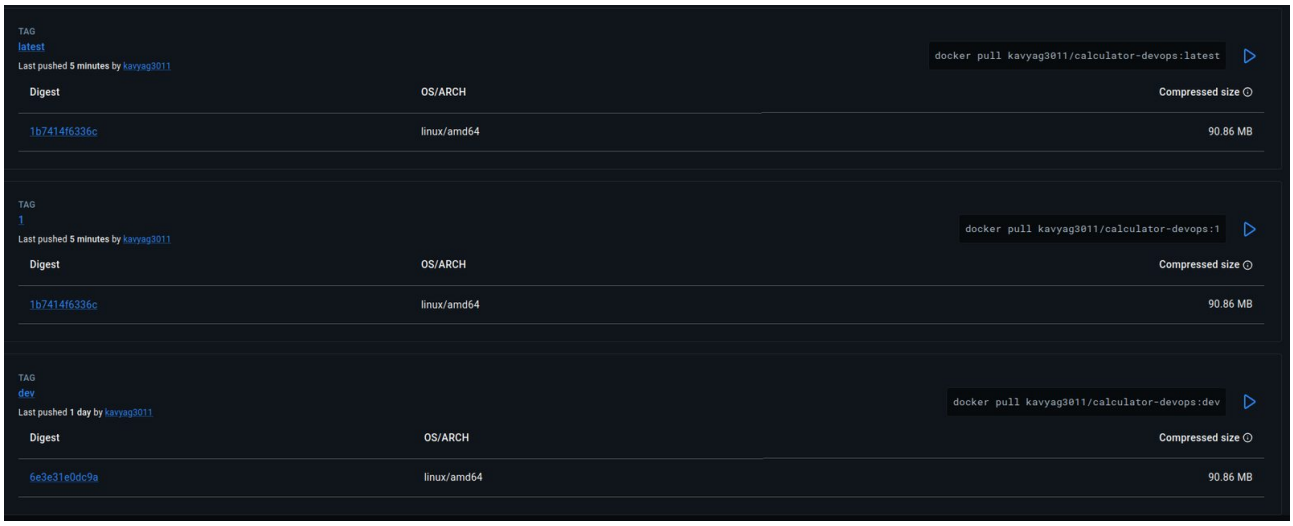


Figure 125: Docker Hub: `1` and `latest` share the same digest.

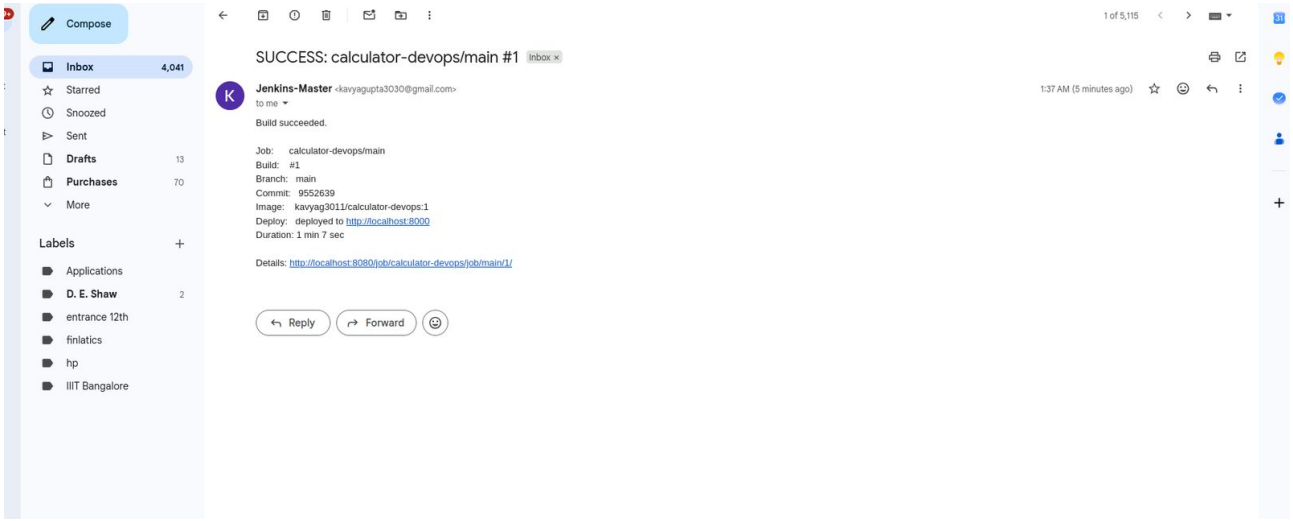


Figure 126: Success email with the deployed URL.

7.6 Re-running the pipeline

Merging the README pull request (#16) triggered `main` build #2, and *Build Now* then started build #3 from the same commit with nothing changed. Both completed every stage. In build #3, Compose recreated only the `calculator` container for the new image tag while `nginx` kept running, the health check passed on the first attempt and the API kept answering. Docker Hub holds one tag per `main` build plus `latest`.

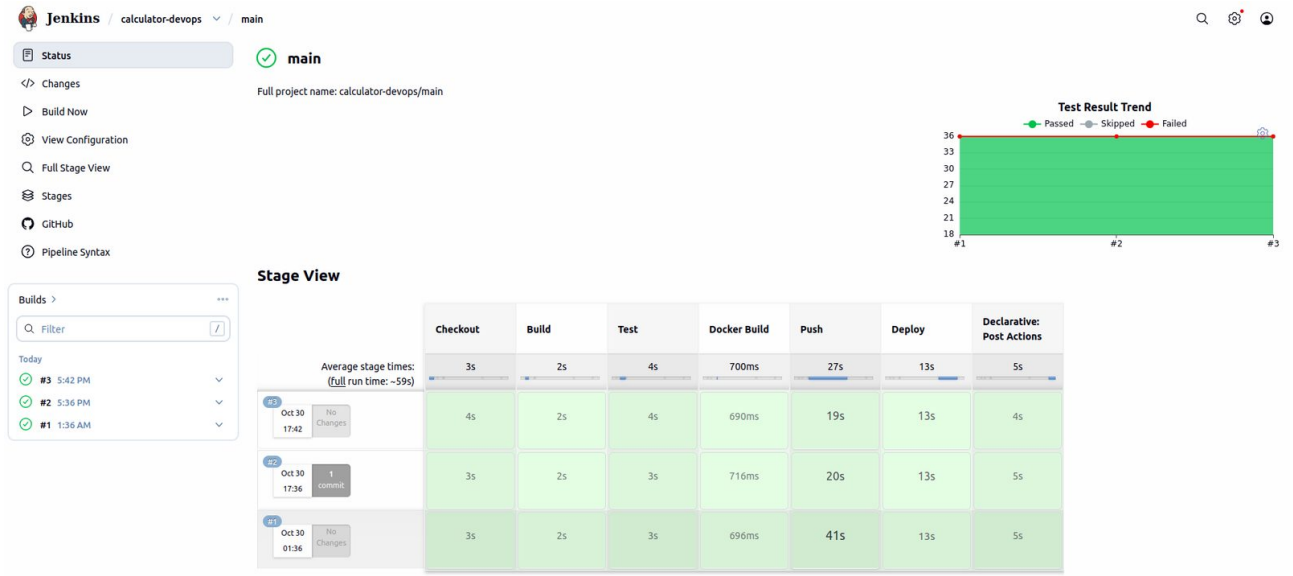


Figure 127: `main` builds #1, #2 and #3 all green.

```
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] sh
+ mkdir -p /var/lib/jenkins/deploy/calculator/nginx
+ cp docker-compose.yml /var/lib/jenkins/deploy/calculator/
+ cp nginx/nginx.conf /var/lib/jenkins/deploy/calculator/nginx/
+ cd /var/lib/jenkins/deploy/calculator
+ docker compose up -d --no-build --remove-orphans
 Container calculator-nginx-1 Running
 Container calculator-calculator-1 Recreate
 Container calculator-calculator-1 Recreated
 Container calculator-calculator-1 Starting
 Container calculator-calculator-1 Started
 Container calculator-calculator-1 Waiting
 Container calculator-calculator-1 Healthy
+ docker compose exec -T nginx nginx -s reload
2026/10/30 12:13:20 [notice] 2050#2050: signal process started
+ docker compose ps
NAME                    IMAGE                          COMMAND                  SERVICE      CREATED          STATUS                    PORTS
calculator-calculator-1   kavyag3011/calculator-devops:3   "java -jar /app/app.…"   calculator   11 seconds ago   Up 10 seconds (healthy)
calculator-nginx-1        nginx:1.27-alpine                "/docker-entrypoint.…"   nginx        16 hours ago     Up 48 minutes (healthy)   0.0.0.0:8000->80/tcp,
[::]:8000->80/tcp
[Pipeline] sh
+ seq 1 30
+ curl -fsS http://localhost:8000/actuator/health
+ grep -q "status":"UP"
+ echo Health check passed (attempt 1)
Health check passed (attempt 1)
+ break
+ docker inspect -f {{.Config.Image}} calculator-calculator-1
+ RUNNING=kavyag3011/calculator-devops:3
+ echo Running image: kavyag3011/calculator-devops:3
```

Figure 128: Build #3 deploy: `nginx` left running, only `calculator` recreated, health check passed, running image `calculator-devops:3`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker compose -p calculator ps
NAME                    IMAGE                          COMMAND                  SERVICE      CREATED         STATUS                    PORTS
calculator-calculator-1   kavyag3011/calculator-devops:3   "java -jar /app/app.…"   calculator   2 minutes ago   Up 2 minutes (healthy)
calculator-nginx-1        nginx:1.27-alpine                "/docker-entrypoint.…"   nginx        16 hours ago    Up 50 minutes (healthy)   0.0.0.0:8000->80/tcp, [::]:8000->80/tcp
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker inspect -f '{{.Config.Image}}' calculator-calculator-1
kavyag3011/calculator-devops:3
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' "http://localhost:8000/api/v1/add?a=2&b=3"
{"operation":"add","a":2,"b":3,"result":5}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ curl -s -w '\n%{http_code}\n' http://localhost:8000/actuator/health
{"status":"UP","groups":["liveness","readiness"]}
200
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 129: After the re-run: both containers healthy on image tag 3; API and health endpoint respond.

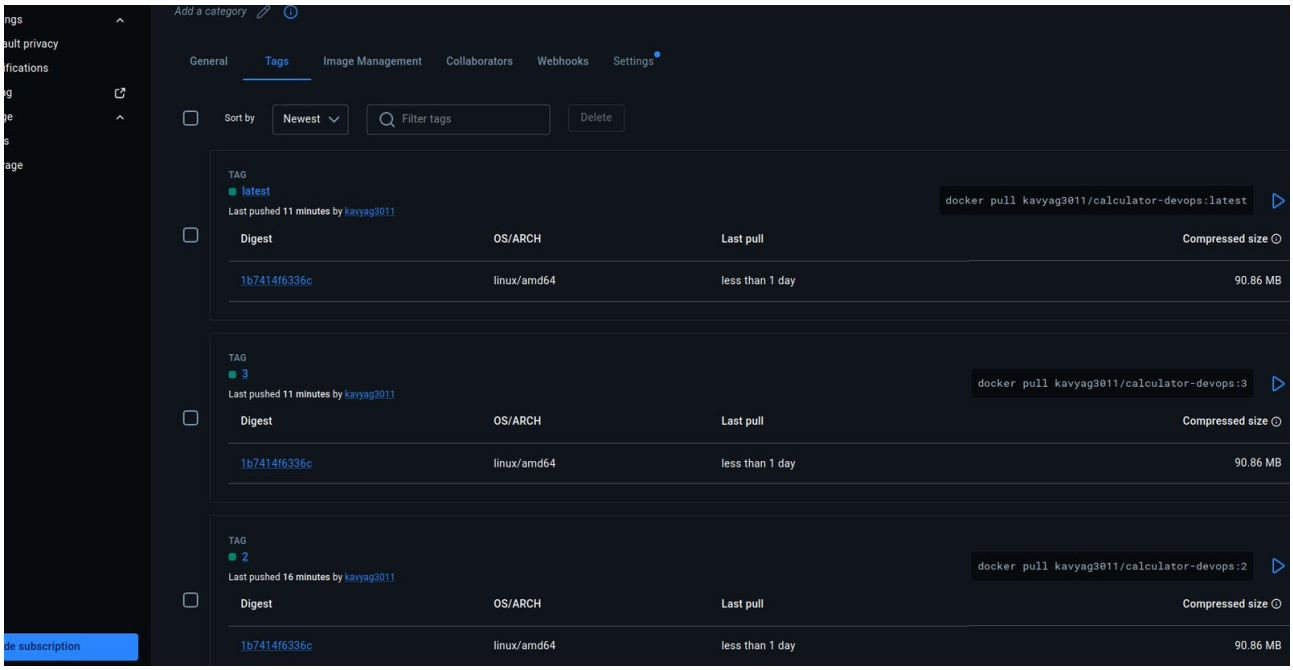


Figure 130: Docker Hub tags for each `main` build and `latest`.

8 Documentation

The README was completed on `docs/readme-report` and merged through pull request #16, which closed issue #6. It covers the architecture (as a Mermaid diagram that GitHub renders), build and test commands, the test classes, running with Java, Docker and Compose, the pipeline stages and Jenkins setup, the API with success and error examples, the Git workflow and the project structure.

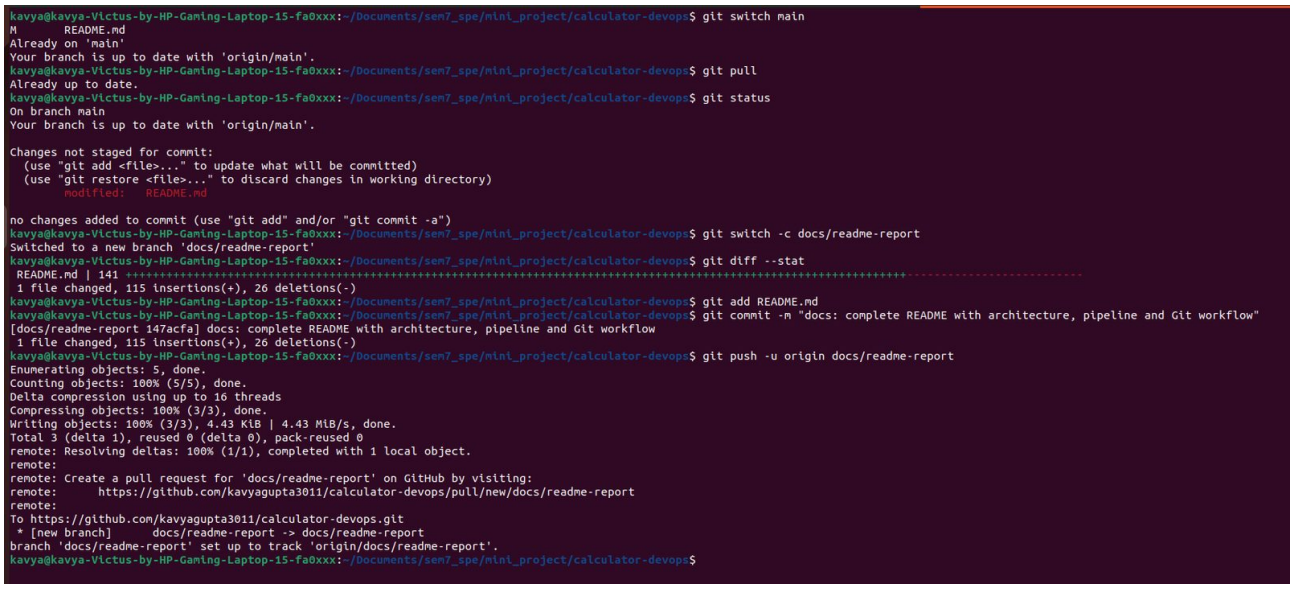


Figure 131: README committed on its own branch.

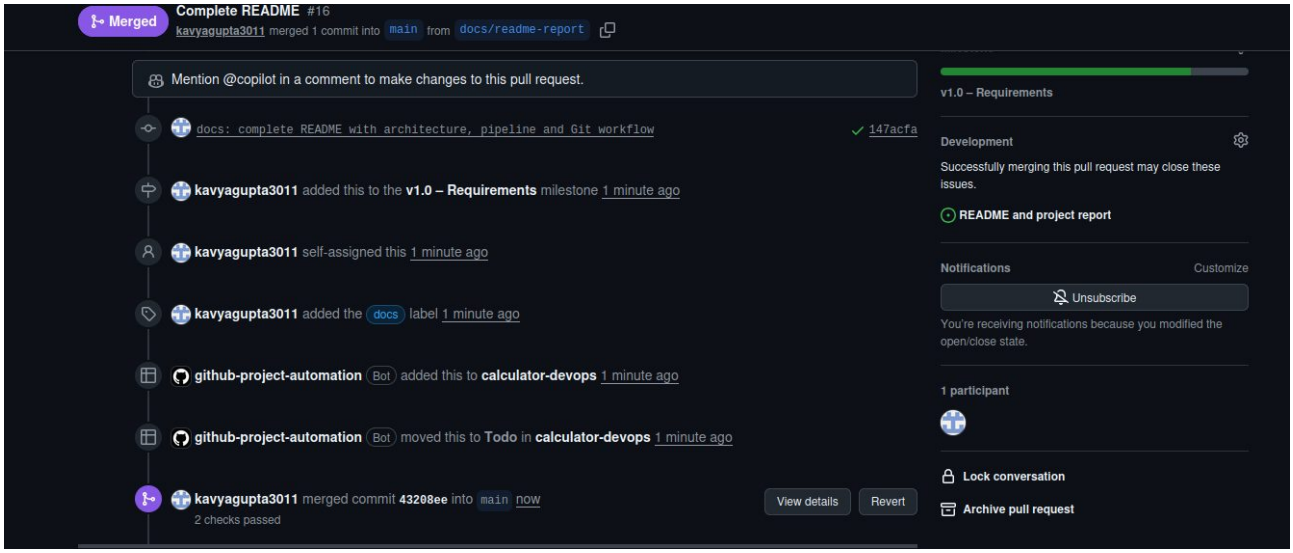


Figure 132: Pull request #16 merged with both checks passing, closing issue #6.

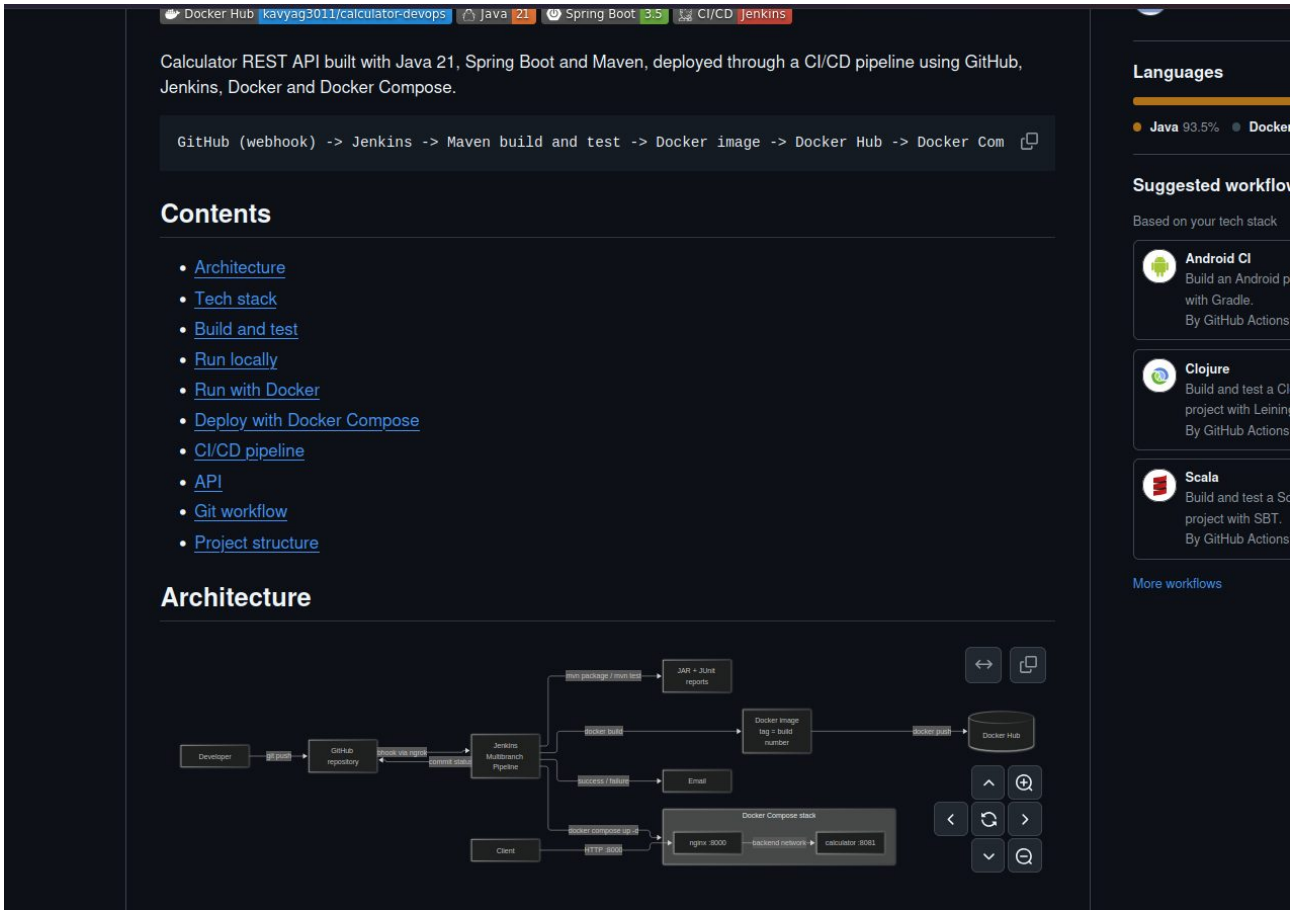
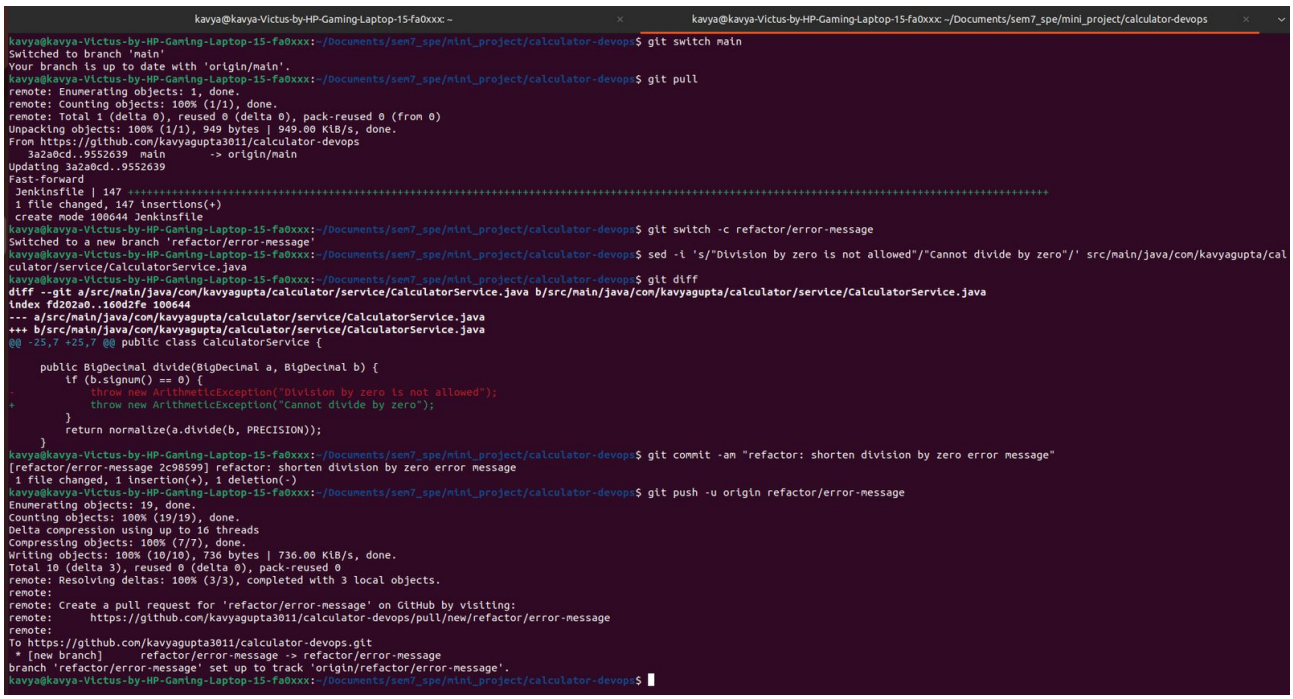


Figure 133: README on GitHub with badges, contents and the architecture diagram.

9 Failure handling and revert

To confirm that the pipeline protects `main`, a change that broke the API's error contract was pushed on `refactor/error-message`: the division-by-zero message was shortened. The Test stage failed (2 of 36 tests), no image was built or deployed, a failure email was sent and the pull request could not be merged. A bug was filed with the bug template, the commit was undone with `git revert` – which adds a new commit rather than rewriting pushed history, so no force push was needed – and the next build passed. The pull request was then closed without merging because it no longer changed anything.


kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~/Documents/sem7_spe/mini_project/calculator-devops
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git pull
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 949 bytes | 949.00 KiB/s, done.
From https://github.com/kavyagupta3011/calculator-devops
3a2a0cd..9552639 main -> origin/main
Updating 3a2a0cd..9552639
Fast-forward
Jenkinsfile | 147 ++++++

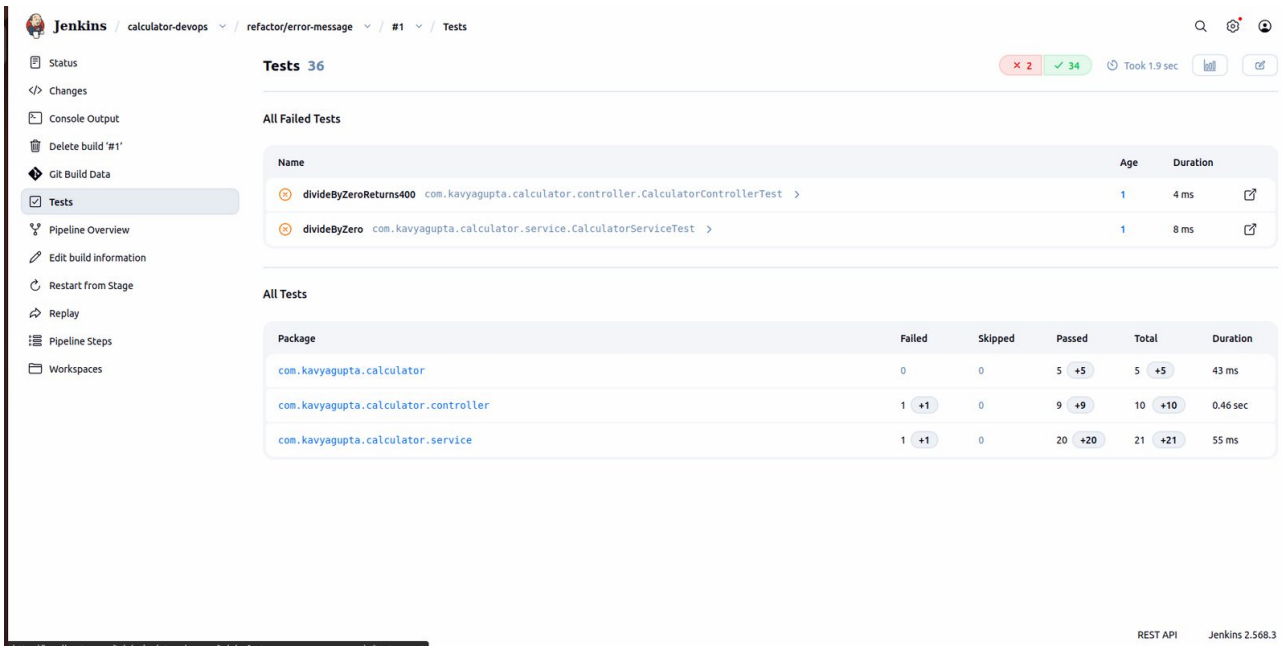


Figure 136: The two failing tests.



Figure 137: `Tests run: 36, Failures: 2`, BUILD FAILURE.

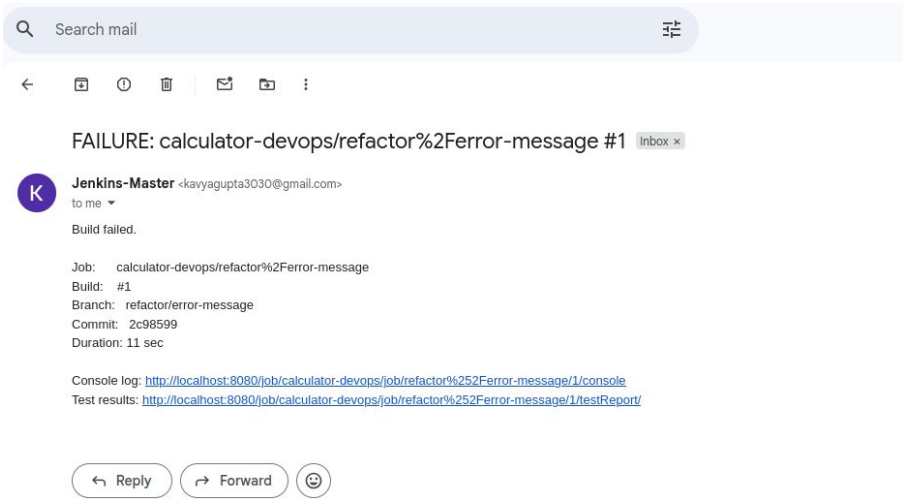


Figure 138: Failure email with links to the console and test results.

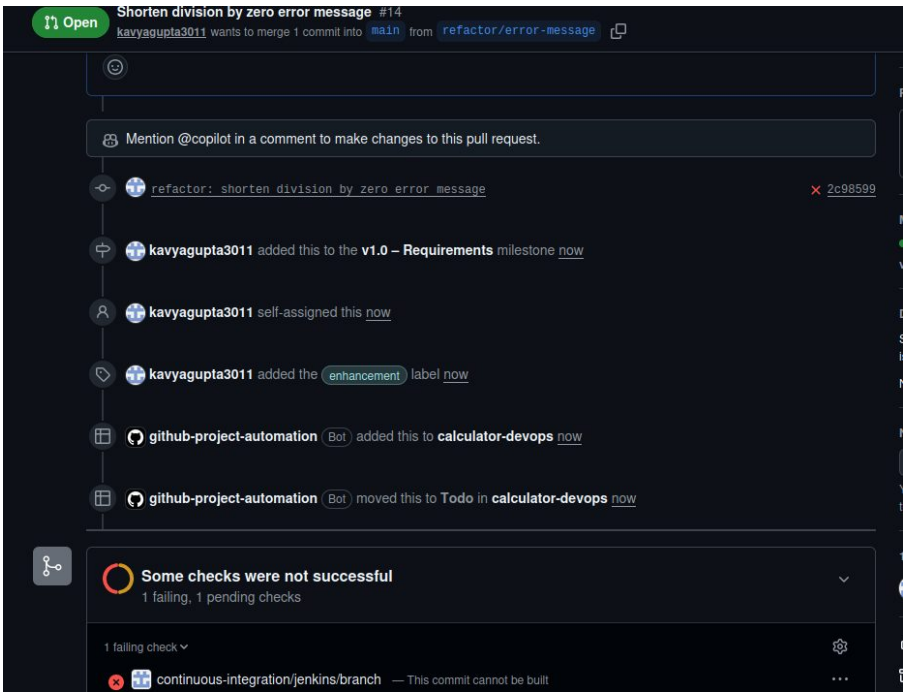


Figure 139: Pull request blocked by the failing check.

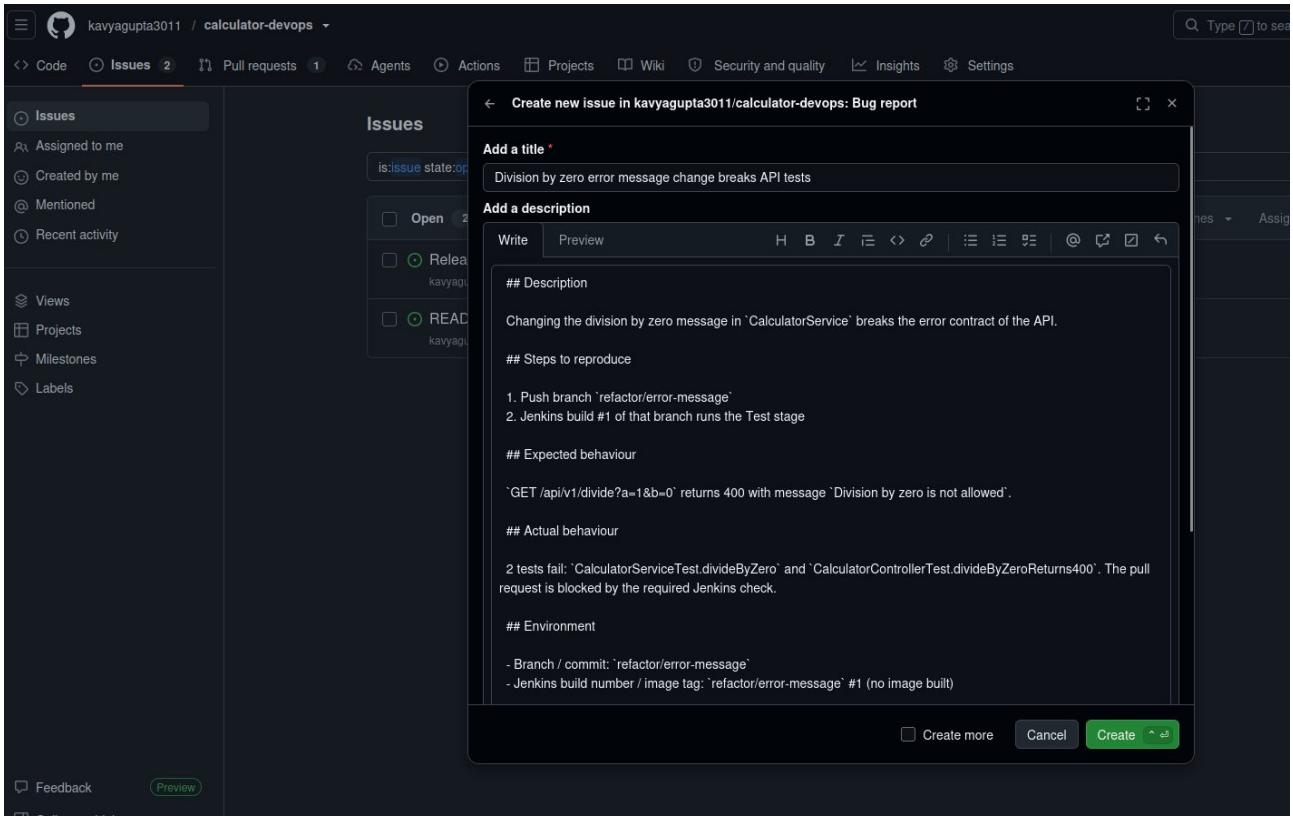


Figure 140: Bug report filed from the template.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline -3
2c98599 (HEAD -> refactor/error-message, origin/refactor/error-message) refactor: shorten division by zero error message
9552639 (origin/main, main) Merge pull request #13 from kavyagupta3011/feature/jenkins-pipeline
f049731 (origin/feature/jenkins-pipeline, feature/jenkins-pipeline) ci: push image, deploy with Compose and send email notifications
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git revert HEAD --no-edit
[refactor/error-message 956e493] Revert "refactor: shorten division by zero error message"
 Date: Fri Oct 30 01:51:50 2026 +0530
 1 file changed, 1 insertion(+), 1 deletion(-)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline -4
956e493 (HEAD -> refactor/error-message) Revert "refactor: shorten division by zero error message"
2c98599 (origin/refactor/error-message) refactor: shorten division by zero error message
9552639 (origin/main, main) Merge pull request #13 from kavyagupta3011/feature/jenkins-pipeline
f049731 (origin/feature/jenkins-pipeline, feature/jenkins-pipeline) ci: push image, deploy with Compose and send email notifications
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git show --stat HEAD
commit 956e493b15ded6b849ff2fe7236904e368044660 (HEAD -> refactor/error-message)
Author: kavyagupta3011 <Kavya.Gupta@iiitb.ac.in>
Date:   Fri Oct 30 01:51:50 2026 +0530

    Revert "refactor: shorten division by zero error message"

    This reverts commit 2c985998a17178bc54ab4bf4b1138cb7d8a1bc6e.

 src/main/java/com/kavyagupta/calculator/service/CalculatorService.java | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git diff main --stat
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git push
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (10/10), 775 bytes | 775.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/kavyagupta3011/calculator-devops.git
   2c98599..956e493  refactor/error-message -> refactor/error-message
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 141: `git revert HEAD`: new commit, history kept, branch identical to `main`.



Figure 142: Pull request build #1 red, #2 green after the revert.

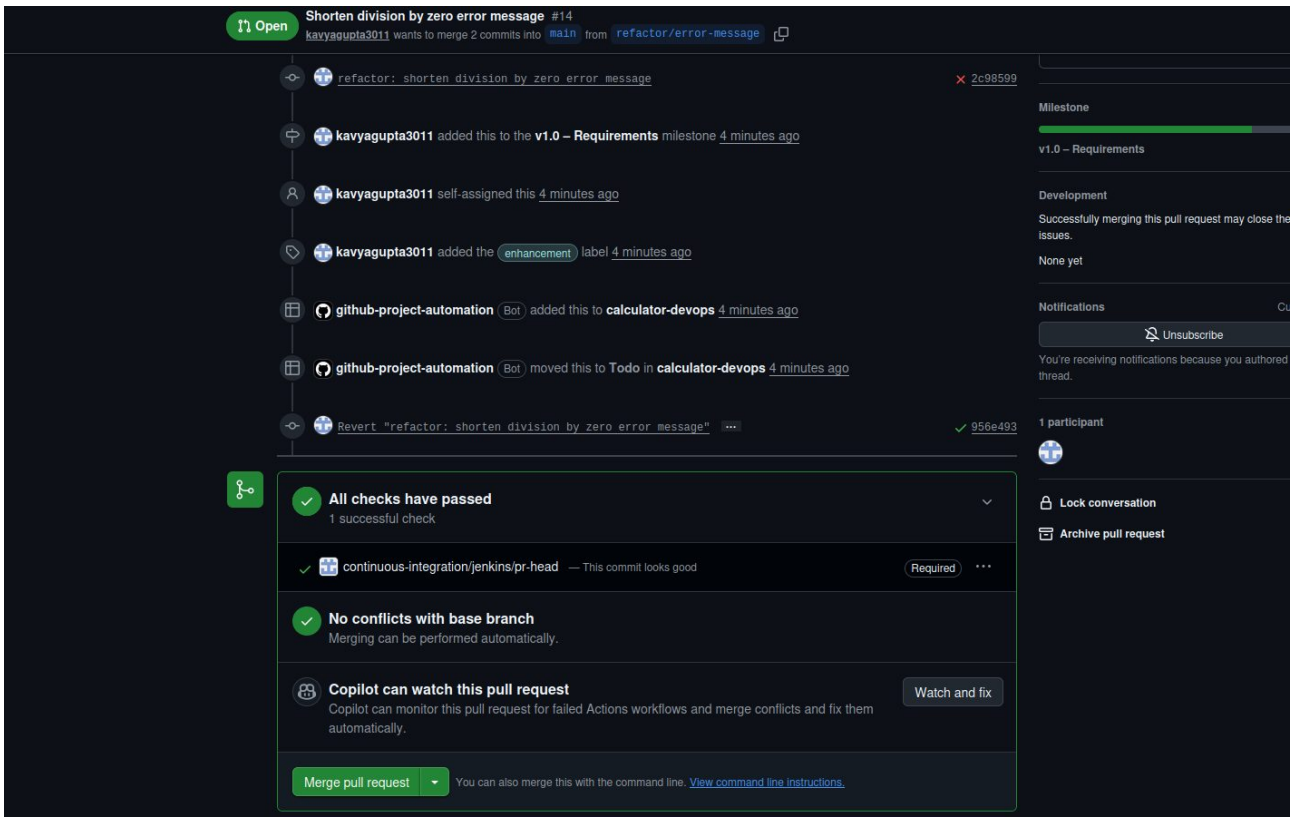


Figure 143: Required check passing again.

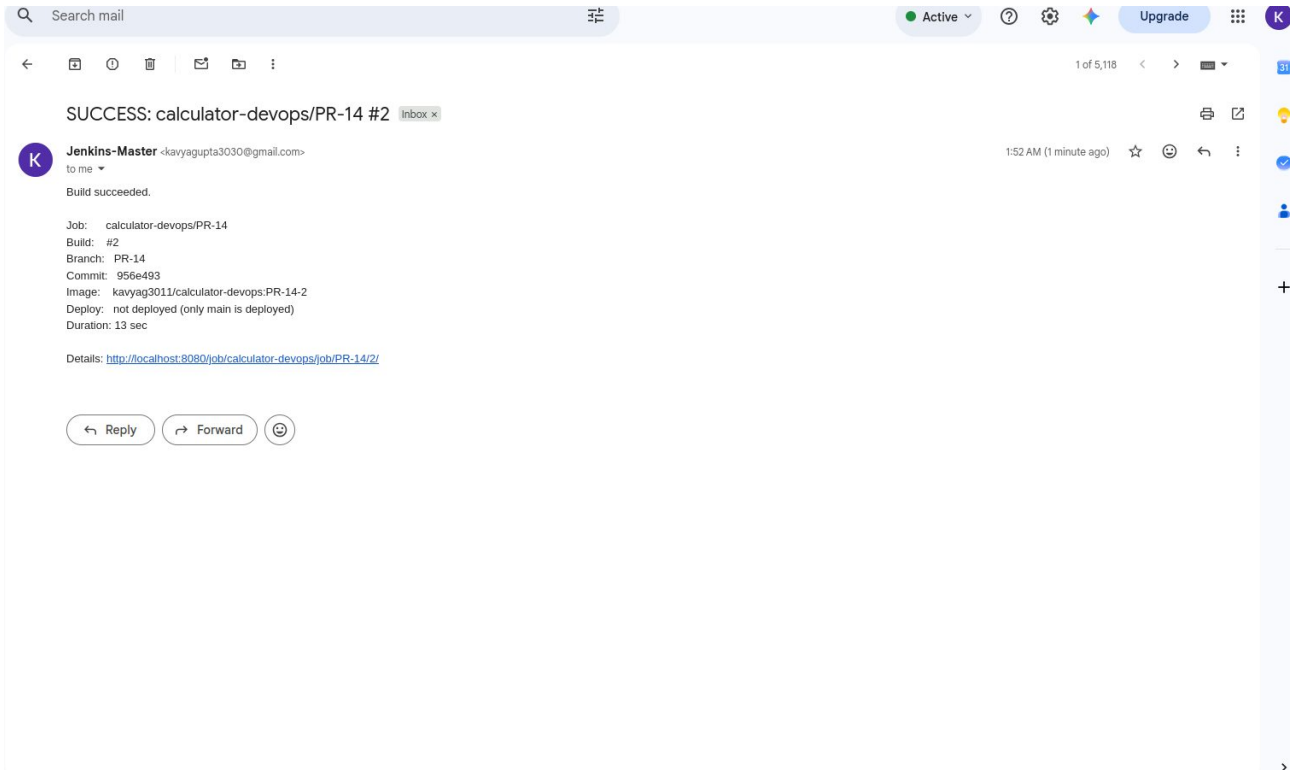


Figure 144: Success email after the revert.

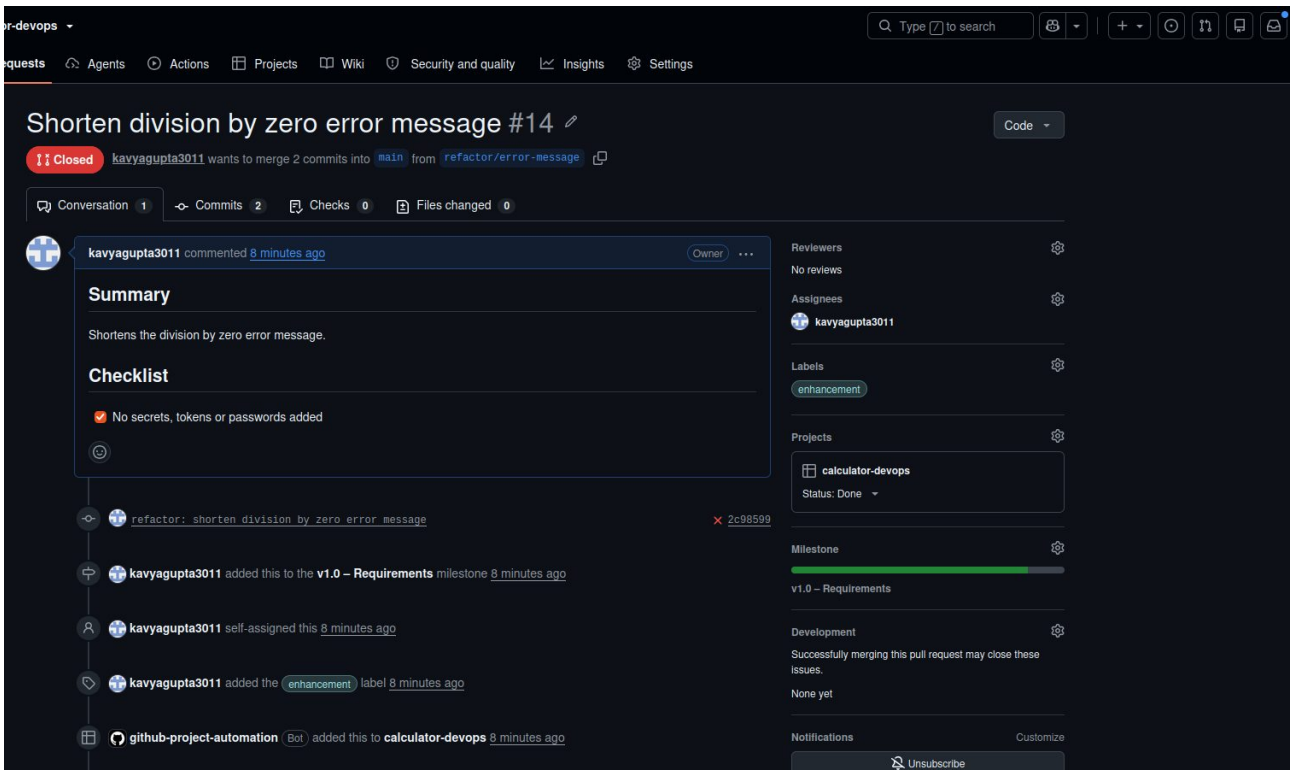


Figure 145: Pull request #14 closed without merging (no files changed).

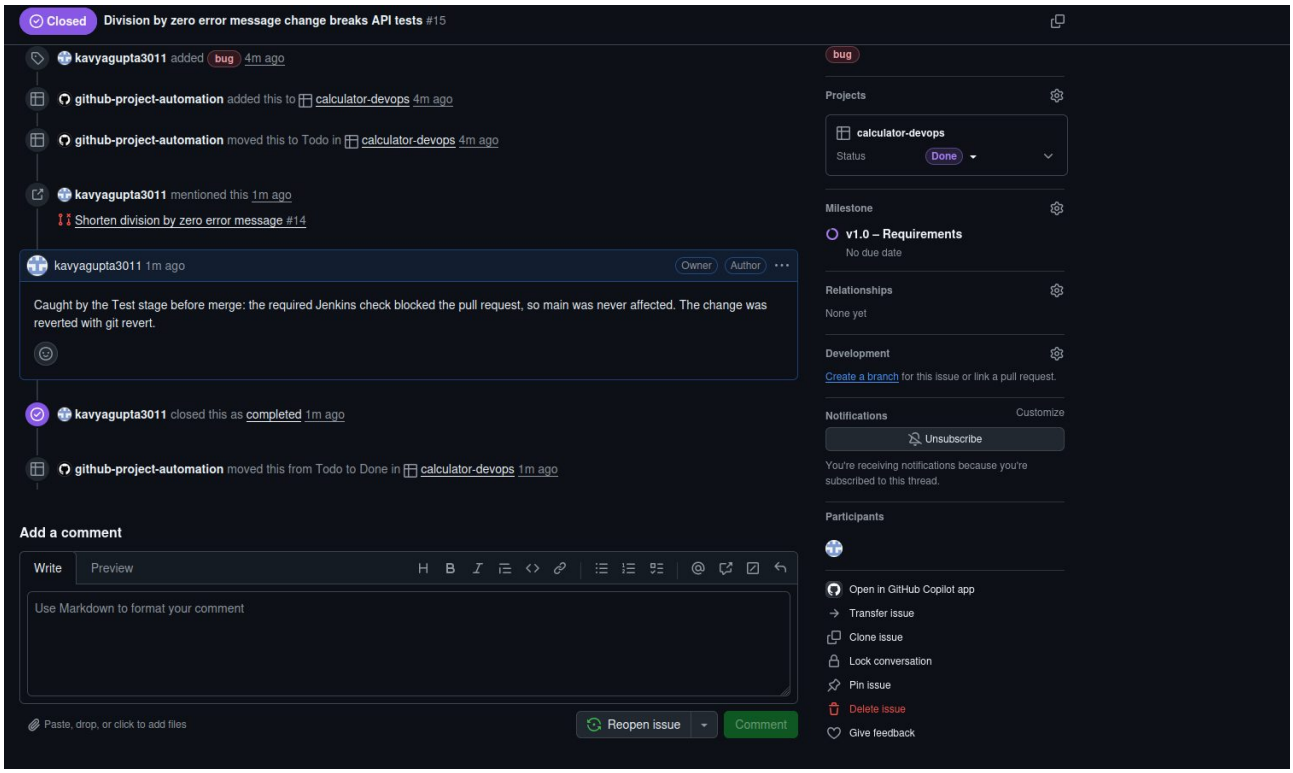


Figure 146: Bug #15 closed and moved to *Done*.

10 Release v1.0.0

With every requirement met and all sixteen issues of milestone `v1.0 -- Requirements` closed, commit `43208ee` on `main` was tagged `v1.0.0` and published as a GitHub release with notes on the application, container, deployment and pipeline. The image built for it (`main` build #3) was pulled, tagged `v1.0.0` and pushed to Docker Hub; every layer already existed, so only the new tag was added and the digest stayed the same.

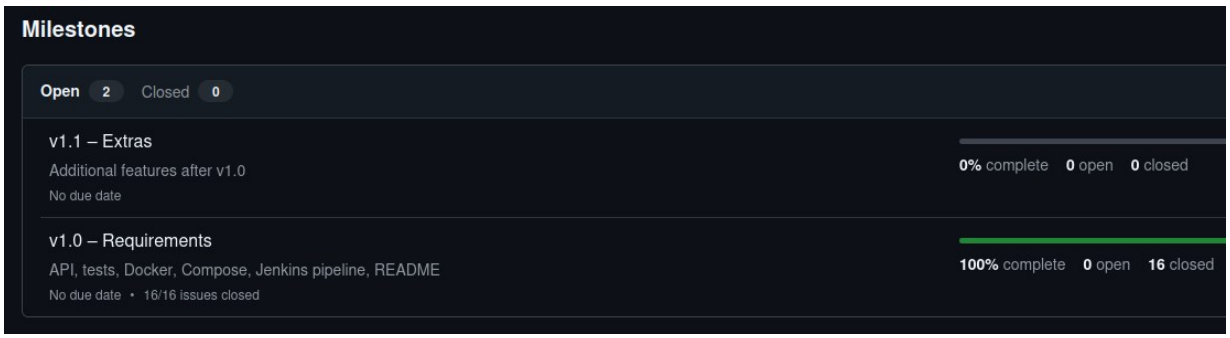


Figure 147: Milestone `v1.0 -- Requirements` complete: 16 of 16 issues closed.

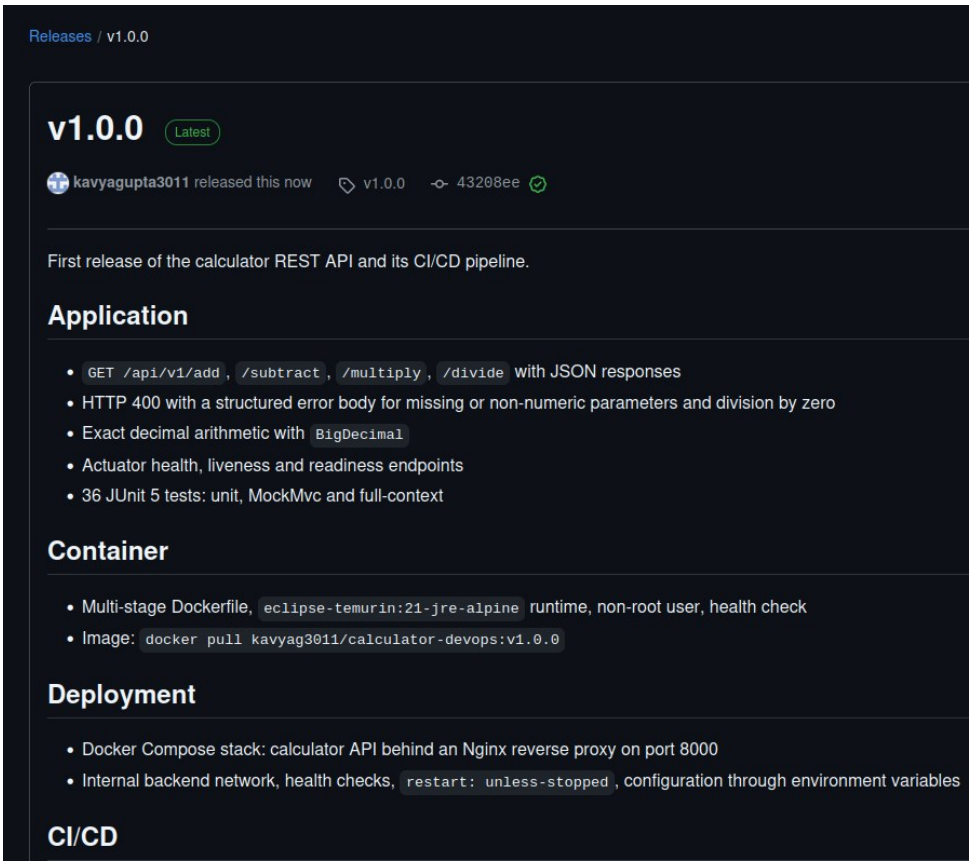


Figure 148: GitHub release `v1.0.0` on commit `43208ee`.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker pull kavyag3011/calculator-devops:3
3: Pulling from kavyag3011/calculator-devops
Digest: sha256:1b7414f6336c1edba8981d5dec197ac49297b31b817a2ebbe446d58f07f9965b
Status: Image is up to date for kavyag3011/calculator-devops:3
docker.io/kavyag3011/calculator-devops:3
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker tag kavyag3011/calculator-devops:3 kavyag3011/calculator-devops:v1.0.0
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker loginn -u kavyag3011
Password:
WARNING! Your password will be stored unencrypted in /home/kavya/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker push kavyag3011/calculator-devops:v1.0.0
The push refers to repository [docker.io/kavyag3011/calculator-devops]
f2bd8d034d12: Layer already exists
65cddfdb3351: Layer already exists
7adeec26ad24: Layer already exists
7ca81c7d98e6: Layer already exists
51791efe7ec1: Layer already exists
33b18757a0db: Layer already exists
b65808c42991: Layer already exists
75c137aedfb2: Layer already exists
v1.0.0: digest: sha256:1b7414f6336c1edba8981d5dec197ac49297b31b817a2ebbe446d58f07f9965b size: 1992
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ docker logout
Removing login credentials for https://index.docker.io/v1/
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 149: Image from build #3 re-tagged `v1.0.0` and pushed; all layers already present.

```
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$ git log --oneline --graph --decorate --all | head -60
* 43208ee (HEAD -> main, tag: v1.0.0, origin/main) docs: complete README with architecture, pipeline and Git workflow (#16)
*   9552639 Merge pull request #13 from kavyagupta3011/feature/jenkins-pipeline
|\
| * f049731 ci: push image, deploy with Compose and send email notifications
| * e2c94e6 ci: build Docker image tagged with build number
| * e9017d2 ci: run tests and publish JUnit results
| * 3f0fdfc ci: trigger build via webhook
| * 7c8eaca ci: trigger build via webhook
| * be3b2ad ci: add Jenkinsfile with checkout and build stages
|/
* 3a2a0cd docs: document Docker Compose deployment in README
* 9203ea9 chore: add .env.example documenting Compose variables
* 9a0e568 feat: add Docker Compose stack with health checks, restart policy and networks
* e2ec88c feat: add Nginx reverse proxy configuration
* 525a60d docs: mention webhook trigger in pipeline overview (#11)
* a7b0fac build: add multi-stage Dockerfile with non-root user and health check (#NN)
* 89bcb67 docs: add Docker usage and Docker Hub link to README (#9)
*   96c1308 Merge pull request #8 from kavyagupta3011/feature/calculator-api
|\
| * 1d38198 docs: document build, run and API usage in README
| * bd2a881 test: verify actuator health endpoints in full application context
| * 58e0329 feat: expose actuator health endpoint with liveness and readiness probes
| * 7b86484 test: add MockMvc tests for calculator endpoints
| * 9589a23 feat: return structured 400 errors for invalid input and division by zero
| * 67dec6d feat: add calculation response DTO and REST controller
| * 6ebebd6 test: add unit tests for CalculatorService
| * 3b2cf6f feat: add CalculatorService with arithmetic operations
| * c1a343a build: scaffold Spring Boot project with Maven
|/
* 1bc4057 docs: simplify README overview
* 6958639 chore: add issue templates for features and bugs
* 78d734c chore: add pull request template
* 5fa3099 docs: add initial README with project overview and API endpoints
* db1264f chore: add .gitignore for Java/Maven, IDE and secret files
kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/Documents/sem7_spe/mini_project/calculator-devops$
```

Figure 150: Full history at `v1.0.0`: merge commits for PR #8 and #13, squashed and rebased commits in between.

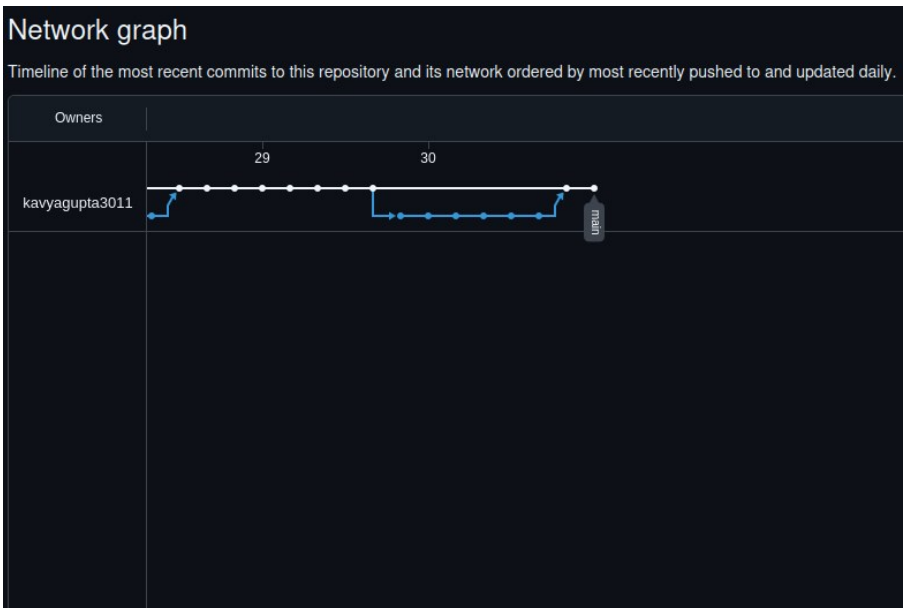


Figure 151: GitHub network graph of the same history.

11 Adding a feature after the release: modulo

To check the whole workflow on a fresh change, a modulo operation was added after the release, following the same path as every other feature: issue, branch, pull request, required check, merge, automatic deployment.

11.1 Issue and code

Issue #17 *Modulo operation* was created from the feature template with tasks and acceptance criteria, labelled `feature`, assigned to milestone `v1.1 -- Extras` and added to the board automatically. Before any change, the deployed API returned 404 for the new path.

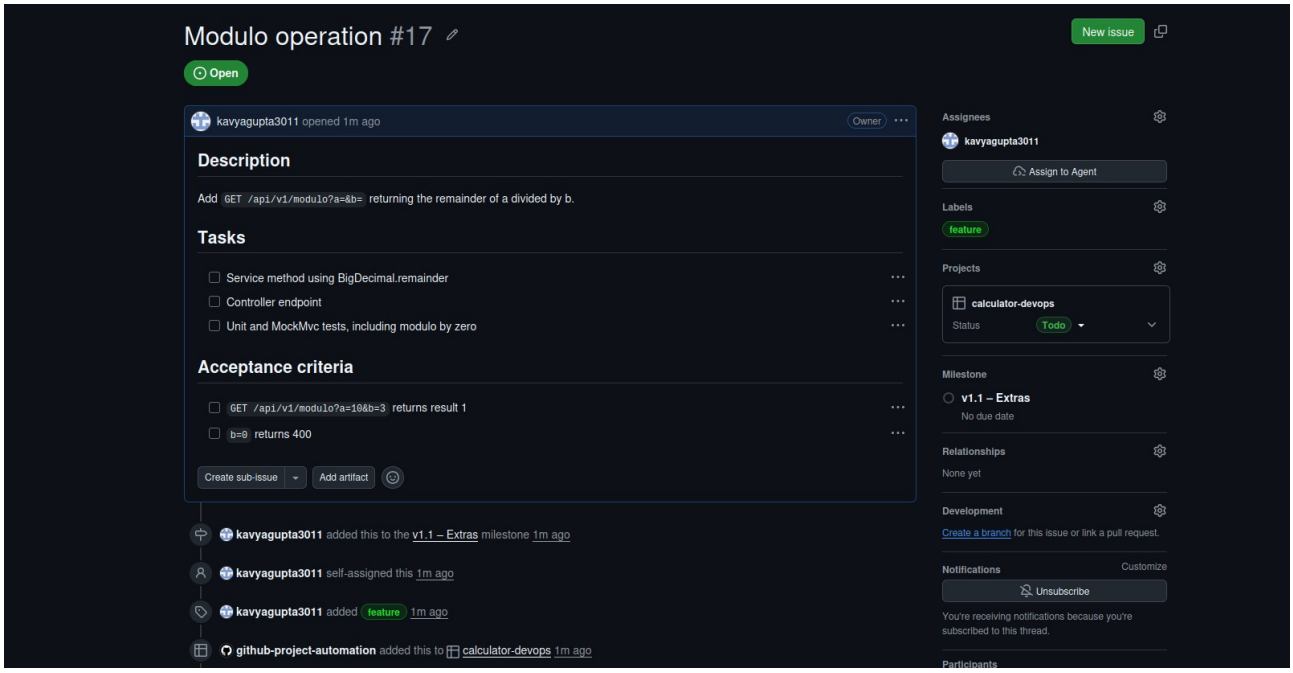


Figure 152: Issue #17 with label, milestone and board status.



Figure 153: Before the change: `/api/v1/modulo` returns 404 through Nginx.

`CalculatorService.modulo` uses `BigDecimal.remainder` with the same `DECIMAL64` precision and rejects a zero divisor with an `ArithmeticException`, which the existing exception handler already turns into a 400. The controller maps it to `GET /api/v1/modulo`. Tests were added at both levels: a parameterized unit test with four cases (10 mod 3, −10 mod 3, 7.5 mod 2, 6 mod 3), a modulo-by-zero unit test, and two MockMvc tests for the 200 and 400 responses. The total went from 36 to 43 tests. The code and tests were committed separately (`feat:` and `test:`).

11.2 Fixing a wrongly named branch

The two commits were made on a branch called `feature/modulo_new`, but a leftover empty branch `feature/modulo` (pointing at `main`) was pushed instead. Since that branch held no work, it was deleted on GitHub (`git push origin --delete feature/modulo`) and locally, the correct branch was renamed with `git branch -m feature/modulo` and pushed with `-u`. No history was rewritten on any shared branch.

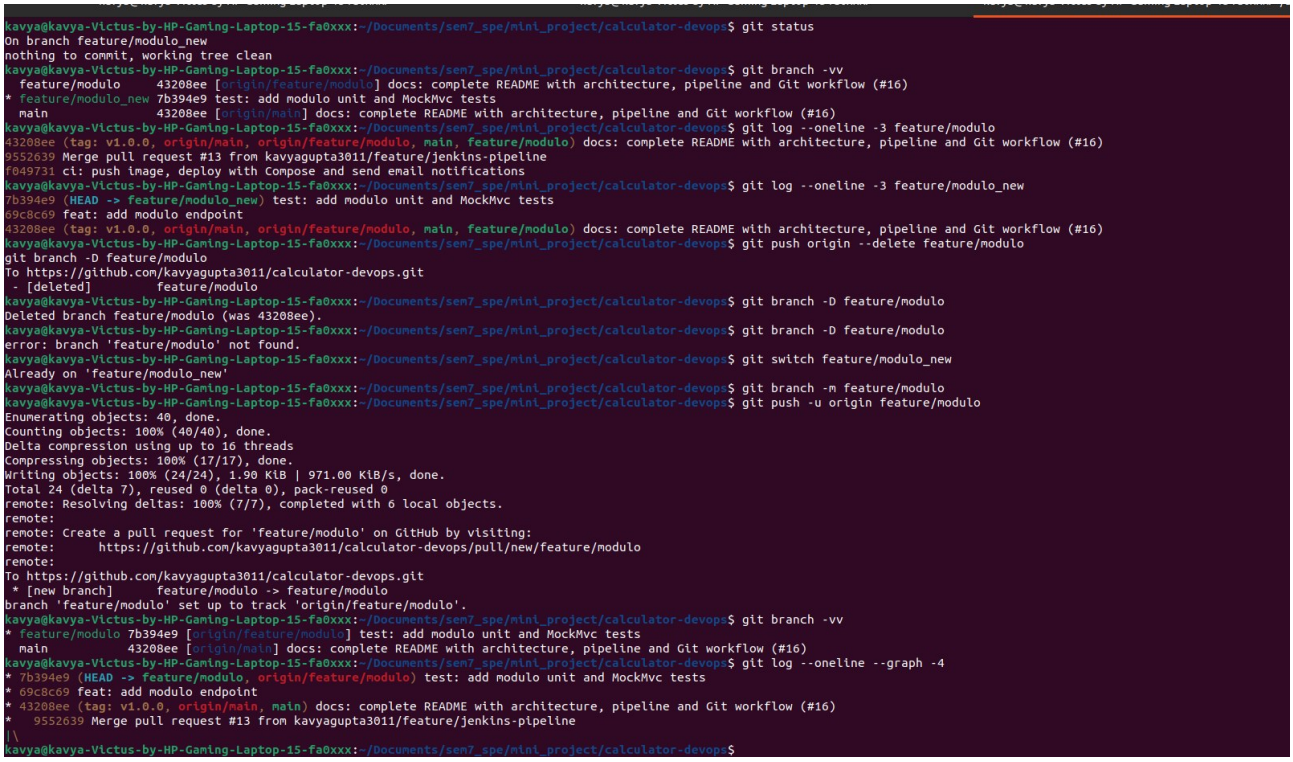


Figure 154: Inspecting both branches, deleting the empty one, renaming and pushing the correct branch; two commits on top of `v1.0.0`.

11.3 Pipeline

The push started a branch build that ran Checkout, Build, Test and Docker Build with all 43 tests passing; Push and Deploy were skipped because the branch is not `main`. The pull request (`Closes #17`) passed the required Jenkins check and was merged with *squash and merge*, closing the issue. The webhook then started `main` build #4, which pushed image tag `4` and `latest` and redeployed the stack.

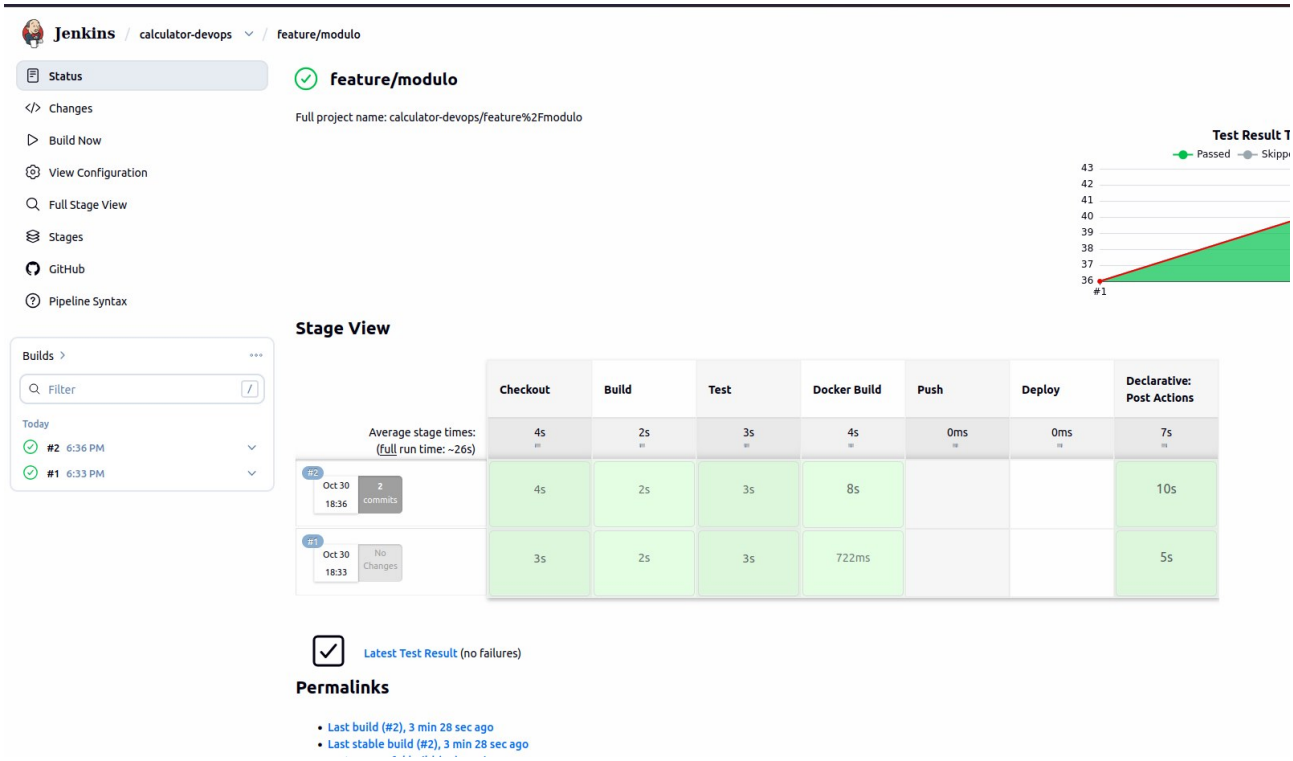


Figure 155: Branch build: test count rises from 36 to 43; Push and Deploy skipped.

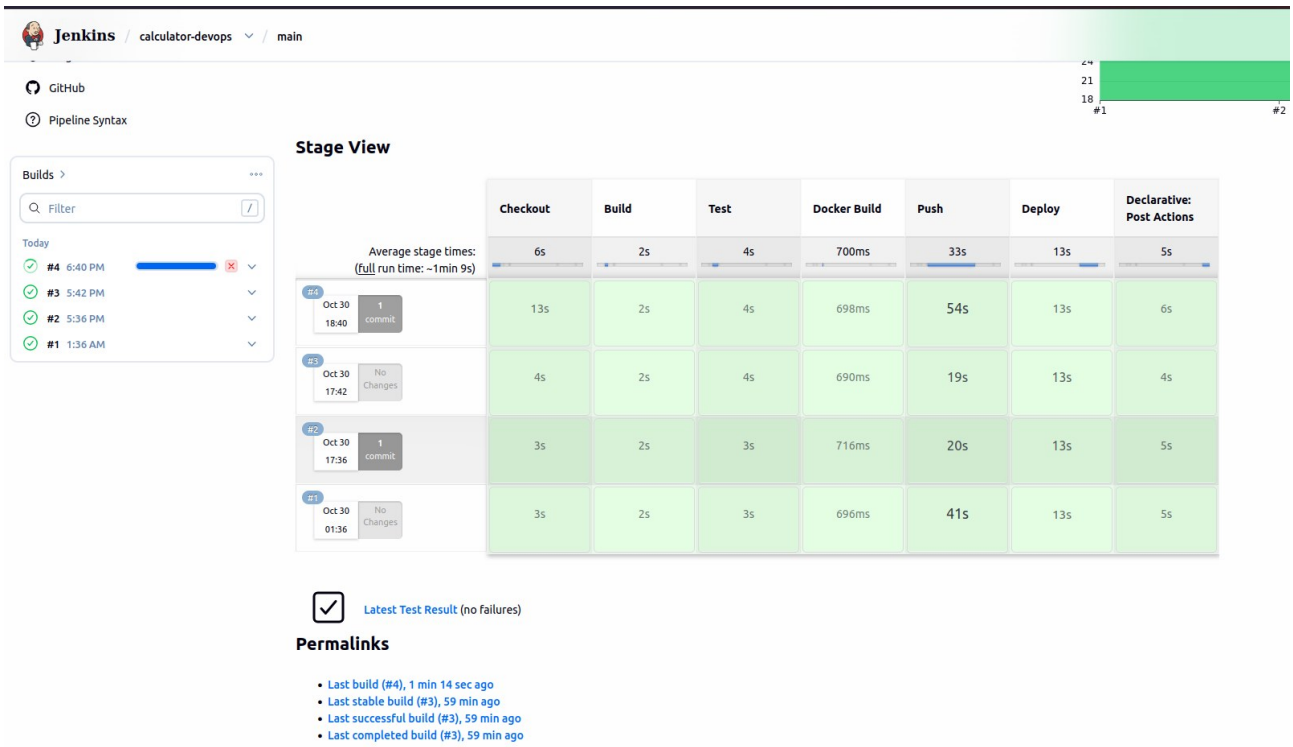


Figure 156: `main` build #4 after the merge: all stages green.

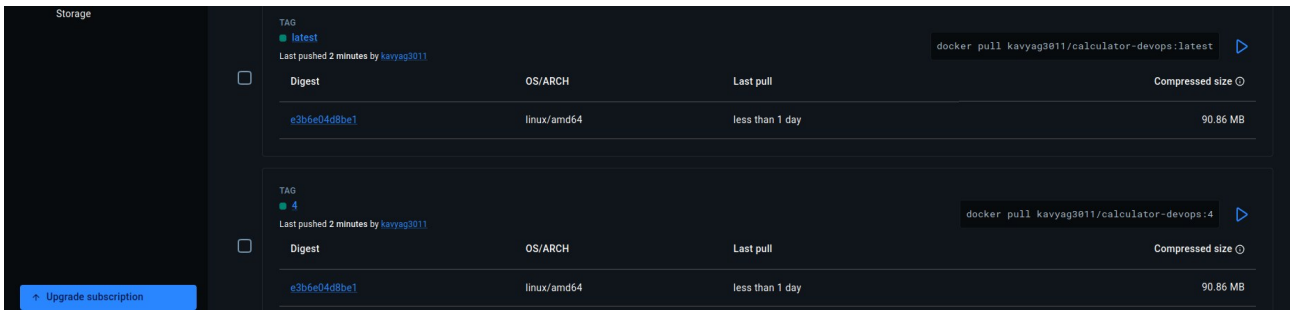


Figure 157: Docker Hub: tag `4` and `latest` with the same digest.

11.4 Result

After the deployment the new endpoint answers through Nginx: 10 mod 3 returns `1` with HTTP 200, and a zero divisor returns HTTP 400 with the message *Modulo by zero is not allowed* in the usual error format. The API container was recreated on `calculator-devops:4` while Nginx kept running.



Figure 158: After build #4: modulo returns 200 and 400 as expected; stack running image tag 4.

12 Summary of Git practices

Practice	Where	Figure
Issues, labels, milestones, board automation	every phase	26, 28
Feature branch per change, linked to an issue	every phase	36
Conventional commit messages	every commit	58
Pull request template, `Closes #N`, line reviews	PR #8, #10, #12, #13, #16	56
Merge commit	PR #8, #13, #16	57
Squash and merge	PR #9, #10, #11, modulo PR	75
Rebase and merge	PR #12	94
`git rebase`, `--force-with-lease`	Phase 3	74
Merge conflict resolution	Phase 4	92
`git stash`	Phase 2	46
`git restore`, `git reset` on unpushed work	Phases 1, 2	51
`git revert` on pushed work	Phase 5	141
Branch protection with required Jenkins check	Phases 1, 5	117
No secrets in history	Phase 6	35
Commit statuses from Jenkins	Phase 5	104
Release tag and GitHub release	`v1.0.0`	148
Deleting a remote branch, renaming a local branch	modulo	154

Table 5: Git and GitHub practices used.

13 Challenges faced

Problem	Cause and fix
`docker compose` not found	Only the standalone binary was installed; linked into the system-wide `cli-plugins` directory.
`permission denied` on `docker.sock`	Jenkins runs as `jenkins`; added the user to the `docker` group and restarted Jenkins.
Builds sent to a node without tools	Lab agent set to "use as much as possible"; restricted to label-matching jobs.
Port 8080 already in use	Jenkins uses 8080; the API defaults to 8081 and Nginx is published on 8000.
HTTP 500 for division by zero	Exceptions reached Spring's default handler; added a `@RestControllerAdvice` returning 400.
Nginx configuration deleted after builds	`cleanWs()` removed the bind-mounted file; Compose now runs from a fixed deploy directory.
Stale upstream IP after redeploy	Nginx caches DNS at start-up; used Docker's resolver with a variable in `proxy_pass`.
Webhook not triggering	Webhook not saved and the free ngrok URL changes; re-added and verified in *Recent Deliveries*.
Push authentication failure	Expired token cached by `credential.helper store`; replaced with a new scoped token.
Docker Hub pull timeout	Retried the pull.
Image tag collisions between branches	Branch builds number from 1; non-`main` tags are prefixed with the branch name.
Wrong branch pushed	Commits were on `feature/modulo_new` but an empty `feature/modulo` was pushed; deleted the empty branch, renamed the correct one and pushed it.

Table 6: Problems encountered and their fixes.

A Source listings

A.1 Jenkinsfile

```
def appImage

pipeline {
    agent any

    options {
        skipDefaultCheckout(true)
        disableConcurrentBuilds()
        buildDiscarder(logRotator(numToKeepStr: '15'))
        timeout(time: 30, unit: 'MINUTES')
    }

    environment {
        IMAGE_NAME = 'kavyag3011/calculator-devops'
        // main builds are tagged with the plain build number, other branches get a prefix
        IMAGE_TAG = "${env.BRANCH_NAME == 'main' ? env.BUILD_NUMBER : env.BRANCH_NAME.replaceAll('[^A-Za-z0-9_.-]', '-') + '-'
 + env.BUILD_NUMBER}"
        // Compose files are copied here so the running stack does not depend on the build workspace
        DEPLOY_DIR = "${env.JENKINS_HOME}/deploy/calculator"
        APP_URL = 'http://localhost:8000'
        NOTIFY_TO = 'kavyagupta3030@gmail.com'
    }

    stages {
        stage('Checkout') {
            steps {
                checkout scm
                sh 'git log -1 --oneline'
                script {
                    env.GIT_SHORT = sh(script: 'git rev-parse --short HEAD', returnStdout: true).trim()
                }
            }
        }

        stage('Build') {
            steps {
                sh 'mvn -B clean package -DskipTests'
            }
        }

        stage('Test') {
```

```
            steps {
                sh 'mvn -B test'
            }
            post {
                always {
                    junit testResults: 'target/surefire-reports/*.xml', allowEmptyResults: false
                }
            }
        }

        stage('Docker Build') {
            steps {
                script {
                    appImage = docker.build("${IMAGE_NAME}:${IMAGE_TAG}")
                }
                sh 'docker images "$IMAGE_NAME" --format "{{.Repository}}:{{.Tag}}  {{.Size}}" | head -5'
            }
        }

        stage('Push') {
            when { branch 'main' }
            steps {
                script {
                    docker.withRegistry('', 'DockerHubCred') {
                        appImage.push()
                        appImage.push('latest')
                    }
                }
            }
        }

        stage('Deploy') {
            when { branch 'main' }
            steps {
                sh '''
                    mkdir -p "$DEPLOY_DIR/nginx"
                    cp docker-compose.yml "$DEPLOY_DIR/"
                    cp nginx/nginx.conf "$DEPLOY_DIR/nginx/"
                    cd "$DEPLOY_DIR"

                    # Recreates only the services whose image or configuration changed
                    docker compose up -d --no-build --remove-orphans
                    docker compose exec -T nginx nginx -s reload
                    docker compose ps
                '''
                sh '''
                    for i in $(seq 1 30); do
                        if curl -fsS "$APP_URL/actuator/health" | grep -q '"status":"UP"'; then
                            echo "Health check passed (attempt $i)"
                            break
                        fi
                        if [ "$i" -eq 30 ]; then
                            echo "Health check failed after 60 seconds"
                            cd "$DEPLOY_DIR" && docker compose ps && docker compose logs --tail=50
                            exit 1
                        fi
                        sleep 2
                    done

                    RUNNING=$(docker inspect -f '{{.Config.Image}}' calculator-calculator-1)
                    echo "Running image: $RUNNING"
                    test "$RUNNING" = "$IMAGE_NAME:$IMAGE_TAG"

                    curl -fsS "$APP_URL/api/v1/add?a=2&b=3"
                    echo
                '''
            }
        }
    }

    post {
        success {
            mail to: "${env.NOTIFY_TO}",
                 subject: "SUCCESS: ${env.JOB_NAME} #${env.BUILD_NUMBER}",
                 body: """Build succeeded.

Job:      ${env.JOB_NAME}
Build:    #${env.BUILD_NUMBER}
Branch:   ${env.BRANCH_NAME}
Commit:   ${env.GIT_SHORT ?: 'n/a'}
Image:    ${env.IMAGE_NAME}:${env.IMAGE_TAG}
Deploy:   ${env.BRANCH_NAME == 'main' ? 'deployed to ' + env.APP_URL : 'not deployed (only main is deployed)'}
Duration: ${currentBuild.durationString.replace(' and counting', '')}

Details: ${env.BUILD_URL}
"""
        }
        failure {
            mail to: "${env.NOTIFY_TO}",
                 subject: "FAILURE: ${env.JOB_NAME} #${env.BUILD_NUMBER}",
                 body: """Build failed.
```

```
132
133 Job:      ${env.JOB_NAME}
134 Build:    #${env.BUILD_NUMBER}
135 Branch:   ${env.BRANCH_NAME}
136 Commit:   ${env.GIT_SHORT ?: 'n/a'}
137 Duration: ${currentBuild.durationString.replace(' and counting', '')}
138
139 Console log: ${env.BUILD_URL}console
140 Test results: ${env.BUILD_URL}testReport/
141 """
142         }
143         cleanup {
144             cleanWs()
145         }
146     }
147 }
```

A.2 Dockerfile and .dockerignore

```
1  # Stage 1: build the JAR with Maven
2  FROM maven:3.9.11-eclipse-temurin-21 AS build
3  WORKDIR /build
4
5  # Download dependencies first so this layer is cached until pom.xml changes
6  COPY pom.xml .
7  RUN mvn -B -q dependency:go-offline
8
9  # Then copy the source and package (tests run in the Jenkins Test stage)
10 COPY src ./src
11 RUN mvn -B -q package -DskipTests
12
13 # Stage 2: run the JAR on a small JRE image
14 FROM eclipse-temurin:21-jre-alpine
15
16 LABEL org.opencontainers.image.title="calculator" \
17       org.opencontainers.image.description="Calculator REST API" \
18       org.opencontainers.image.source="https://github.com/kavyagupta3011/calculator-devops"
19
20 RUN addgroup -S app && adduser -S app -G app
21
22 WORKDIR /app
23 COPY --from=build --chown=app:app /build/target/calculator.jar app.jar
24
25 USER app
26
27 ENV SERVER_PORT=8081
28 EXPOSE 8081
29
30 HEALTHCHECK --interval=15s --timeout=3s --start-period=30s --retries=3 \
31   CMD wget -qO- "http://localhost:${SERVER_PORT}/actuator/health" || exit 1
32
33 ENTRYPOINT ["java", "-jar", "/app/app.jar"]
```

```
1  # Build output (the image builds its own JAR)
2  target/
3
4  # Version control and CI files
5  .git/
6  .github/
7  .gitignore
8  Jenkinsfile
9
10 # Deployment files
11 docker-compose.yml
12 nginx/
13
14 # IDE and OS files
15 .idea/
16 .vscode/
17 *.iml
18 .DS_Store
19
20 # Logs and secrets
21 *.log
22 .env
23 *.env
```

A.3 docker-compose.yml

```
1  name: calculator
2
3  services:
```

```
  calculator:
    image: ${IMAGE_NAME:-kavyag3011/calculator-devops}:${IMAGE_TAG:-dev}
    build: .
    environment:
      SERVER_PORT: "8081"
      JAVA_TOOL_OPTIONS: "-XX:MaxRAMPercentage=75"
    expose:
      - "8081"
    healthcheck:
      test: ["CMD-SHELL", "wget -qO- http://localhost:8081/actuator/health || exit 1"]
      interval: 10s
      timeout: 3s
      retries: 5
      start_period: 30s
    restart: unless-stopped
    networks:
      - backend

  nginx:
    image: nginx:1.27-alpine
    ports:
      - "${NGINX_PORT:-8000}:80"
    volumes:
      - ./nginx/nginx.conf:/etc/nginx/conf.d/default.conf:ro
    depends_on:
      calculator:
        condition: service_healthy
    healthcheck:
      test: ["CMD-SHELL", "wget -qO- http://localhost/nginx-health || exit 1"]
      interval: 10s
      timeout: 3s
      retries: 3
    restart: unless-stopped
    networks:
      - frontend
      - backend

networks:
  frontend:
  backend:
    internal: true
```

A.4 nginx/nginx.conf

```
server {
    listen 80;
    server_name _;

    # Docker's internal DNS. Re-resolving the service name means a redeployed
    # calculator container (with a new IP) is picked up without restarting Nginx.
    resolver 127.0.0.11 valid=10s ipv6=off;
    set $calculator http://calculator:8081;

    proxy_set_header Host              $host;
    proxy_set_header X-Real-IP         $remote_addr;
    proxy_set_header X-Forwarded-For   $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_connect_timeout 5s;
    proxy_read_timeout    10s;

    location /api/ {
        proxy_pass $calculator;
    }

    location /actuator/health {
        proxy_pass $calculator;
    }

    location = /nginx-health {
        access_log off;
        default_type text/plain;
        return 200 "ok\n";
    }

    location / {
        default_type application/json;
        return 404 '{"status":404,"error":"Not Found","message":"Use /api/v1/... or /actuator/health"}\n';
    }
}
```

A.5 CalculatorService.java

```
package com.kavyagupta.calculator.service;
```

```
import java.math.BigDecimal;
import java.math.MathContext;

import org.springframework.stereotype.Service;

@Service
public class CalculatorService {

    // 16 significant digits, same precision as a double, but in decimal.
    private static final MathContext PRECISION = MathContext.DECIMAL64;

    public BigDecimal add(BigDecimal a, BigDecimal b) {
        return normalize(a.add(b, PRECISION));
    }

    public BigDecimal subtract(BigDecimal a, BigDecimal b) {
        return normalize(a.subtract(b, PRECISION));
    }

    public BigDecimal multiply(BigDecimal a, BigDecimal b) {
        return normalize(a.multiply(b, PRECISION));
    }

    public BigDecimal divide(BigDecimal a, BigDecimal b) {
        if (b.signum() == 0) {
            throw new ArithmeticException("Division by zero is not allowed");
        }
        return normalize(a.divide(b, PRECISION));
    }

    public BigDecimal modulo(BigDecimal a, BigDecimal b) {
        if (b.signum() == 0) {
            throw new ArithmeticException("Modulo by zero is not allowed");
        }
        return normalize(a.remainder(b, PRECISION));
    }

    // 5.000 -> 5, 2.50 -> 2.5
    private BigDecimal normalize(BigDecimal value) {
        return value.signum() == 0 ? BigDecimal.ZERO : value.stripTrailingZeros();
    }
}
```

A.6 CalculatorController.java

```
package com.kavyagupta.calculator.controller;

import java.math.BigDecimal;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

import com.kavyagupta.calculator.dto.CalculationResponse;
import com.kavyagupta.calculator.service.CalculatorService;

@RestController
@RequestMapping("/api/v1")
public class CalculatorController {

    private final CalculatorService calculatorService;

    public CalculatorController(CalculatorService calculatorService) {
        this.calculatorService = calculatorService;
    }

    @GetMapping("/add")
    public CalculationResponse add(@RequestParam BigDecimal a, @RequestParam BigDecimal b) {
        return new CalculationResponse("add", a, b, calculatorService.add(a, b));
    }

    @GetMapping("/subtract")
    public CalculationResponse subtract(@RequestParam BigDecimal a, @RequestParam BigDecimal b) {
        return new CalculationResponse("subtract", a, b, calculatorService.subtract(a, b));
    }

    @GetMapping("/multiply")
    public CalculationResponse multiply(@RequestParam BigDecimal a, @RequestParam BigDecimal b) {
        return new CalculationResponse("multiply", a, b, calculatorService.multiply(a, b));
    }

    @GetMapping("/divide")
    public CalculationResponse divide(@RequestParam BigDecimal a, @RequestParam BigDecimal b) {
        return new CalculationResponse("divide", a, b, calculatorService.divide(a, b));
    }
```

```
42
43      @GetMapping("/modulo")
44      public CalculationResponse modulo(@RequestParam BigDecimal a, @RequestParam BigDecimal b) {
45          return new CalculationResponse("modulo", a, b, calculatorService.modulo(a, b));
46      }
47  }
```

A.7 GlobalExceptionHandler.java

```
1   package com.kavyagupta.calculator.exception;
2
3   import java.time.Instant;
4
5   import org.springframework.http.HttpStatus;
6   import org.springframework.http.ResponseEntity;
7   import org.springframework.web.bind.MissingServletRequestParameterException;
8   import org.springframework.web.bind.annotation.ExceptionHandler;
9   import org.springframework.web.bind.annotation.RestControllerAdvice;
10  import org.springframework.web.method.annotation.MethodArgumentTypeMismatchException;
11
12  import com.kavyagupta.calculator.dto.ErrorResponse;
13
14  import jakarta.servlet.http.HttpServletRequest;
15
16  @RestControllerAdvice
17  public class GlobalExceptionHandler {
18
19      // ?a=5 (b not given)
20      @ExceptionHandler(MissingServletRequestParameterException.class)
21      public ResponseEntity<ErrorResponse> handleMissingParameter(
22              MissingServletRequestParameterException ex, HttpServletRequest request) {
23          return badRequest("Required parameter '" + ex.getParameterName() + "' is missing", request);
24      }
25
26      // ?a=abc, ?a=NaN, ?a=Infinity
27      @ExceptionHandler(MethodArgumentTypeMismatchException.class)
28      public ResponseEntity<ErrorResponse> handleTypeMismatch(
29              MethodArgumentTypeMismatchException ex, HttpServletRequest request) {
30          return badRequest("Parameter '" + ex.getName() + "' must be a number, but was '" + ex.getValue() + "'",
31                  request);
32      }
33
34      // division by zero
35      @ExceptionHandler(ArithmeticException.class)
36      public ResponseEntity<ErrorResponse> handleArithmetic(ArithmeticException ex, HttpServletRequest request) {
37          return badRequest(ex.getMessage(), request);
38      }
39
40      private ResponseEntity<ErrorResponse> badRequest(String message, HttpServletRequest request) {
41          HttpStatus status = HttpStatus.BAD_REQUEST;
42          ErrorResponse body = new ErrorResponse(
43                  Instant.now(),
44                  status.value(),
45                  status.getReasonPhrase(),
46                  message,
47                  request.getRequestURI());
48          return ResponseEntity.status(status).body(body);
49      }
50  }
```